

THE TETRAHEDRAL HORN PROBLEM AND ASYMPTOTICS OF $U(N)$ $6j$ SYMBOLS

ABSTRACT. Horn's problem is concerned with characterizing the eigenvalues (a, b, c) of a Hermitian matrices (A, B, C) satisfying the constraint $A + B = C$ and has deep connections to tensor product invariants, Littlewood-Richardson coefficients, geometric invariant theory and the intersection theory of Schubert varieties. This paper concerns a generalization of Horn's problem we call the *tetrahedral Horn problem* that aims to characterize the tuples of eigenvalues (a, b, c, d, e, f) of Hermitian matrices (A, B, C, D, E, F) forming the edges of a tetrahedron in the space of Hermitian matrices and thus satisfying the constraints $A + B = C$, $B + D = F$, $D + C = E$ and $A + F = E$. We derive new inequalities satisfied by the Schur-polynomials of such eigenvalues and, using eigenvalue estimation techniques from quantum information theory, prove their satisfaction up to degree k implies the existence of approximate solutions with error $O(\sqrt{\frac{\log k}{k}})$. Moreover, the existence of such a matrices is connected to the semiclassical asymptotics of the $6j$ -symbols for the unitary group $U(n)$, which are maps between multiplicity spaces which encode the associativity constraint $(V_\alpha \otimes V_\beta) \otimes V_\delta \cong V_\alpha \otimes (V_\beta \otimes V_\delta)$ for unitary irreducible representations. Using our techniques, we prove the decay rate of the norms of such $6j$ -symbols is either inverse-polynomial or exponential depending on whether there exists such tetrahedra of Hermitian matrices.

CONTENTS

1. Introduction	2
2. Background material	6
2.1. Representation theory	6
2.2. Probability measures	13
3. Inequalities for tetrahedral eigenvalues	17
3.1. Necessary conditions	19
3.2. Distance bound & sufficiency	20
References	23
4. [tc] Introduction	31
Acknowledgments	41
5. [tc] Inequalities for tetrahedral eigenvalues	42
5.1. [tc] Inequalities for the tetrahedral Horn problem	42
6. [tc] Distance bounds via eigenvalue concentration	47
6.1. [tc] Distance bound for the tetrahedral Horn problem	47
6.2. [tc] Asymptotics of the $6j$ symbols	50

Date: September 1, 2025.

Appendix A. Symmetries of tetrahedral eigenvalues	53
Appendix B. Results for the original Horn problem	58
B.1. Necessary conditions	59
B.2. Distance bound & sufficiency	61
Appendix C. The large n limit & an entropic inequality	62

1. INTRODUCTION

Denote by

$$(1) \quad \text{eig} : \mathcal{H}_n = \{A \in \text{Mat}_n(\mathbb{C}); A^\dagger = A\} \rightarrow \mathbb{R}_{\text{ord}}^n = \{(a_1 \geq a_2 \cdots \geq a_n) \in \mathbb{R}^n\}$$

the map assigning to a Hermitian matrix the set of its ordered eigenvalues. For $a \in \mathbb{R}_{\text{ord}}^n$, it is convenient to use notation $\text{Tr}[a] = \sum_{i=1}^n a_i$, where $\text{Tr}[a]$ coincides with the trace of a diagonal matrix with eigenvalues a_i .

The Horn problem is a classical Linear Algebra problem asking to determine the set

$$(2) \quad \text{Horn}(n) := \left\{ (a, b, c) \in (\mathbb{R}_{\text{ord}}^n)^3 \left| \begin{array}{l} \exists (A, B, C) \in (\mathcal{H}_n)^3, \\ \text{eig}(A) = a, \text{eig}(B) = b, \text{eig}(C) = c, \\ A + B = C \end{array} \right. \right\}.$$

The necessary and sufficient conditions for $(a, b, c) \in \text{Horn}(n)$ consists of the trace equality

$$(3) \quad \text{Tr}[c] = \text{Tr}[a] + \text{Tr}[b],$$

and of a system of linear inequalities on (a, b, c) . These inequalities were first conjectured by Horn [Hor62], and they were established by Klyachko [Kly98], and by Knutson-Tao [KT99]. In the case of $n = 2$, the Horn problem reduces to the question of existence of triangles with given length sides in $\mathbb{R}^3$, and Horn inequalities reduce to triangle inequalities.

The fact that Horn inequalities are linear in (a, b, c) has several explanations stemming from different fields in Mathematics. One approach is as follows: for fixed (a, b) , the possible values of c such that $(a, b, c) \in \text{Horn}(n)$ form a convex polytope, and this follows from the Kirwan Convexity Theorem in symplectic geometry [kirwan???]. Another approach [APS17] is by relating the Horn problem to a similar problem for weighted planar network, in which case the counterpart of the eigenvalue map is piece-wise linear. Yet another approach is through representation theory of the group $U(n)$ to complete.

In this paper, we consider the following generalization of the set $\text{Horn}(n)$:

$$(4) \quad \text{Tetra}(n) := \left\{ (a, b, c, d, e, f) \in (\mathbb{R}_{\text{ord}}^n)^6 \left| \begin{array}{l} \exists (A, B, C, D, E, F) \in (\mathcal{H}_n)^6, \\ \text{eig}(A) = a, \text{eig}(B) = b, \text{eig}(C) = c, \\ \text{eig}(D) = d, \text{eig}(E) = e, \text{eig}(F) = f, \\ A + B = C, B + D = F, C + D = E, A + F = E \end{array} \right. \right\}.$$

We say that 6-tuples $(a, b, c, d, e, f) \in \text{Tetra}(n)$ are tetrahedral, and we refer to the problem of determining $\text{Tetra}(n)$ as the tetrahedral Horn problem. This name is motivated by the fact that for $n = 2$ the set $\text{Tetra}(2)$ is closely related to the set of tetrahedra in $\mathbb{R}^3$. The tetrahedral Horn problem was stated in [CŞW18] (see Section 6, Problem 8) motivated by the quantum marginal problem in quantum information theory (**is it true?**).

As for the Horn problem, there are necessary trace conditions for the tetrahedral Horn problem:

$$(5) \quad \text{Tr}[c] = \text{Tr}[a] + \text{Tr}[b], \quad \text{Tr}[e] = \text{Tr}[a] + \text{Tr}[b] + \text{Tr}[d], \quad \text{Tr}[f] = \text{Tr}[b] + \text{Tr}[d].$$

Denote by $\mathcal{H}_n^+ \subset \mathcal{H}_n$ the set of non-negative Hermitian matrices, and by $\mathbb{R}_+^n = \{a \in \mathbb{R}_{\text{ord}}^n; a_n \geq 0\}$ the corresponding set of non-negative ordered tuples of non-negative real numbers. One can then define

$$(6) \quad \text{Tetra}^+(n) := \{(a, b, c, d, e, f) \in (\mathbb{R}_+^n)^6 \cap \text{Tetra}(n)\}.$$

Without loss of generality, the tetrahedral Horn problem can be reduced to the problem of characterizing $\text{Tetra}^+(n)$. The reason for this is because $(a, b, c, d, e, f) \in \text{Tetra}(n)$ if and only if $(\tilde{a}, \tilde{b}, \tilde{c}, \tilde{d}, \tilde{e}, \tilde{f}) \in \text{Tetra}^+(n)$ where $(\tilde{a}, \tilde{b}, \tilde{c}, \tilde{d}, \tilde{e}, \tilde{f})$ is obtained by a shifting of the components of (a, b, c, d, e, f) by particular positive real numbers (for more details see Corollary 30 in Appendix A).

Besides the trace conditions, other necessary conditions for the tetrahedral Horn problem are available in literature. First, observe that the ordinary Horn inequalities apply to triples $(a, b, c), (b, d, f), (a, f, e), (c, d, e)$. In addition to these linear inequalities, there are many non-linear ones, see *e.g.* [BS15; TCL11; Sag13; BGS08; FG13; LS14; Sra], [NS23, Section 6.2], [Sch52], [SJS18, Equation 5.2]. As an example, we will single out the following entropic inequality [????]:

$$(7) \quad s(c) + s(f) \geq s(b) + s(e), \quad s(x) = - \sum_{i=1}^n x_i \log x_i.$$

The non-linear inequalities are needed in the description of the set $\text{Tetra}^+(n)$ since this set is not a polyhedral cone (in contrast to the set $\text{Horn}(n)$). We illustrate this fact by the Figure 1 depicts a three-dimensional projection of the space $\text{Tetra}^+(2)$ for fixed values of a, b, d .

In order to describe our main results, we will need the following notation. For $a \in \mathbb{R}_+^n$, we denote by $O_a = \{A \in \mathcal{H}_n^+; \text{eig}(A) = a\}$ the set of non-negative Hermitian matrices with eigenvalues a . We denote by

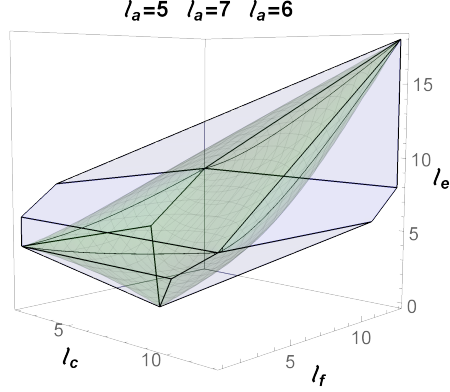


FIGURE 1. A comparison between the values for (ℓ_c, ℓ_e, ℓ_f) with fixed $(\ell_a, \ell_b, \ell_d) = (5, 7, 6)$ which merely satisfy the triangle inequalities (exterior blue polytope) and those which additionally satisfy the Cayley-Menger determinant inequality (interior green). The interior green region depicts (a slice of) the space of Euclidean tetrahedra with side lengths $(\ell_a, \ell_b, \ell_c, \ell_d, \ell_e, \ell_f)$ and thus depicts the solution to the tetrahedral Horn problem for 2×2 traceless Hermitian matrices; remark that it is neither a polytope nor convex.

$\Lambda_+^* = \{(\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n); \lambda_i \in \mathbb{Z}\}$ the set of dominant weights for $G = U(n)$, and by $\Lambda_{++}^* = \{\lambda \in \Lambda_+^*; \lambda_n \geq 0\}$ the set of positive dominant weights. The irreducible representations π_λ corresponding to positive weights are polynomial, and they extend to the matrix algebra $\text{Mat}_n(\mathbb{C})$. For $\lambda \in \Lambda_{++}^*$, we denote by $|\lambda| = \sum_{i=1}^n \lambda_i$, and by H_λ the dimension of the representation for the symmetric group $S_{|\lambda|}$ associated to λ by the Schur-Weyl duality. It will be convenient to use rescaled characters of representations

$$(8) \quad \varphi_\lambda(X) = \frac{1}{H_\lambda} \chi_\lambda(X)$$

extended to $X \in \text{Mat}_n(\mathbb{C})$. Irreducible representations V_λ of $U(n)$ are equipped with Hermitian structures, and the decomposition of their tensor products

$$(9) \quad V_\alpha \otimes V_\beta = \oplus_\gamma C_{\alpha\beta}^\gamma \otimes V_\gamma$$

induces Hermitian structures of multiplicity spaces $C_{\alpha\beta}^\gamma$. The isomorphism between triple tensor products

$$(10) \quad (V_\alpha \otimes V_\beta) \otimes V_\delta \cong V_\alpha \otimes (V_\beta \otimes V_\delta)$$

induces maps

$$(11) \quad \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} : C_{\alpha\beta}^\gamma \otimes C_{\gamma\delta}^\epsilon \rightarrow C_{\alpha\phi}^\epsilon \otimes C_{\beta\delta}^\phi$$

called Wigner's 6j-symbols.

Our first result is a necessary and sufficient condition for tetrahedral 6-tuples:

thm:tet_iff_intro

Theorem 1. *Let $(a, b, c, d, e, f) \in (\mathbb{R}_+^n)^6$ be a 6-tuple satisfying the trace conditions. Then (a, b, c, d, e, f) is tetrahedral if and only if for all triples of positive weights $(\alpha, \beta, \delta) \in (\Lambda_{++}^*)^3$ the following inequalities are satisfied*

{eq:tet_iff_intro}

$$(12) \quad \varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) \leq \sum_{\gamma, \epsilon, \phi} \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty (\varphi_\alpha(a)\varphi_\delta(d)\varphi_\phi(f)\varphi_\gamma(c)\varphi_\epsilon(e))^{\frac{1}{3}}.$$

Furthermore, if inequalities ?? hold true for all triples $(\alpha, \beta, \delta) \in (\Lambda_{++}^*)^3$ with $|\alpha| + |\beta| + |\delta| = k$, then

$$(13) \quad \frac{1}{\text{Tr}[e]} D \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \leq 36n(n+1) \sqrt{\frac{\log(k+1)}{k}},$$

where

$$(14) \quad D \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} := \min_{(A, B, D) \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d} \left(\|\text{eig}(A+B) - c\|_1^2 + \|\text{eig}(B+D) - f\|_1^2 + \|\text{eig}(A+B+D) - e\|_1^2 \right).$$

Note that in Eq. (12) the sum over γ, ϵ, f on the right hand side is finite since there is only a finite number of triples (γ, ϵ, f) for which the 6j-symbol is nonvanishing. Theorem 1 establishes an infinite set of polynomial (up to taking power 1/3) inequalities which characterize the set $\text{Tetra}^+(n)$. To the best of our knowledge, this the first system of polynomial inequalities which provides not only necessary but also sufficient conditions for $(a, b, c, d, e, f) \in \text{Tetra}^+(n)$. Furthermore, for each integer k we provide a finite set of inequalities, and the set defined by these inequalities is getting arbitrarily close to $\text{Tetra}^+(n)$ with the increase of k .

Our second result concerns the behavior of 6j-symbols for $G = U(n)$ for increasing weights:

thm:6j_estimates

Theorem 2. *Let $(a, b, c, d, e, f) \in \text{Tetra}^+(n)$. Then, there is a sequence of 6-tuples of positive weights $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k) \in (\Lambda_{++}^*)^6$ with $|\epsilon_k| = k$ such that*

{eq:intro converge}

$$(15) \quad \frac{1}{k} (\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k) \rightarrow_{k \rightarrow \infty} \frac{1}{\text{Tr}[e]} (a, b, c, d, e, f),$$

and

$$(16) \quad \left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\| \geq \frac{1}{\text{poly}(k)}.$$

for some positive polynomial of k . Furthermore, if $(a, b, c, d, e, f) \notin \text{Tetra}^+(n)$, then for any sequence $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k) \in (\Lambda_{++}^*)^6$ converging to (a, b, c, d, e, f) we have

$$(17) \quad \left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\| \leq C e^{-kr}$$

for some $C, r > 0$.

Theorem 2 is an instance of a correspondence between geometry and representation theory. Indeed, the asymptotic behavior of representation theoretic objects (6j-symbols) is controlled by solutions of a geometric problem (the tetrahedral Horn problem). The statement of Theorem 2 should be compared with a similar statement available for $G = SU(2)$, see [Rob99]. There, the exponential decay statement is similar to our statement, but the polynomial bound statement is more precise with the sequence

$$(18) \quad (\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k) = k(\alpha_1, \beta_1, \gamma_1, \delta_1, \epsilon_1, \phi_1).$$

While being a weaker statement for $G = SU(2)$, Theorem 2 applies to all n . Implicitly, Theorem 2 is contained in [CSW18]. In this paper, we give a direct proof.

The methods used in the proofs include the Schur-Weyl duality and the non-commutative binomial formula based on it. These features distinguish the case of $G = U(n)$ discussed in this paper and versions of the tetrahedral Horn problem that one may define for other compact groups. Another key observation used in the proofs is the eigenvalue estimation which shows that positive functions $\varphi_\alpha(a)$ are small outside the region near

$$(19) \quad \alpha \sim a \Leftrightarrow \frac{1}{|\alpha|} \alpha = \frac{1}{\text{Tr}[a]} a.$$

This observation together with Theorem 2 explain the meaning of mysterious inequalities in Eq. (12) in Theorem 1. Indeed, by choosing (α, β, δ) such that $\alpha \sim a, \beta \sim b, \delta \sim d$ we maximize the left hand side. The choice of (γ, ϵ, ϕ) with $\gamma \sim c, \epsilon \sim e, \phi \sim f$ maximizes the second factor on the right hand side. If the 6-tuple (a, b, c, d, e, f) is not tetrahedral, the 6j-symbol is exponentially decaying, and Eq. (12) is unlikely to hold for all k .

2. BACKGROUND MATERIAL

[sec:background](#)

In Section 2.1, we review material on the representation theory of compact connected Lie groups, specifically the group $U(n)$, and in Section 2.2 we review some measure theory associated to Haar integrals of such groups. Along the way, we introduce a series of the polynomial functions $\varphi_\lambda(X)$, $\varphi_\gamma^{\alpha\beta}(X, Y)$ and $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ of one, two and three $n \times n$ matrices respectively, and provide a formula for the representation of a matrix sum $X + Y$ which we refer to as the binomial theorem.

[sec:background_rep_theory](#)

2.1. Representation theory. Even though we will primarily deal with $G = U(n)$, we start with a more general story.

2.1.1. G compact. Let G be a compact connected Lie group, $T \subset G$ a maximal torus, $\mathfrak{t} = \text{Lie}(T)$ its Lie algebra, $\Lambda = \ker(\exp) \subset \mathfrak{t}$ the integral lattice, $\Lambda^* \subset \mathfrak{t}^*$ the weight lattice, and $\Lambda_+^* \subset \Lambda^*$ the set of dominant weights of G . For $\lambda \in \Lambda^*$, we denote by $\pi_\lambda : G \rightarrow \text{End}(V_\lambda)$ the corresponding irreducible representation of G . We will always assume that the space V_λ is equipped with an invariant Hermitian structure such that π_λ becomes a

unitary representation. All representations π_λ extend to the corresponding complex Lie group $G^\mathbb{C}$.

The tensor product of two irreducible representations admits a unique decomposition of the form

$$\text{\texttt{\{eq:tensor_product\}}} \quad (20) \quad V_\alpha \otimes V_\beta = \bigoplus_{\gamma} C_{\alpha\beta}^\gamma \otimes V_\gamma,$$

where $C_{\alpha\beta}^\gamma$ are multiplicity spaces with dimensions $c_{\alpha\beta}^\gamma = \dim C_{\alpha\beta}^\gamma$. Hermitian structures on V_λ 's induces Hermitian structures on the multiplicity spaces $C_{\alpha\beta}^\gamma$. In particular, the decomposition in Eq. (20) is an orthogonal decomposition. We let $\iota_{\alpha\beta}^\gamma : C_{\alpha\beta}^\gamma \otimes V_\gamma \rightarrow V_\alpha \otimes V_\beta$ denote the corresponding isometry and $P_{\alpha\beta}^\gamma = \iota_{\alpha\beta}^\gamma \iota_{\alpha\beta}^{\gamma\dagger}$ the corresponding orthogonal projection operator. Using this notation, we have for all $g \in G$

$$\text{\texttt{\{eq:operator_product\}}} \quad (21) \quad \pi_\alpha(g) \otimes \pi_\beta(g) = \sum_{\gamma} \iota_{\alpha\beta}^\gamma (\text{id}_{C_{\alpha\beta}^\gamma} \otimes \pi_\gamma(g)) \iota_{\alpha\beta}^{\gamma\dagger},$$

Characters of irreducible representations are denoted by

$$\text{\texttt{\{eq:character\}}} \quad (22) \quad \chi_\lambda(g) = \text{Tr}_{V_\lambda}[\pi_\lambda(g)].$$

Taking a trace of Eq. (21) we obtain the standard product rule for characters:

$$\text{\texttt{\{eq:product_rule\}}} \quad (23) \quad \chi_\alpha(g) \chi_\beta(g) = \sum_{\gamma} c_{\alpha\beta}^\gamma \chi_\gamma(g).$$

Triple tensor products can be decomposed in two alternative ways:

$$\begin{aligned} (24) \quad (V_\alpha \otimes V_\beta) \otimes V_\delta &= \bigoplus_{\gamma, \epsilon} (C_{\alpha\beta}^\gamma \otimes C_{\gamma\delta}^\epsilon) \otimes V_\epsilon, \\ V_\alpha \otimes (V_\beta \otimes V_\delta) &= \bigoplus_{\phi, \epsilon} (C_{\alpha\phi}^\epsilon \otimes C_{\beta\delta}^\phi) \otimes V_\epsilon. \end{aligned}$$

We will denote the corresponding orthogonal projectors by $\Pi_{\alpha\beta\delta}^{\epsilon;\gamma}$ and $\Pi_{\alpha\beta\delta}^{\epsilon;\phi}$, respectively (the letters γ and ϕ will always be associated to these decompositions):

$$\text{\texttt{\{eq:double_projs\}}} \quad (25) \quad \begin{aligned} \Pi_{\alpha\beta\delta}^{\epsilon;\gamma} &:= P_{\alpha\beta\delta}^\epsilon (P_{\alpha\beta}^\gamma \otimes \text{id}_{V_\delta^n}) = (\iota_{\alpha\beta}^\gamma \otimes \text{id}_{V_\delta^n}) (P_{\gamma\delta}^\epsilon \otimes \text{id}_{C_{\alpha\beta}^\gamma}) (\iota_{\alpha\beta}^\gamma \otimes \text{id}_{V_\delta^n})^\dagger, \\ \Pi_{\alpha\beta\delta}^{\epsilon;\phi} &:= P_{\alpha\beta\delta}^\epsilon (\text{id}_{V_\alpha^n} \otimes P_{\beta\delta}^\phi) = (\text{id}_{V_\alpha^n} \otimes \iota_{\beta\delta}^\phi) (P_{\alpha\phi}^\epsilon \otimes \text{id}_{C_{\beta\delta}^\phi}) (\text{id}_{V_\alpha^n} \otimes \iota_{\beta\delta}^\phi)^\dagger. \end{aligned}$$

For fixed α, β, δ and γ, ϕ, ϵ , consider the composition of these two projectors $\Pi_{\alpha\beta\delta}^{\epsilon;\gamma}$ and $\Pi_{\alpha\beta\delta}^{\epsilon;\phi}$. This composition induces a G -equivariant linear map of the type

$$(26) \quad (C_{\alpha\beta}^\gamma \otimes C_{\gamma\delta}^\epsilon) \otimes V_\epsilon \rightarrow (C_{\alpha\phi}^\epsilon \otimes C_{\beta\delta}^\phi) \otimes V_\epsilon.$$

Since π_ϵ is an irreducible representation, by Schur's Lemma this map is a tensor product of the identity on V_ϵ , and a map

$$(27) \quad \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} : C_{\alpha\beta}^\gamma \otimes C_{\gamma\delta}^\epsilon \rightarrow C_{\alpha\phi}^\epsilon \otimes C_{\beta\delta}^\phi$$

called the $6j$ -symbol for G . Concretely, we will define $\left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\}$ by the for-

`{eq:sixj}`

$$(28) \quad \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} := \frac{1}{\dim(V_\epsilon)} \operatorname{Tr}_{V_\epsilon}[(\iota_{\alpha\phi}^\epsilon \otimes \operatorname{id}_{C_{\beta\delta}^\phi})^\dagger (\operatorname{id}_{V_\alpha^n} \otimes \iota_{\beta\delta}^\phi)^\dagger (\iota_{\alpha\beta}^\gamma \otimes \operatorname{id}_{V_\delta^n}) (\operatorname{id}_{C_{\alpha\beta}^\gamma} \otimes \iota_{\gamma\delta}^\epsilon)],$$

which ensures

(29)

$$\Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Pi_{\alpha\beta\delta}^{\epsilon;\phi} = (\operatorname{id}_{V_\alpha^n} \otimes \iota_{\beta\delta}^\phi) (\iota_{\alpha\phi}^\epsilon \otimes \operatorname{id}_{C_{\beta\delta}^\phi}) \left(\left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \otimes V_\epsilon^n \right) (\operatorname{id}_{C_{\alpha\beta}^\gamma} \otimes \iota_{\gamma\delta}^\epsilon)^\dagger (\iota_{\alpha\beta}^\gamma \otimes \operatorname{id}_{V_\delta^n})^\dagger.$$

Recall that for Hermitian vector spaces V_1, V_2 and an operator $O : V_1 \rightarrow V_2$ one can introduce the norms

$$(30) \quad \begin{aligned} \|O\|_\infty &= \max_{v \in V_1, \|v\|_{V_1}=1} \|O(v)\|_{V_2}, \\ \|O\|_2 &= \operatorname{Tr}_{V_1}[O^\dagger O]^{\frac{1}{2}} = \operatorname{Tr}_{V_2}[OO^\dagger]^{\frac{1}{2}}, \end{aligned}$$

which satisfy the following estimates:

`{eq:norm_estimates}`

$$(31) \quad \|O\|_\infty^2 \leq \|O\|_2^2 \leq \dim(V_1) \|O\|_\infty^2.$$

In particular, for $6j$ -symbols, we have

(32)

$$\left\| \Pi_{\alpha\beta\delta}^{\epsilon;\phi} \Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \right\|_\infty^2 = \left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_\infty^2 \leq \left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_2^2 = \frac{1}{\dim(V_\epsilon)} \operatorname{Tr}(\Pi_{\alpha\beta\delta}^{\epsilon;\phi} \Pi_{\alpha\beta\delta}^{\epsilon;\gamma}).$$

In the above formula, we have used the fact that for two orthogonal Hermitian projectors Π_1 and Π_2 one has

$$(33) \quad \|\Pi_1 \Pi_2\|_2^2 = \operatorname{Tr}((\Pi_1 \Pi_2)(\Pi_1 \Pi_2)^\dagger) = \operatorname{Tr}(\Pi_1 \Pi_2^2 \Pi_1) = \operatorname{Tr}(\Pi_1 \Pi_2).$$

For future use, we will need the following fact:

`prop: Pi Omega Pi`

Proposition 3. *Let Π_1, Π_2 be two orthogonal projections, and Ω be a non-negative Hermitian operator. Then,*

`{eq:Pi Omega Pi}`

$$(34) \quad |\operatorname{Tr}(\Pi_1 \Omega \Pi_2)| \leq \|\Pi_1 \Pi_2\|_\infty (\operatorname{Tr} \Pi_1 \Omega)^{\frac{1}{2}} (\operatorname{Tr} \Omega \Pi_2)^{\frac{1}{2}}.$$

Proof. We compute,

$$(35) \quad \begin{aligned} |\operatorname{Tr}(\Pi_1 \Omega \Pi_2)| &= |\operatorname{Tr}(\Pi_1 \Pi_2 \Omega \Pi_1)|, \\ &= |\operatorname{Tr}(\Pi_1 \Pi_2 \Omega^{\frac{1}{2}})(\Omega^{\frac{1}{2}} \Pi_1)|, \\ &\leq (\operatorname{Tr} \Pi_1 \Pi_2 \Omega \Pi_2 \Pi_1)^{\frac{1}{2}} (\operatorname{Tr} \Pi_1 \Omega \Pi_1)^{\frac{1}{2}}, \\ &= (\operatorname{Tr} \Pi_1 \Pi_2 \Omega \Pi_2 \Pi_1)^{\frac{1}{2}} (\operatorname{Tr} \Pi_1 \Omega)^{\frac{1}{2}}, \end{aligned}$$

where in the third line we have used the Cauchy-Schwarz inequality for the Hilbert-Schmidt norm. Let $U = \operatorname{im}(\Pi_2)$ and choose a basis of eigenvectors

$\{e_j\}$ of the operator $\Pi_2 \Omega \Pi_2$ with eigenvalues $\omega_j \geq 0$. Then,

$$\begin{aligned}
 \text{Tr}(\Pi_1 \Pi_2 \Omega \Pi_2 \Pi_1) &= \text{Tr}(\Pi_2 \Pi_1^2 \Pi_2)(\Pi_2 \Omega \Pi_2), \\
 &= \sum_j \omega_j \|\Pi_1 \Pi_2 e_j\|^2, \\
 &\leq \|\Pi_1 \Pi_2\|_\infty^2 \text{Tr}(\Pi_2 \Omega \Pi_2), \\
 &= \|\Pi_1 \Pi_2\|_\infty^2 \text{Tr}(\Omega \Pi_2).
 \end{aligned}
 \tag{36}$$

This completes the proof. $\square$

2.1.2. $G = U(n)$ and functions φ_λ . For $G = U(n)$, dominant weights are parametrized by non-increasing arrays of integers, $\lambda = (\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n)$, where $\lambda_i \in \mathbb{Z}$ and $\lambda_i \geq \lambda_{i+1}$ for all i . In particular, the first fundamental weight $\varpi_1 = (1, 0, \dots, 0)$ corresponds to the defining representation of $U(n)$ on $V_{\varpi_1} = \mathbb{C}^n$.

We say that a dominant weight λ is positive if each of its components is positive (i.e., $\lambda_n \geq 0$), and we denote the set of positive weights by Λ_{++}^* . For positive weights, we let

$$|\lambda| := \sum_{i=1}^n \lambda_i, \quad \ell(\lambda) := |\{i; \lambda_i \neq 0\}|,
 \tag{37}$$

where $|\lambda|$ is called the size of λ and $\ell(\lambda)$ is called the depth of λ . Henceforth we use the notation $\lambda \vdash_n k$ to denote the set of positive weights with size k and length at most n , and $\lambda \vdash k$ to denote the set of positive weights with size k and arbitrary length. Positive weights λ are then equivalent to integer partitions of $|\lambda|$ into $\ell(\lambda) \leq n$ parts, or as Young diagrams with $|\lambda|$ boxes and at most $\ell(\lambda)$ rows. Representations $\pi_\lambda(g)$ corresponding to positive weights are polynomial in matrix elements of g , and thus they extend to matrix algebras $\text{Mat}_n(\mathbb{C})$. That is,

$$\pi_\lambda(X) \pi_\lambda(Y) = \pi_\lambda(XY)
 \tag{38}$$

for all $X, Y \in \text{Mat}_n(\mathbb{C})$. If X is a Hermitian matrix with eigenvalues $x = (x_1 \geq \dots \geq x_n)$ then the character of X for the representation π_λ , denoted by $\chi_\lambda(X)$, is equal to the Schur-polynomial of its eigenvalues x , denoted by $s_\lambda(x)$. Moreover, the dimension for the irreducible representation V_λ is given by Weyl's dimension formula for $U(n)$ [Hal08; IN66],

$$\dim(V_\lambda) = \frac{\Delta(\lambda + \rho)}{\Delta(\rho)} = \frac{\prod_{1 \leq i < j \leq n} (\lambda_i - i - \lambda_j + j)}{\prod_{1 \leq i < j \leq n} (j - i)},
 \tag{39}$$

where $\rho \simeq (n-1, n-2, \dots, 1, 0)$ is known as the Weyl vector and where

$$\Delta(x_1, \dots, x_n) = \prod_{1 \leq i < j \leq n} (x_i - x_j).
 \tag{40}$$

is known as the Vandermonde determinant. Tensor products of representations with positive weights decompose into direct sums of representations

of the same type. Note that in the decomposition of the tensor product representation $V_\alpha \otimes V_\beta$ appearing in Eq. (20), the only positive weights $|\gamma|$ which appear are those which satisfy the size condition

$$\{eq:size_condition_tri\} \quad (41) \quad |\gamma| = |\alpha| + |\beta|.$$

Recall that for Young diagrams (or partitions) λ with fixed size $|\lambda| = k$ label isomorphism classes of irreducible representations W_λ of the symmetric group S_k . The dimensions of these representations are given by the formula

$$\{eq:dim_W_hook_product\} \quad (42) \quad \dim(W_\lambda) = \frac{|\lambda|!}{H_\lambda}, \quad H_\lambda = \prod_{b \in \lambda} h_\lambda(b),$$

where $h_\lambda(b)$ is the hook length of box b and is defined as the number of boxes to the right and below b in λ plus one for itself, and H_λ is referred to as the hook product of λ . For the future use, we introduce rescaled representations

$$(43) \quad \Phi_\lambda(X) = \frac{1}{H_\lambda} \pi_\lambda(X),$$

and rescaled characters

$$(44) \quad \varphi_\lambda(X) = \text{Tr}_{V_\lambda} \Phi_\lambda(X) = \frac{1}{H_\lambda} \chi_\lambda(X).$$

Note that

$$\{eq:varphi_normalization\} \quad (45) \quad \sum_{\lambda \vdash k} \varphi_\lambda(X) = \frac{[\text{Tr}(X)]^k}{k!}, \quad \sum_{\lambda} \varphi_\lambda(X) = \exp(\text{Tr}(X)).$$

Indeed, by the Schur-Weyl duality, there is an isomorphism of $(S_k \otimes U(n))$ -modules

$$\{eq:schur_weyl\} \quad (46) \quad (\mathbb{C}^n)^{\otimes k} \cong \bigoplus_{\lambda \vdash_n k} W_\lambda \otimes V_\lambda,$$

where the symmetric group S_k acts by permutations of tensor factors in $(\mathbb{C}^n)^{\otimes k}$. For each λ , we let

$$(47) \quad \iota_\lambda : W_\lambda \otimes V_\lambda^n \rightarrow (\mathbb{C}^n)^{\otimes k},$$

denote the unique isometric $S_k \times U(n)$ -equivariant map from $W_\lambda \otimes V_\lambda^n$ to $(\mathbb{C}^n)^{\otimes k}$. Since all $U(n)$ representations entering the decomposition in Eq. (46) are positive, the Schur-Weyl duality extends to an isomorphism of modules for monoids $(S_k \otimes \text{Mat}_n(\mathbb{C}))$. Applying it to an element $X \in \text{Mat}_n(\mathbb{C})$, we obtain

$$\{eq:X^{\otimes k}\} \quad (48) \quad X^{\otimes k} = \bigoplus_{\lambda \vdash_n k} \iota_\lambda(\text{id}_{W_\lambda} \otimes \pi_\lambda(X)) \iota_\lambda^\dagger,$$

Taking a trace of this equation over $(\mathbb{C}^n)^{\otimes k}$ yields Eq. (45).

Henceforth we denote the space of Hermitian matrices of size n by $\mathcal{H}_n$ and the space of positive-semidefinite Hermitian matrices by $\mathcal{H}_n^+$. Given a Hermitian matrix $X \in \mathcal{H}_n$, $\text{eig}(X)$ denotes the eigenvalues of X .

Proposition 4. For $X \in \mathcal{H}_n^+$, $\lambda \in \Lambda_{++}^*$, operators $\pi_\lambda(X)$ and $\Phi_\lambda(X)$ are non-negative Hermitian operators, and functions $\varphi_\lambda(X)$ take non-negative real values on $\mathcal{H}_n^+$.

Proof. By Eq. (48), operators $\text{id}_{W_\lambda} \otimes \pi_\lambda(X)$ are orthogonal restrictions of the non-negative Hermitian operator $X^{\otimes k}$. Hence, they are also non-negative Hermitian operators, and so are $\pi_\lambda(X)$ and $\Phi_\lambda(X)$. Since $\varphi_\lambda(X)$ is the trace of $\Phi_\lambda(X)$, it takes values in $\mathbb{R}_{\geq 0}$. $\square$

2.1.3. *Functions $\varphi_\gamma^{\alpha\beta}$ and $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}$.* It is convenient to introduce the following functions of two matrices,

$$(49) \quad \varphi_\gamma^{\alpha\beta}(X, Y) = \text{Tr}_{V_\alpha \otimes V_\beta}(\Phi_\alpha(X) \otimes \Phi_\beta(Y)) P_{\alpha\beta}^\gamma.$$

Proposition 5. For $\alpha, \beta \in \Lambda_{++}^*$ and $X, Y \in \text{Mat}_n(\mathbb{C})$, we have

{eq:varphi_product_sum}

$$(50) \quad \varphi_\alpha(X) \varphi_\beta(Y) = \sum_\gamma \varphi_\gamma^{\alpha\beta}(X, Y).$$

For $X, Y \in \mathcal{H}_n^+$, the function $\varphi_\gamma^{\alpha\beta}(X, Y)$ takes non-negative real values.

Proof. For the first statement, note that

$$(51) \quad \sum_\gamma P_{\alpha\beta}^\gamma = \text{id}_{V_\alpha} \otimes \text{id}_{V_\beta},$$

which then implies

$$(52) \quad \begin{aligned} \sum_\gamma \varphi_\gamma^{\alpha\beta}(X, Y) &= \sum_\gamma \text{Tr}_{V_\alpha \otimes V_\beta}(\Phi_\alpha(X) \otimes \Phi_\alpha(Y)) P_{\alpha\beta}^\gamma, \\ &= \text{Tr}_{V_\alpha \otimes V_\beta}(\Phi_\alpha(X) \otimes \Phi_\alpha(Y)), \\ &= \varphi_\alpha(X) \varphi_\beta(Y). \end{aligned}$$

For the second statement, we have

$$(53) \quad \begin{aligned} \varphi_\gamma^{\alpha\beta}(X, Y) &= \text{Tr}_{V_\alpha \otimes V_\beta}(\Phi_\alpha(X) \otimes \Phi_\alpha(Y)) P_{\alpha\beta}^\gamma, \\ &= \text{Tr}_{V_\alpha \otimes V_\beta} P_{\alpha\beta}^\gamma (\Phi_\alpha(X) \otimes \Phi_\alpha(Y)) P_{\alpha\beta}^\gamma. \end{aligned}$$

Hence, $\varphi_\gamma^{\alpha\beta}(X, Y)$ is the trace of the orthogonal restriction of the non-negative Hermitian operator $(\Phi_\alpha(X) \otimes \Phi_\alpha(Y))$ to the subspace $(C_{\alpha\beta} \otimes V_\gamma) \subseteq V_\alpha \otimes V_\beta$, meaning $\varphi_\gamma^{\alpha\beta}(X, Y) \geq 0$ as claimed. $\square$

thm:the_binomial_theorem

Theorem 6 (Binomial theorem). For $X, Y \in \text{Mat}_n(\mathbb{C})$ and $\gamma \in \Lambda_{++}^*$, we have

{eq:varphi_sum_sum}

$$(54) \quad \varphi_\gamma(X + Y) = \sum_{\alpha, \beta} \varphi_\gamma^{\alpha\beta}(X, Y).$$

Proof. Let $k = |\gamma|$ and let ρ_k denote the representation of the monoid $(S_k \times \text{Mat}_n(\mathbb{C}))$ on the space $(\mathbb{C}^n)^{\otimes k}$, and $P^\gamma = \iota_\gamma \iota_\gamma^\dagger$ the orthogonal projector to

the summand $W_\gamma \otimes V_\gamma$ in the Schur-Weyl decomposition Eq. (46). Here we use the fact that

$$(55) \quad (X + Y)^{\otimes k} = \sum_{l=0}^k \frac{k!}{l!(k-l)!} \sum_{\sigma \in S_k} \rho_k(\sigma) (X^{\otimes l} \otimes Y^{\otimes(k-l)}) \rho_k(\sigma^{-1})$$

which is a non-commutative counterpart of the binomial decomposition. Recalling that $\dim(W_\lambda) = |\lambda|!/H_\lambda$, we compute

$$(56) \quad \begin{aligned} & \varphi_\gamma(X + Y) \\ &= H_\gamma^{-1} \operatorname{Tr}_{V_\lambda} \pi_\gamma(X + Y), \\ &= (H_\gamma \dim(W_\gamma))^{-1} \operatorname{Tr}_{(\mathbb{C}^n)^{\otimes k}} P^\gamma \rho_k(X + Y), \\ &= k!^{-1} \operatorname{Tr}_{(\mathbb{C}^n)^{\otimes k}} P^\gamma (X + Y)^{\otimes k}, \\ &= k!^{-1} \operatorname{Tr}_{(\mathbb{C}^n)^{\otimes k}} P^\gamma \sum_{l=0}^k \sum_{\sigma \in S_k} \frac{k!}{l!(k-l)!} \rho_k(\sigma) (X^{\otimes l} \otimes Y^{\otimes(k-l)}) \rho_k(\sigma^{-1}), \\ &= \sum_{l=0}^k \sum_{\sigma \in S_k} (l!(k-l)!)^{-1} \operatorname{Tr}_{(\mathbb{C}^n)^{\otimes k}} P^\gamma \rho_k(\sigma) (X^{\otimes l} \otimes Y^{\otimes(k-l)}) \rho_k(\sigma^{-1}), \\ &= \sum_{l=0}^k \sum_{\sigma \in S_k} (l!(k-l)!)^{-1} \operatorname{Tr}_{(\mathbb{C}^n)^{\otimes k}} P^\gamma (X^{\otimes l} \otimes Y^{\otimes(k-l)}), \\ &= \sum_{\alpha, \beta} (|\alpha|!|\beta|!)^{-1} \operatorname{Tr}_{(W_\alpha \otimes V_\alpha) \otimes (W_\beta \otimes V_\beta)} P_{\alpha\beta}^\gamma \pi_\alpha(X) \otimes \pi_\beta(Y), \\ &= \sum_{\alpha, \beta} (H_\alpha H_\beta)^{-1} \operatorname{Tr}_{V_\alpha \otimes V_\beta} P_{\alpha\beta}^\gamma (\pi_\alpha(X) \otimes \pi_\beta(Y)), \\ &= \sum_{\alpha, \beta} \operatorname{Tr}_{V_\alpha \otimes V_\beta} P_{\alpha\beta}^\gamma (\varphi_\alpha(X) \otimes \varphi_\beta(Y)), \\ &= \sum_{\alpha, \beta} \phi_{\alpha\beta}^\gamma(X, Y). \end{aligned}$$

□

For a 6-tuple of positive weights, we introduce a function of three matrices,

$$(57) \quad \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) = \operatorname{Tr}_{V_\alpha \otimes V_\beta \otimes V_\delta} \Pi_{\alpha\beta\delta}^{\epsilon;\phi}(\Phi_\alpha(X) \otimes \Phi_\beta(Y) \otimes \Phi_\delta(Z)) \Pi_{\alpha\beta\delta}^{\epsilon;\gamma}.$$

Note that the following size equalities are necessary conditions for $\left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\}$ being non-zero and thus $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}$ being non-zero:

$$(58) \quad |\alpha| + |\beta| = |\gamma|, \quad |\beta| + |\delta| = |\phi|, \quad |\alpha| + |\phi| = |\epsilon|, \quad |\gamma| + |\delta| = |\epsilon|.$$

Even when restricted to $(\mathcal{H}_n^+)^{\times 3}$, functions $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}$ take complex values, in general. However, the following is true:

Proposition 7. For $X, Y, Z \in \mathcal{H}_n^+$, the functions

$$(59) \quad \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) = \sum_{\gamma} \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z), \quad \varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z) = \sum_{\phi} \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$$

take non-negative real values.

Proof. We compute,

$$(60) \quad \begin{aligned} \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) &= \sum_{\gamma} \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z), \\ &= \sum_{\gamma} \text{Tr}_{V_{\alpha} \otimes V_{\beta} \otimes V_{\delta}} \Pi_{\alpha\beta\delta}^{\epsilon;\phi}(\Phi_{\alpha}(X) \otimes \Phi_{\beta}(Y) \otimes \Phi_{\delta}(Z)) \Pi_{\alpha\beta\delta}^{\epsilon;\gamma}, \\ &= \text{Tr}_{V_{\alpha} \otimes V_{\beta} \otimes V_{\delta}} \Pi_{\alpha\beta\delta}^{\epsilon;\phi}(\Phi_{\alpha}(X) \otimes \Phi_{\beta}(Y) \otimes \Phi_{\delta}(Z)). \end{aligned}$$

Here we have used the fact that $\sum_{\gamma} \Pi_{\alpha\beta\delta}^{\epsilon;\gamma} = P_{\alpha\beta\delta}^{\epsilon}$, where $P_{\alpha\beta\delta}^{\epsilon}$ is the orthogonal projector to the isotypical component of ϵ in the triple tensor product. We have also used the fact that $P_{\alpha\beta\delta}^{\epsilon} \Pi_{\alpha\beta\delta}^{\epsilon;\phi} = \Pi_{\alpha\beta\delta}^{\epsilon;\phi}$. The right hand side in the equation above is the trace of the orthogonal restriction of the non-negative Hermitian operator $(\Phi_{\alpha}(X) \otimes \Phi_{\beta}(Y) \otimes \Phi_{\delta}(Z))$ to the image of the projector $\Pi_{\alpha\beta\delta}^{\epsilon;\phi}$. Hence, it is a non-negative real number. The proof for functions $\varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z)$ is similar. $\square$

[sec:background_probability](#)

2.2. Probability measures. In this Section, we collect information about probability measures on finite sets and on conjugacy classes of Hermitian matrices needed later in the paper.

2.2.1. Measures on finite sets. Recall that for a finite set $[r] = \{1, \dots, r\}$ a probability measure is defined by an array of non-negative real numbers $(p_1, \dots, p_r)$ such that $\sum_{i=1}^r p_i = 1$. For two measures p and q on $[r]$, the Kolmogorov distance $K(p, q)$ between them is defined by:

$$(61) \quad K(p, q) := \frac{1}{2} \|p - q\|_1 = \frac{1}{2} \sum_{i=1}^r |p_i - q_i|.$$

Another measure of similarity between distributions p and q is the Bhattacharyya coefficient [\[Bha46; FV02\]](#):

[{eq:Bhattacharyya_coefficient}](#)

$$(62) \quad \text{BC}(p, q) = \sum_{i=1}^r \sqrt{p_i q_i}.$$

For all distributions p, q , the Bhattacharyya coefficient $\text{BC}(p, q)$ lies in the interval $[0, 1]$, and $\text{BC}(p, q) = 1$ if and only if $p = q$. Recall the following relation between the Kolmogorov Distance and the Bhattacharyya coefficient (see *e.g.* [\[FV02, Proposition 5, Equation \(41\)\]](#)):

$$(63) \quad 1 - \text{BC}(p, q) \leq K(p, q) \leq \sqrt{1 - \text{BC}(p, q)^2}.$$

This implies¹
(64)

$$\text{BC}(p, q) \leq \sqrt{1 - K(p, q)^2} \leq \exp\left(-\frac{1}{2}K(p, q)^2\right) = \exp\left(-\frac{1}{8}\|p - q\|_1^2\right).$$

We extend the Bhattacharyya coefficient to non-normalized non-negative tuples with $\sum_i p_i = \sum_i q_i = \tau > 0$. In that case, the *unnormalized* Bhattacharyya coefficient is also denoted by

$$(65) \quad \text{BC}(p, q) = \tau B\left(\frac{p}{\tau}, \frac{q}{\tau}\right),$$

and the estimate in Eq. (64) then reads

$$(66) \quad \frac{\text{BC}(p, q)}{\tau} \leq \exp\left(-\frac{1}{8\tau^2}\|p - q\|_1^2\right).$$

Finally, the Kullback-Leibler (KL) divergence is defined as

$$(67) \quad D(p\|q) = \sum_{i=1}^n p_i \log \frac{p_i}{q_i}.$$

It takes values between 0 and $+\infty$, and it is equal to zero if and only if $p = q$.

2.2.2. *Measures on partitions.* We define the set

$$(68) \quad \mathbb{R}_n^+ = \{x = (x_1 \geq x_2 \geq \dots \geq x_n) \in \mathbb{R}^n; x \neq 0, x_n \geq 0\}.$$

One can associate to $x \in \mathbb{R}_n^+$ a probability measure p^x on the set $[n]$ by assigning

$$(69) \quad p_i^x = \frac{x_i}{|x|}, \quad \text{where} \quad |x| = \sum_{i=1}^n x_i.$$

Let $X = \text{diag}(x_1, \dots, x_n)$ be a diagonal non-negative Hermitian matrix with x_i 's on the diagonal. By abuse of notation, we write $\varphi_\lambda(x) = \varphi_\lambda(X)$. For each x and each $k \geq 1$, using Eq. (45), one can define a probability measure on the set of positive weights of $U(n)$ with size $|\lambda| = k$:

$$(70) \quad P_x(\lambda) := \frac{k!}{|x|^k} \varphi_\lambda(x) = \frac{\text{Tr}[P_\lambda X^{\otimes k}]}{\text{Tr}[X^{\otimes k}]}.$$

This probability distribution is known as the *Schur-Weyl* distribution for x of degree $k = |\lambda|$ [OW16]. A key property of the Schur-Weyl distribution $P_x(\lambda)$ is that it behaves similarly to the multinomial distribution $m_x(\lambda)$ where

$$(71) \quad m_x(\lambda) := \frac{|\lambda|!}{\text{Tr}[x]^k} \frac{x_1^{\lambda_1}}{\lambda_1!} \frac{x_2^{\lambda_2}}{\lambda_2!} \dots \frac{x_n^{\lambda_n}}{\lambda_n!}. \quad \text{prop:multinomial_bound}$$

Proposition 8. *Given a Young diagram λ and non-negative eigenvalues x ,*

$$(72) \quad P_x(\lambda) \leq \dim V_\lambda^n m_x(\lambda).$$

¹Here we used the inequality $\sqrt{1 - x^2} \leq \exp(-x^2/2)$ which holds for all $x \in [0, 1]$.

Proof. The proof follows from (i) an upper bound on the Schur-polynomial $s_\lambda(x)$,

$$(73) \quad s_\lambda(x) \leq \dim V_\lambda^n x_1^{\lambda_1} x_2^{\lambda_2} \cdots x_n^{\lambda_n},$$

which follows because $s_\lambda(x)$ can be expressed as a sum over $\dim V_\lambda^n$ multinomials of the form $x_1^{\mu_1} x_2^{\mu_2} \cdots x_n^{\mu_n}$ with $x_1^{\lambda_1} x_2^{\lambda_2} \cdots x_n^{\lambda_n}$ the largest, and (ii) a lower bound on the Hook-product,

$$(74) \quad H_\lambda \geq \lambda_1! \cdots \lambda_n!,$$

which follows because $h_\lambda(b) \geq \lambda_i - j + 1$ holds for the box b in row i and column j . $\square$

Remark 9. We will henceforth relax the dimension prefactor $\dim V_\lambda^n$ using the following dimension bound which holds for all Young diagrams with k boxes:

`{eq:unitary_irrep_dim_bound_2}`

$$(75) \quad \dim V_\lambda^n \leq (k+1)^{\binom{n}{2}} := (k+1)^{\frac{n(n-1)}{2}}.$$

The following is known result is known as eigenvalue or spectrum estimation [ARS88; KW01; HM02; CM06; OW16].

prop:eig_est

Proposition 10 (Eigenvalue estimation). *Let $x \in \mathbb{R}_+^n$, $\lambda \in \Lambda_{++}^*$ with $|\lambda| = k$. Then the Schur-Weyl distribution $P_x(\lambda)$ satisfies*

$$(76) \quad P_x(\lambda) \leq (k+1)^{\binom{n}{2}} \exp\left(-kD\left(p^\lambda \parallel p^x\right)\right).$$

Proof. From Proposition 8 we have the bound $P_x(\lambda) \leq (k+1)^{\binom{n}{2}} m_x(\lambda)$. What remains is to bound the multinomial distribution $m_x(\lambda)$ using the Kullback-Leibler divergence. To this end, note that $k = |\lambda| = \lambda_1 + \cdots + \delta_n$ which implies

$$(77) \quad \begin{aligned} m_x(\lambda) &= \frac{k!}{\text{Tr}[x]^k} \frac{x_1^{\lambda_1}}{\lambda_1!} \cdots \frac{x_n^{\lambda_n}}{\lambda_n!}, \\ &= \left(\frac{k!}{k^k} \frac{\lambda_1^{\lambda_1}}{\lambda_1!} \cdots \frac{\lambda_n^{\lambda_n}}{\lambda_n!} \right) \left(\frac{k^k}{\text{Tr}[x]^k} \frac{x_1^{\lambda_1}}{\lambda_1^{\lambda_1}} \cdots \frac{x_n^{\lambda_n}}{\lambda_n^{\lambda_n}} \right), \\ &= m_\lambda(\lambda) \exp \left[-k \sum_{i=1}^n \frac{\lambda_i}{k} \log \left(\frac{\lambda_i}{k} \frac{\text{Tr}[x]}{x_i} \right) \right], \\ &= m_\lambda(\lambda) \exp \left[-kD \left(\frac{\lambda}{k} \parallel \frac{x}{\text{Tr}[x]} \right) \right]. \end{aligned}$$

Since $m_\lambda(\lambda) \in [0, 1]$ is a probability, we have the bound $m_\lambda(\lambda) \leq 1$ and therefore we obtain the claimed result:

$$(78) \quad P_x(\lambda) \leq (k+1)^{\binom{n}{2}} m_x(\lambda) \leq (k+1)^{\binom{n}{2}} \exp \left[-kD \left(\frac{\lambda}{k} \parallel \frac{x}{\text{Tr}[x]} \right) \right].$$

$\square$

In summary, the eigenvalue estimation result captures the fact that, like the multinomial distribution $m_x(\lambda)$, the Schur-Weyl distribution $P_x(\lambda)$ for large k is sharply concentrated around Young diagrams λ whose distribution, $p^\lambda = \lambda/|\lambda|$, is similar to the eigenvalue distribution $p^x = x/\text{Tr}[x]$.

The following proposition compares Schur-Weyl measures P_x and P_y for different $x, y \in \mathbb{R}_n^+$.

lem:eig_sep

Lemma 11 (Eigenvalue separation). *Let $x, y \in \mathbb{R}_n^+$ satisfy $\tau = \text{Tr}[x] = \text{Tr}[y]$. For any non-negative integer $k \geq 1$,*

{eq:eig_sep}

$$(79) \quad \text{BC}(P_x, P_y) \leq (k+1) \binom{n}{2} \text{BC}(p^x, p^y)^k.$$

Proof. From Proposition 8 and Eq. (75), we have

$$(80) \quad \text{BC}(P_x, P_y) \leq (k+1) \binom{n}{2} \text{BC}_k(m_x, m_y).$$

where $\text{BC}_k(m_x, m_y)$ is the Bhattacharyya coefficient between the multinomial distributions m_x and m_y . Since Bhattacharyya coefficients are multiplicative with respect to multinomial distributions, i.e.,

$$(81) \quad \text{BC}_k(m_x, m_y) = \text{BC}(p^x, p^y)^k,$$

we conclude Eq. (79). $\square$

In subsequent sections we will frequently use of the *weaker* bound

{eq:weak_eig_sep}

$$(82) \quad \forall \lambda \vdash k : \quad \sqrt{P_x(\lambda)P_y(\lambda)} \leq (k+1) \binom{n}{2} \text{BC}(p^x, p^y)^k,$$

which follows from Lemma 11 because $\sqrt{P_x(\lambda)P_y(\lambda)} \leq \text{BC}(P_x, P_y)$ by definition.

2.2.3. Measures on conjugacy classes. For $x \in \mathbb{R}_n^+$, we define the conjugacy class $\mathcal{O}_x$ of the diagonal matrix $[x] = \text{diag}(x_1, \dots, x_n)$ under the action of the unitary group $U(n)$:

$$(83) \quad \mathcal{O}_x = \{X = U[x]U^{-1}; U \in U(n)\} \subset \mathcal{H}_n^+.$$

Each conjugacy class carries a unique $U(n)$ -invariant probability measure μ_x . This measure can be constructed as the pushforward of the probability Haar measure on $U(n)$. Also, conjugacy classes are isomorphic to coadjoint orbits of $U(n)$ and they carry canonical Duistermaat-Heckman measure proportional to μ_x . In what follows, we will need expectation values $\mathbb{E}_x$ on conjugacy classes $\mathcal{O}_x$ with respect to the measure μ_x , $\mathbb{E}_{x,y}$ on $\mathcal{O}_x \times \mathcal{O}_y$ with the measure $\mu_x \times \mu_y$, and $\mathbb{E}_{x,y,z}$ on $\mathcal{O}_x \times \mathcal{O}_y \times \mathcal{O}_z$ with the measure $\mu_x \times \mu_y \times \mu_z$. *prop:estimate E_x_y_z*

Proposition 12. *We have,*

$$\begin{aligned}
 \mathbb{E}_x \Phi_\lambda(X) &= \frac{\varphi_\lambda(x)}{\dim V_\lambda} \text{id}_{V_\lambda}, \\
 \mathbb{E}_{x,y} \frac{\varphi_\gamma^{\alpha\beta}(X, Y)}{\dim V_\gamma} &= c_{\alpha\beta}^\gamma \frac{\varphi_\alpha(x)}{\dim V_\alpha} \frac{\varphi_\beta(y)}{\dim V_\beta}, \\
 \mathbb{E}_{x,y,z} \frac{\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)}{\dim V_\epsilon} &= \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_2^2 \frac{\varphi_\alpha(x)}{\dim V_\alpha} \frac{\varphi_\beta(y)}{\dim V_\beta} \frac{\varphi_\delta(y)}{\dim V_\delta}.
 \end{aligned}
 \tag{84}$$

Proof. For the first equality, by Schur's Lemma we have

$$\mathbb{E}_x \Phi_\lambda(X) = \int_{O_x} \Phi_\lambda(X) d\mu_x = c(x) \text{id}_{V_\lambda}
 \tag{85}$$

for some $c(x) \in \mathbb{C}$. By taking a trace of the equality above, we obtain

$$\varphi_\lambda(x) = c(x) \dim V_\lambda \Rightarrow c(x) = \frac{\varphi_\lambda(x)}{\dim V_\lambda},
 \tag{86}$$

as required. For the second equality, we compute

$$\begin{aligned}
 \mathbb{E}_{x,y} \frac{\varphi_\gamma^{\alpha\beta}(X, Y)}{\dim V_\gamma} &= \frac{1}{\dim V_\gamma} \mathbb{E}_{x,y} \text{Tr}(\Phi_\alpha(X) \otimes \Phi_\beta(Y)) P_{\alpha\beta}^\gamma, \\
 &= \frac{\varphi_\alpha(x)}{\dim V_\alpha} \frac{\varphi_\beta(y)}{\dim V_\beta} \frac{1}{\dim V_\gamma} \text{Tr} P_{\alpha\beta}^\gamma, \\
 &= c_{\alpha\beta}^\gamma \frac{\varphi_\alpha(x)}{\dim V_\alpha} \frac{\varphi_\beta(y)}{\dim V_\beta}.
 \end{aligned}
 \tag{87}$$

Finally, for the third equality we compute

$$\begin{aligned}
 \mathbb{E}_{x,y,z} \frac{\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)}{\dim V_\epsilon} &= \frac{1}{\dim V_\epsilon} \mathbb{E}_{x,y,z} \text{Tr} \Pi_{\alpha\beta\delta}^{\epsilon;\phi} (\Phi_\alpha(X) \otimes \Phi_\beta(Y) \otimes \Phi_\delta(Z)) \Pi_{\alpha\beta\delta}^{\epsilon;\gamma}, \\
 &= \frac{1}{\dim V_\epsilon} \frac{\varphi_\alpha(x)}{\dim V_\alpha} \frac{\varphi_\beta(y)}{\dim V_\beta} \frac{\varphi_\delta(y)}{\dim V_\delta} \text{Tr} \Pi_{\alpha\beta\delta}^{\epsilon;\phi} \Pi_{\alpha\beta\delta}^{\epsilon;\gamma}, \\
 &= \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_2^2 \frac{\varphi_\alpha(x)}{\dim V_\alpha} \frac{\varphi_\beta(y)}{\dim V_\beta} \frac{\varphi_\delta(y)}{\dim V_\delta}.
 \end{aligned}
 \tag{88}$$

□

Note that even though functions $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ are complex values, their expectation values are real and positive.

3. INEQUALITIES FOR TETRAHEDRAL EIGENVALUES

In this Section, we state and prove one of the main results of this paper. To set up notation, we will be considering 6-tuples of non-negative arrays $(a, b, c, d, e, f) \in (\mathbb{R}_+^n)^6$ satisfying the trace conditions

$$\begin{aligned}
 \text{Tr}[a] + \text{Tr}[b] &= \text{Tr}[c], & \text{Tr}[b] + \text{Tr}[d] &= \text{Tr}[f], & \text{Tr}[a] + \text{Tr}[b] + \text{Tr}[d] &= \text{Tr}[e].
 \end{aligned}
 \tag{89}$$

On the space of such 6-tuples, we introduce a function

$$(90) \quad D \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} := \min_{(A,B,D) \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d} \left(\|\text{eig}(A+B) - c\|_1^2 + \|\text{eig}(B+D) - f\|_1^2 + \|\text{eig}(A+B+D) - e\|_1^2 \right).$$

Note that $D \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} = 0$ if and only if there exists non-negative Hermitian matrices

`{eq:six_matrices}`

$$(91) \quad (A, B, C = A + B, D, E = A + B + C, F = B + D) \in (\mathcal{H}_n^+)^6$$

with ordered eigenvalues given by (a, b, c, d, e, f) . We will also need 6-tuples of positive highest weights

$$(92) \quad (\alpha, \beta, \gamma, \delta, \phi, \epsilon) \in (\Lambda_{++}^*)^6$$

satisfying the size conditions

$$(93) \quad |\alpha| + |\beta| = |\gamma|, \quad |\beta| + |\delta| = |\phi|, \quad |\alpha| + |\beta| + |\delta| = |\epsilon|.$$

We denote $|\epsilon| = k$. The following is a restatement of ?? from the introduction.

`thm:tet_iff_body`

Theorem 13. *Let $(a, b, c, d, e, f) \in (\mathbb{R}_+^n)^6$ be a 6-tuple satisfying the trace conditions. Then, Hermitian matrices satisfing Eq. (91) with eigenvalues (a, b, c, d, e, f) exist if and only if for all triples of positive wieghts $(\alpha, \beta, \delta) \in (\Lambda_{++}^*)^3$, the following inequalities are satisfied*

`feq:tet_iff_body`

$$(94) \quad \varphi_\alpha(a) \varphi_\beta(b) \varphi_\delta(d) \leq \sum_{\gamma, \epsilon, \phi} \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty (\varphi_\alpha(a) \varphi_\delta(d) \varphi_\phi(f) \varphi_\gamma(c) \varphi_\epsilon(e))^{\frac{1}{3}}.$$

Furthermore, if inequalities ?? hold true for all triples $(\alpha, \beta, \delta) \in (\Lambda_{++}^*)^3$ with $|\alpha| + |\beta| + |\delta| = k$, then

`feq:tet_dist_bound_body`

$$(95) \quad \frac{1}{\text{Tr}[e]} D \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \leq O \left(n \sqrt{\frac{\log k}{k}} \right).$$

Our proof of ?? is divided into two parts. First we prove the necessity of Eq. (94) in Section 3.1. Then afterwards we prove the distance bound in Eq. (95). Providing proofs for these two claims yields the whole theorem because the satisfication of Eq. (94) for all (α, β, δ) with $|\alpha| + |\beta| + |\delta| = k$ implies Eq. (95) holds for all k , and thus the $k \rightarrow \infty$ limit implies $D \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} = 0$ (because $\lim_{k \rightarrow \infty} \log(k+1)/k = 0$) which means (a, b, c, d, e, f) are tetrahedral.

`sec:tet_necess`

3.1. Necessary conditions. First, we assume that there are matrices (A, B, C, D, E, F) with eigenvalues (a, b, c, d, e, f) satisfying Eq. (91). We then compute,

$$\begin{aligned}
 (96) \quad \varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) &= \varphi_\alpha(A)\varphi_\beta(B)\varphi_\delta(D), \\
 &= \sum_{\gamma, \epsilon, \phi} \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(A, B, D), \\
 &\leq \sum_{\gamma, \epsilon, \phi} \left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(A, B, D) \right|, \\
 &\leq \sum_{\gamma, \epsilon, \phi} \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty (\varphi_\alpha(A)\varphi_\delta(D)\varphi_\phi(F)\varphi_\gamma(C)\varphi_\epsilon(E))^{\frac{1}{3}}, \\
 &= \sum_{\gamma, \epsilon, \phi} \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty (\varphi_\alpha(a)\varphi_\delta(d)\varphi_\phi(f)\varphi_\gamma(c)\varphi_\epsilon(e))^{\frac{1}{3}}.
 \end{aligned}$$

This calculation follows from the estimate:

$$\text{\texttt{\{eq:vp_ABD_abs_bound\}}} \quad (97) \quad \left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(A, B, D) \right| \leq \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty (\varphi_\alpha(A)\varphi_\delta(D)\varphi_\phi(F)\varphi_\gamma(C)\varphi_\epsilon(E))^{\frac{1}{3}}.$$

To prove this inequality, use Proposition 3 to obtain

$$\begin{aligned}
 (98) \quad \left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(A, B, D) \right| &= \left| \text{Tr } \Pi_{\alpha\beta\delta}^{\epsilon; \phi}(\Phi_\alpha(A) \otimes \Phi_\beta(B) \otimes \Phi_\delta(D)) \Pi_{\alpha\beta\delta}^{\epsilon; \gamma} \right| \\
 &\leq \left\| \Pi_{\alpha\beta\delta}^{\epsilon; \phi} \Pi_{\alpha\beta\delta}^{\epsilon; \gamma} \right\|_\infty (\text{Tr } \Pi_{\alpha\beta\delta}^{\epsilon; \phi}(\Phi_\alpha(A) \otimes \Phi_\beta(B) \otimes \Phi_\delta(D)))^{\frac{1}{2}} \\
 &\quad \times (\text{Tr } (\Phi_\alpha(A) \otimes \Phi_\beta(B) \otimes \Phi_\delta(D)) \Pi_{\alpha\beta\delta}^{\epsilon; \gamma})^{\frac{1}{2}} \\
 &= \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty \varphi_{\epsilon\phi}^{\alpha\beta\delta}(A, B, D)^{\frac{1}{2}} \varphi_{\gamma\epsilon}^{\alpha\beta\delta}(A, B, D)^{\frac{1}{2}}.
 \end{aligned}$$

Then we have the following upper bounds on $\varphi_{\epsilon\phi}^{\alpha\beta\delta}(A, B, D)$:

$$\begin{aligned}
 (99) \quad \varphi_{\epsilon\phi}^{\alpha\beta\delta}(A, B, D) &\leq \sum_{\epsilon\beta\delta} \varphi_{\epsilon\phi}^{\alpha\beta\delta}(A, B, D), \\
 &= \sum_{\epsilon} \text{Tr } P_{\alpha\beta\delta}^{\epsilon} \left(\Phi_\alpha(A) \otimes \sum_{\beta\delta} \Phi_\phi^{\beta\delta}(B, D) \right), \\
 &= \text{Tr } \Phi_\alpha(A) \otimes \Phi_\phi(B + D), \\
 &= \varphi_\alpha(A)\varphi_\phi(B + D),
 \end{aligned}$$

and

$$\begin{aligned}
 \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(A, B, D) &\leq \sum_{\alpha\beta\delta\phi} \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(A, B, D), \\
 &= \sum_{\alpha\phi} \text{Tr } P_{\alpha\beta\delta}^\epsilon \left(\Phi_\alpha(A) \otimes \sum_{\beta\delta} \Phi_\phi^{\beta\delta}(B, D) \right), \\
 (100) \quad &= \sum_{\alpha\phi} \text{Tr } P_{\alpha\beta\delta}^\epsilon (\Phi_\alpha(A) \otimes \Phi_\phi(B + D)), \\
 &= \sum_{\alpha\phi} \varphi_\epsilon^{\alpha\phi}(A, B + D), \\
 &= \varphi_\epsilon(A + B + D).
 \end{aligned}$$

In both estimates, we are using the fact that $\varphi_{-\epsilon\phi}^{\alpha\beta\delta}(A, B, D) \geq 0$. Together, they imply

$$\begin{aligned}
 (101) \quad \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(A, B, D) &\leq (\varphi_\alpha(A) \varphi_\phi(B + D))^{\frac{2}{3}} \varphi_\epsilon(A + B + D)^{\frac{1}{3}} = (\varphi_\alpha(A) \varphi_\phi(F))^{\frac{2}{3}} \varphi_\epsilon(E)^{\frac{1}{3}}.
 \end{aligned}$$

In an analogous fashion, we conclude the following bound on $\varphi_{\gamma\epsilon_-}^{\alpha\beta\delta}(A, B, D)$:

$$(102) \quad \varphi_{\gamma\epsilon_-}^{\alpha\beta\delta}(A, B, D) \leq (\varphi_\gamma(C) \varphi_\delta(D))^{\frac{2}{3}} \varphi_\epsilon(E)^{\frac{1}{3}}.$$

and this completes the proof for Eq. (97).

3.2. Distance bound & sufficiency. Let $(a, b, c, d, e, f) \in (\mathbb{R}_+^n)^6$ which satisfy the trace conditions and also

$$\begin{aligned}
 \text{\texttt{eq:abd_multiplied_2}} \quad (103) \quad \varphi_\alpha(a) \varphi_\beta(b) \varphi_\delta(d) &\leq \sum_{\gamma, \epsilon, \phi} \left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_\infty (\varphi_\alpha(a) \varphi_\phi(f) \varphi_\gamma(c) \varphi_\delta(d) \varphi_\epsilon(e))^{\frac{1}{3}}
 \end{aligned}$$

for all $(\alpha, \beta, \delta) \in (\Lambda_{++}^*)^3$. We will consider triples with $|\alpha| + |\beta| + |\gamma| = k$ for some fixed integer k . Denote by $(\gamma^*, \epsilon^*, \phi^*)$ the triple for which the corresponding term on the right hand side is maximal. We have,

$$\begin{aligned}
 \text{\texttt{eq:first_abd_bound_2}} \quad (104) \quad \varphi_\alpha(a) \varphi_\beta(b) \varphi_\delta(d) &\leq (k+1)^{3(n-1)} \left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma^* \\ \delta & \epsilon^* & \phi^* \end{smallmatrix} \right\} \right\|_\infty (\varphi_\alpha(a) \varphi_{\phi^*}(f) \varphi_{\gamma^*}(c) \varphi_\delta(d) \varphi_{\epsilon^*}(e))^{\frac{1}{3}},
 \end{aligned}$$

where $(k+1)^{3(n-1)}$ serves as an upper bound on the number of terms in the sum in Eq. (103). Equation (225) then implies

$$(105) \quad \varphi_\alpha(a) \varphi_\beta(b) \varphi_\delta(d) > 0 \implies \left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma^* \\ \delta & \epsilon^* & \phi^* \end{smallmatrix} \right\} \right\|_\infty > 0.$$

Next, consider the bound

{eq:second_abd_bound}
(106)

$$\begin{aligned} \left\| \left\{ \begin{smallmatrix} \alpha & \beta \\ \delta & \epsilon^* \end{smallmatrix} \gamma^* \right\} \right\|_2^2 \varphi_\alpha(a) \varphi_\beta(b) \varphi_\delta(d) &= \frac{\dim(V_\alpha^n \otimes V_\beta^n \otimes V_\delta^n)}{\dim(V_{\epsilon^*}^n)} \mathbb{E}_{(A,B,D) \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d} \varphi_{\gamma^* \epsilon^* \phi^*}^{\alpha\beta\delta}(A, B, D), \\ &\leq (k+1)^{3\binom{n}{2}} \left| \varphi_{\gamma^* \epsilon^* \phi^*}^{\alpha\beta\delta}(A^*, B^*, D^*) \right|, \\ &\leq (k+1)^{3\binom{n}{2}} \left\| \left\{ \begin{smallmatrix} \alpha & \beta \\ \delta & \epsilon^* \end{smallmatrix} \gamma^* \right\} \right\|_\infty (\varphi_\alpha(A^*) \varphi_{\phi^*}(F^*) \varphi_{\gamma^*}(C^*) \varphi_\delta(D^*) \varphi_{\epsilon^*}(E^*))^{\frac{1}{3}}. \end{aligned}$$

Here in the first line we have used Proposition 12. In the second line we denote by $(A^*, B^*, D^*) \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d$ a triple of matrices which realizes a maximum of the functions $\left| \varphi_{\gamma^* \epsilon^* \phi^*}^{\alpha\beta\delta}(A, B, D) \right|$, and we estimate the dimensions of representations by ???. In the third line, we denote $C^* = A^* + B^*$, $E^* = A^* + B^* + D^*$, $F^* = B^* + D^*$, and use the estimate in Eq. (97).

By multiplying the estimate in Eq. (104) by the estimate in Eq. (160), we obtain

$$\begin{aligned} (107) \quad \left\| \left\{ \begin{smallmatrix} \alpha & \beta \\ \delta & \epsilon^* \end{smallmatrix} \gamma^* \right\} \right\|_2^2 (\varphi_\alpha(a) \varphi_\beta(b) \varphi_\delta(d))^2 &\leq (k+1)^{3\binom{n}{2}+3n} \left\| \left\{ \begin{smallmatrix} \alpha & \beta \\ \delta & \epsilon^* \end{smallmatrix} \gamma^* \right\} \right\|_\infty^2 \\ &\quad \times (\varphi_\alpha(a) \varphi_{\phi^*}(f) \varphi_{\gamma^*}(c) \varphi_\delta(d) \varphi_{\epsilon^*}(e))^{\frac{1}{3}} \\ &\quad \times (\varphi_\alpha(A^*) \varphi_{\phi^*}(F^*) \varphi_{\gamma^*}(C^*) \varphi_\delta(D^*) \varphi_{\epsilon^*}(E^*))^{\frac{1}{3}}. \end{aligned}$$

This then implies

{eq:10varphi}
(108)

$$\begin{aligned} \varphi_\alpha(a) \varphi_\beta(b) \varphi_\delta(d) &\leq (k+1)^{\frac{3}{2}\binom{n}{2}+\frac{3}{2}n} \\ &\quad \times (\varphi_\alpha(a) \varphi_{\phi^*}(f) \varphi_{\gamma^*}(c) \varphi_\delta(d) \varphi_{\epsilon^*}(e))^{\frac{1}{6}} \\ &\quad \times (\varphi_\alpha(A^*) \varphi_{\phi^*}(F^*) \varphi_{\gamma^*}(C^*) \varphi_\delta(D^*) \varphi_{\epsilon^*}(E^*))^{\frac{1}{6}}. \end{aligned}$$

Indeed, if $\varphi_\alpha(a) \varphi_\beta(b) \varphi_\delta(d) = 0$ the inequality is verified since the right hand side is non-negative. Otherwise, if $\varphi_\alpha(a) \varphi_\beta(b) \varphi_\delta(d) > 0$, then the norms of the 6j-symbol are nonvanishing and we can use the standard operator norm estimates from Eq. (31).

Since $\text{eig}(A^*) = a$, we have $\varphi_\alpha(A^*) = \varphi_\alpha(a)$, and

$$(109) \quad (\varphi_\alpha(a) \varphi_\alpha(A^*))^{\frac{1}{2}} = \varphi_\alpha(a)^{\frac{1}{3}} \leq \sum_\alpha \varphi_\alpha(a) = \frac{|a|^{|a|}}{|\alpha|!}.$$

Similarly, we obtain

$$(110) \quad (\varphi_\delta(d) \varphi_\delta(D^*))^{\frac{1}{2}} \leq \frac{|d|^{|d|}}{|\delta|!}.$$

Next, we consider

$$\begin{aligned}
 (\varphi_{\phi^*}(f)\varphi_{\phi^*}(F^*))^{\frac{1}{2}} &\leq \sum_{\phi} \sqrt{\varphi_{\phi}(f)\varphi_{\phi}(F^*)}, \\
 &= \frac{|f|^{| \phi |}}{| \phi |!} \text{BC}(P_f, P_{\text{eig}(F^*)}), \\
 (111) \quad &\leq (| \phi | + 1)^{\binom{n}{2}} \frac{1}{| \phi |!} \text{BC}(f, \text{eig}(F^*))^{| \phi |}, \\
 &\leq (k + 1)^{\binom{n}{2}} \frac{1}{| \phi |!} \text{BC}(f, \text{eig}(F^*))^{| \phi |}.
 \end{aligned}$$

Here we have used ?? and the fact that $| \phi | \leq k$. Putting two estimates above together, we arrive at

$$\begin{aligned}
 (112) \quad (\varphi_{\alpha}(a)\varphi_{\alpha}(A^*)\varphi_{\phi^*}(f)\varphi_{\phi^*}(F^*))^{\frac{1}{2}} &\leq \frac{1}{| \alpha |! | \phi |!} (k + 1)^{\binom{n}{2}} |a|^{| \alpha |} \text{BC}(f, \text{eig}(F^*))^{| \phi |}, \\
 &\leq \frac{1}{k!} (k + 1)^{\binom{n}{2}} (|a| + \text{BC}(f, \text{eig}(F^*))^{| \phi |}), \\
 &= \frac{1}{k!} (k + 1)^{\binom{n}{2}} \text{BC}(a \oplus f, a \oplus \text{eig}(F^*))^k, \\
 &\leq \frac{1}{k!} (k + 1)^{\binom{n}{2}} |e|^k \exp\left(-\frac{k}{8|e|^2} \|f - \text{eig}(F^*)\|_1^2\right).
 \end{aligned}$$

Similarly, we obtain

$$(113) \quad (\varphi_{\alpha}(d)\varphi_{\alpha}(D^*)\varphi_{\gamma^*}(c)\varphi_{\gamma^*}(C^*))^{\frac{1}{2}} \leq \frac{1}{k!} (k + 1)^{\binom{n}{2}} |e|^k \exp\left(-\frac{k}{8|e|^2} \|c - \text{eig}(C^*)\|_1^2\right),$$

and

$$(114) \quad (\varphi_{\epsilon}(e)\varphi_{\epsilon}(E^*))^{\frac{1}{2}} \leq \frac{1}{k!} (k + 1)^{\binom{n}{2}} |e|^k \exp\left(-\frac{k}{8|e|^2} \|e - \text{eig}(E^*)\|_1^2\right).$$

Putting things together, we obtain the following estimate for the right hand side of the Eq. (108):

$$\begin{aligned}
 (115) \quad &(\varphi_{\alpha}(a)\varphi_{\phi^*}(f)\varphi_{\gamma^*}(c)\varphi_{\delta}(d)\varphi_{\epsilon^*}(e))^{\frac{1}{6}} \times \\
 &(\varphi_{\alpha}(A^*)\varphi_{\phi^*}(F^*)\varphi_{\gamma^*}(C^*)\varphi_{\delta}(D^*)\varphi_{\epsilon^*}(E^*))^{\frac{1}{6}} \leq \frac{1}{k!} (k + 1)^{\binom{n}{2}} |e|^k \exp\left(-\frac{k}{24|e|^2} D\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}\right).
 \end{aligned}$$

Here we have used the fact that

$$(116) \quad \|f - \text{eig}(F^*)\|_1^2 + \|e - \text{eig}(E^*)\|_1^2 + \|e - \text{eig}(E^*)\|_1^2 \geq D\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}.$$

Combining with the inequality in Eq. (108), this implies

$$(117) \quad \varphi_{\alpha}(a)\varphi_{\beta}(b)\varphi_{\delta}(d) \leq \frac{1}{k!} (k + 1)^{\binom{n}{2}} |e|^k \exp\left(-\frac{k}{24|e|^2} D\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}\right).$$

Summing over the triples (α, β, δ) with $|\alpha| + |\beta| + |\delta| = k$ yields

$$(118) \quad \frac{|e|^k}{k!} = \sum_{\alpha, \beta, \delta} \varphi_\alpha(a) \varphi_\beta(b) \varphi_\delta(d) \leq \frac{1}{k!} (k+1)^{\binom{n}{2} + 3n} |e|^k \exp \left(-\frac{k}{24|e|^2} D \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \right),$$

where we have estimated by $(k+1)^3 n$ the number of summands. This implies

$$(119) \quad \frac{1}{|e|^2} D \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \leq 36n(n+1) \frac{\log(k+1)}{k},$$

as required.

REFERENCES

- [AH11] Daiki Akimoto and Masahito Hayashi. “Discrimination of the change point in a quantum setting”. In: *Physical Review A* 83.5 (2011), p. 052328.
- [AI19] Arseniy Akopyan and Ivan Izmistiev. “The Regge symmetry, confocal conics, and the Schläfli formula”. In: *Bulletin of the London Mathematical Society* 51.5 (2019), pp. 765–775.
- [Alc+23] Judith Alcock-Zeilinger et al. “Wigner 6j symbols for SU (N): Symbols with at least two quark-lines”. In: *Journal of Mathematical Physics* 64.2 (2023).
- [Ale+25] Anton Alekseev et al. “Multiple Horn problems for planar networks and invertible matrices”. In: *arXiv preprint arXiv:2503.05277* (2025).
- [AMS20] Victor Alekseev, Andrey Morozov, and Alexey Sleptsov. “Multiplicity-free Uq (slN) 6-j symbols: relations, asymptotics, symmetries”. In: *Nuclear Physics B* 960 (2020), p. 115164.
- [AMW01] Anton Alekseev, Eckhard Meinrenken, and Chris Woodward. “Linearization of Poisson actions and singular values of matrix products”. In: *Annales de l’institut Fourier*. Vol. 51. 6. 2001, pp. 1691–1717.
- [APS17] Anton Alekseev, Masha Podkopaeva, and Andras Szenes. “A symplectic proof of the Horn inequalities”. In: *Advances in Mathematics* 318 (2017), pp. 711–736.
- [Aqu+12] Vincenzo Aquilanti et al. “Semiclassical mechanics of the Wigner 6j-symbol”. In: *Journal of Physics A: Mathematical and Theoretical* 45.6 (2012), p. 065209.
- [ARS88] Robert Alicki, Slawomir Rudnicki, and Slawomir Sadowski. “Symmetry properties of product states for the system of N n-level atoms”. In: *Journal of mathematical physics* 29.5 (1988), pp. 1158–1162.
- [Art23] DV Artamonov. “Classical 6j-symbols of finite-dimensional representations of the algebra $\mathfrak{gl}_3$ ”. In: *Theoretical and Mathematical Physics* 216.1 (2023), pp. 909–923.

- [Art24] Dmitry Artamonov. “Calculation of $6j$ -symbols for the Lie algebra $\mathfrak{gl}_n$ ”. In: *arXiv preprint arXiv:2405.05628* (2024).
- [AW97] Sharad Agnihotri and Chris Woodward. “Eigenvalues of products of unitary matrices and quantum Schubert calculus”. In: *arXiv preprint alg-geom/9712013* (1997).
- [Bac10] Miriam Backens. “The $6j$ symbols and an extended version of Horn’s problem”. In: (2010).
- [Bae96] John C Baez. “Spin networks in gauge theory”. In: *Advances in Mathematics* 117.2 (1996), pp. 253–272.
- [Bae99] John C Baez. “An introduction to spin foam models of quantum gravity and BF theory”. In: (1999). arXiv: [gr-qc/9905087](#).
- [Bar03] John W Barrett. “Geometrical measurements in three-dimensional quantum gravity”. In: *International Journal of Modern Physics A* 18.supp02 (2003), pp. 97–113.
- [BC98] John W Barrett and Louis Crane. “Relativistic spin networks and quantum gravity”. In: *Journal of Mathematical Physics* 39.6 (1998), pp. 3296–3302.
- [Bel02] Prakash Belkale. “Geometric proofs of Horn and saturation conjectures”. In: *arXiv preprint math/0208107* (2002).
- [Bel05] Prakash Belkale. “Geometric proof of a conjecture of Fulton”. In: *arXiv preprint math/0511664* (2005).
- [Bel08] Prakash Belkale. “Quantum generalization of the Horn conjecture”. In: *Journal of the American Mathematical Society* 21.2 (2008), pp. 365–408.
- [BGS08] Mustapha Bouhtou, Stéphane Gaubert, and Guillaume Sagnol. “Optimization of network traffic measurement: a semidefinite programming approach”. In: *Proceedings of the International*. Vol. 230. 2008.
- [Bha01] Rajendra Bhatia. “[Linear algebra to quantum cohomology: the story of Alfred Horn’s inequalities](#)”. In: *The American Mathematical Monthly* 108.4 (2001), pp. 289–318.
- [Bha46] Anil Bhattacharyya. “On a measure of divergence between two multinomial populations”. In: *Sankhyā: the indian journal of statistics* (1946), pp. 401–406.
- [BK16] Prakash Belkale and Shrawan Kumar. “The multiplicative eigenvalue problem and deformed quantum cohomology”. In: *Advances in Mathematics* 288 (2016), pp. 1309–1359.
- [BN23] Bruce Bartlett and Nzaganya Nzaganya. “Coherent loop states and angular momentum”. In: *arXiv preprint arXiv:2306.17293* (2023).
- [Boa07] Philip P Boalch. “[Regge and Okamoto symmetries](#)”. In: *Communications in mathematical physics* 276.1 (2007), pp. 117–130.
- [BS10] John Baez and Mike Stay. “[Physics, topology, logic and computation: a Rosetta Stone](#)”. In: *New structures for physics*. Springer, 2010, pp. 95–172. arXiv: [0903.0340v3](#).

- [BS15] Wolfgang Berndt and Suvrit Sra. “Hlawka–Popoviciu inequalities on positive definite tensors”. In: *Linear Algebra and its Applications* 486 (2015), pp. 317–327.
- [BVW18] Nicole Berline, Michèle Vergne, and Michael Walter. “The Horn inequalities from a geometric point of view”. In: *L’Enseignement Mathématique* 63.3 (2018), pp. 403–470.
- [Chr08] Matthias Christandl. “A Quantum Information-Theoretic Proof of the Relation between Horn’s Problem and the Littlewood–Richardson Coefficients”. In: *Lecture Notes in Computer Science*. Springer Berlin Heidelberg, 2008, pp. 120–128.
- [CHW07] Andrew M Childs, Aram W Harrow, and Paweł Wocjan. “Weak Fourier-Schur sampling, the hidden subgroup problem, and the quantum collision problem”. In: *Annual Symposium on Theoretical Aspects of Computer Science*. Springer. 2007, pp. 598–609.
- [CIS99] Li-Chen Chen, Mourad EH Ismail, and Plamen Simeonov. “Asymptotics of Racah coefficients and polynomials”. In: *Journal of Physics A: Mathematical and General* 32.3 (1999), p. 537.
- [CM06] Matthias Christandl and Graeme Mitchison. “The spectra of quantum states and the Kronecker coefficients of the symmetric group”. In: *Communications in mathematical physics* 261.3 (2006), pp. 789–797.
- [CMZ19] Robert Coquereaux, Colin McSwiggen, and Jean-Bernard Zuber. “Revisiting Horn’s problem”. In: *Journal of Statistical Mechanics: Theory and Experiment* 2019.9 (2019), p. 094018.
- [Coq23] Robert Coquereaux. “About integer-valued variants of the theta and 6j symbols”. In: *Journal of Mathematical Physics* 64.3 (2023).
- [Cou16] Thomas A Courtade. “Strengthening the entropy power inequality”. In: *2016 IEEE International Symposium on Information Theory (ISIT)*. IEEE. 2016, pp. 2294–2298.
- [CŞW18] Matthias Christandl, M Burak Şahinoğlu, and Michael Walter. “Recoupling coefficients and quantum entropies”. In: *Annales Henri Poincaré*. Vol. 19. 2. Springer. 2018, pp. 385–410.
- [CZ18] Robert Coquereaux and Jean-Bernard Zuber. “From orbital measures to Littlewood–Richardson coefficients and hive polytopes”. In: *Annales de l’Institut Henri Poincaré D* 5.3 (2018), pp. 339–386.
- [DE+72] George B Dantzig, B Curtis Eaves, et al. “Fourier-Motzkin elimination and its dual”. In: (1972).
- [DH05] Sumit Daftuar and Patrick Hayden. “Quantum state transformations and the Schubert calculus”. In: *Annals of Physics* 315.1 (2005), pp. 80–122.
- [DJN93] Bergfinnur Durhuus, Hans Plesner Jakobsen, and Ryszard Nest. “Topological quantum field theories from generalized 6j-symbols”. In: *Reviews in Mathematical Physics* 5.01 (1993), pp. 1–67.

- [Edm96] Alan Robert Edmonds. *Angular momentum in quantum mechanics*. Vol. 4. Princeton university press, 1996.
- [Fan49] Ky Fan. “On a theorem of Weyl concerning eigenvalues of linear transformations I”. In: *Proceedings of the National Academy of Sciences* 35.11 (1949), pp. 652–655.
- [FG13] S Friedland and S Gaubert. “Submodular spectral functions of principal submatrices of a hermitian matrix, extensions and applications”. In: *Linear Algebra and its Applications* 438.10 (2013), pp. 3872–3884.
- [FK00] Laurent Freidel and Kirill Krasnov. “Simple spin networks as Feynman graphs”. In: *Journal of Mathematical Physics* 41.4 (2000), pp. 1681–1690.
- [FKL10] Laurent Freidel, Kirill Krasnov, and Etera R Livine. “Holomorphic factorization for a quantum tetrahedron”. In: *Communications in Mathematical Physics* 297.1 (2010), pp. 45–93.
- [FL03] Laurent Freidel and David Louapre. “Asymptotics of 6j and 10j symbols”. In: *Classical and Quantum Gravity* 20.7 (2003), p. 1267.
- [Fra22] Thomas C Fraser. *A sufficient family of necessary inequalities for the compatibility of quantum marginals*. 2022. arXiv: [2211.00685v1](#).
- [Ful00] William Fulton. “Eigenvalues, invariant factors, highest weights, and Schubert calculus”. In: *Bulletin of the American Mathematical Society* 37.3 (2000), pp. 209–249.
- [Ful98] William Fulton. “Eigenvalues of sums of Hermitian matrices”. In: *Séminaire Bourbaki* 40.255-269 (1998), p. 5.
- [FV02] Christopher A Fuchs and Jeroen Van De Graaf. “Cryptographic distinguishability measures for quantum-mechanical states”. In: *IEEE Transactions on Information Theory* 45.4 (2002), pp. 1216–1227.
- [GS82] Victor Guillemin and Shlomo Sternberg. “Geometric quantization and multiplicities of group representations”. In: *Inventiones mathematicae* 67.3 (1982), pp. 515–538.
- [Gur08] Razvan Gurau. “The Ponzano–Regge Asymptotic of the 6j Symbol: An Elementary Proof”. In: *Annales Henri Poincaré*. Vol. 9. Springer. 2008, pp. 1413–1424.
- [GW00] Roe Goodman and Nolan R Wallach. *Representations and invariants of the classical groups*. Cambridge University Press, 2000.
- [Hal08] Matthew John Hall. “Representations of the Symmetric Groups and Combinatorics of the Frobenius-Young Correspondence”. PhD thesis. Citeseer, 2008.
- [Har23] Aram W Harrow. “Approximate orthogonality of permutation operators, with application to quantum information”. In: *Letters in Mathematical Physics* 114.1 (2023), p. 1.

- [Hay17] Masahito Hayashi. *A Group Theoretic Approach to Quantum Information*. Springer, 2017.
- [HK96] Jean-Claude Hausmann and Allen Knutson. “Polygon spaces and Grassmannians”. In: *arXiv preprint dg-ga/9602012* (1996).
- [HM02] Masahito Hayashi and Keiji Matsumoto. “Quantum universal variable-length source coding”. In: *Physical Review A* 66.2 (2002), p. 022311.
- [Hor62] Alfred Horn. “Eigenvalues of sums of Hermitian matrices.” In: (1962).
- [htt] Suvrit (<https://mathoverflow.net/users/8430/suvrit>). *Reverse Minkowski (and related) Determinant Inequalities*. MathOverflow. URL:<https://mathoverflow.net/q/251684>. (version: 2016-10-10). eprint: <https://mathoverflow.net/q/251684>.
- [IN66] Claude Itzykson and Michael Nauenberg. “Unitary groups: Representations and decompositions”. In: *Reviews of Modern Physics* 38.1 (1966), p. 95.
- [IP23] Christian Ikenmeyer and Greta Panova. “All Kronecker coefficients are reduced Kronecker coefficients”. In: *arXiv preprint arXiv:2305.03003* (2023).
- [Kir04] Aleksandr Aleksandrovich Kirillov. *Lectures on the orbit method*. Vol. 64. American Mathematical Soc., 2004.
- [Kly00] Alexander A Klyachko. “Random walks on symmetric spaces and inequalities for matrix spectra”. In: *Linear Algebra and its Applications* 319.1-3 (2000), pp. 37–59.
- [Kly98] Alexander A Klyachko. “Stable bundles, representation theory and Hermitian operators”. In: *Selecta Mathematica* 4 (1998), pp. 419–445.
- [Knu00] Allen Knutson. “The symplectic and algebraic geometry of Horn’s problem”. In: *Linear Algebra and its Applications* 319.1-3 (2000), pp. 61–81.
- [KT99] Allen Knutson and Terence Tao. “The honeycomb model of $GL_n(\mathbb{C})$ tensor products I: Proof of the saturation conjecture”. In: *Journal of the American Mathematical Society* 12.4 (1999), pp. 1055–1090.
- [KW01] Michael Keyl and Reinhard F Werner. “Estimating the spectrum of a density operator”. In: *Physical Review A* 64.5 (2001), p. 052311.
- [Lid53] V B Lidskiĭ. “The proper values of the sum and product of symmetric matrices”. Trans. by Curtis D. Benster. In: (1953). Ed. by George E. Forsythe Wallace Givens.
- [Lit58] Dudley E Littlewood. “Products and plethysms of characters with orthogonal, symplectic and symmetric groups”. In: *Canadian Journal of Mathematics* 10 (1958), pp. 17–32.

- [LS14] Mathieu Lewin and Julien Sabin. “A family of monotone quantum relative entropies”. In: *Letters in Mathematical Physics* 104.6 (2014), pp. 691–705. arXiv: [1309.4046v1](#).
- [LS23] Elena Lanina and Alexey Sleptsov. “Tug-the-hook symmetry for quantum 6j-symbols”. In: *Physics Letters B* 845 (2023), p. 138138.
- [Mou84] J Moussouris. “Quantum models of space-time based on recoupling theory”. PhD thesis. University of Oxford, 1984.
- [MR05] Annalisa Marzuoli and Mario Rasetti. “Computing spin networks”. In: *Annals of Physics* 318.2 (2005), pp. 345–407.
- [MS18] Andrey Morozov and Alexey Sleptsov. “New Symmetries for the $U_q(\mathfrak{sl}_N)$ 6j Symbols from the Eigenvalue Conjecture”. In: *JETP Letters* 108 (2018), pp. 697–704.
- [Mur55] Francis D Murnaghan. “On the analysis of the Kronecker product of irreducible representations of S_n ”. In: *Proceedings of the National Academy of Sciences* 41.7 (1955), pp. 515–518.
- [Nis00] Kyo Nishiyama. “Restriction of the irreducible representations of GL_n to the symmetric group S_n ”. In: *preprint* (2000).
- [NP06] Constantin Niculescu and Lars-Erik Persson. *Convex functions and their applications*. Vol. 23. Springer, 2006.
- [NPZ13] Satoshi Nawata, Ramadevi Pichai, and Zodinmawia. “Multiplicity-free Quantum 6j-Symbols for $U_q(\mathfrak{sl}_N)$ ”. In: *Letters in Mathematical Physics* 103 (2013), pp. 1389–1398.
- [NS23] Constantin P Niculescu and Suvrit Sra. “Functions with Positive Differences on Convex Cones”. In: *Results in Mathematics* 78.6 (2023), p. 217.
- [OW16] Ryan O’Donnell and John Wright. “Efficient quantum tomography”. In: *Proceedings of the forty-eighth annual ACM symposium on Theory of Computing*. 2016, pp. 899–912. arXiv: [1508.01907v2](#).
- [Pen71] Roger Penrose. “Applications of negative dimensional tensors”. In: *Combinatorial mathematics and its applications* 1 (1971), pp. 221–244.
- [PR68] Giorgio Ponzano and Tullio Eugenio Regge. “Semiclassical limit of Racah coefficients”. In: *Spectroscopic and group theoretical methods in physics* (1968).
- [Rac42] Giulio Racah. “Theory of complex spectra. II”. In: *Physical Review* 62.9-10 (1942), p. 438.
- [Reg59] Tullio Regge. “Symmetry properties of Racah’s coefficients”. In: *Il Nuovo Cimento (1955-1965)* 11.1 (1959), pp. 116–117.
- [Res87] N Yu Reshetikhin. *Quantized universal enveloping algebras, the Yang-Baxter equation and invariants of links, 1*. Tech. rep. 1987.
- [Ric02] G Richardson. “The trigonometry of the tetrahedron”. In: *The Mathematical Gazette* 2.32 (1902), pp. 149–158.
- [Rob02] Justin Roberts. “Asymptotics and 6j-symbols”. In: *Geom. Topol. Monogr* 4 (2002), pp. 245–261.

- [Rob55] Herbert Robbins. “A remark on Stirling’s formula”. In: *The American mathematical monthly* 62.1 (1955), pp. 26–29.
- [Rob99] Justin Roberts. “Classical 6j-symbols and the tetrahedron”. In: *Geometry & Topology* 3.1 (1999), pp. 21–66.
- [Rov04] Carlo Rovelli. *Quantum gravity*. Cambridge university press, 2004.
- [Sag13] Guillaume Sagnol. “Approximation of a maximum-submodular-coverage problem involving spectral functions, with application to experimental designs”. In: *Discrete Applied Mathematics* 161.1-2 (2013), pp. 258–276.
- [Sch52] Isaac J Schoenberg. “A remark on MM Day’s characterization of inner-product spaces and a conjecture of LM Blumenthal”. In: *Proceedings of the American Mathematical Society* 3.6 (1952), pp. 961–964.
- [She15] Cass Sherman. “Geometric proof of a conjecture of King, Tollu, and Toumazet”. In: *arXiv preprint arXiv:1505.06551* (2015).
- [SJS18] Meera Sitharam, Audrey John, and Jessica Sidman, eds. *Handbook of geometric constraint systems principles*. Discrete Mathematics and Its Applications. New York, NY: Productivity Press, July 2018.
- [Spe05] David E Speyer. “Horn’s problem, Vinnikov curves, and the hive cone”. In: (2005).
- [Sra] Suvrit Sra. *A determinant inequality*. MathOverflow. URL:<https://mathoverflow.net/q/258595>. (version: 2017-04-24). eprint: <https://mathoverflow.net/q/258595>.
- [TCL11] Lin Tie, Kai-Yuan Cai, and Yan Lin. “Rearrangement inequalities for Hermitian matrices”. In: *Linear algebra and its applications* 434.2 (2011), pp. 443–456.
- [TF70] Robert C. Thompson and Linda J. Freede. “On the Eigenvalues of Sums of Hermitian Matrices. II.” In: *Aequationes mathematicae* 5 (1970), pp. 103–115.
- [TF71] Robert C Thompson and Linda J Freede. “On the eigenvalues of sums of Hermitian matrices”. In: *Linear Algebra and Its Applications* 4.4 (1971), pp. 369–376.
- [TT71] RC Thompson and S Therianos. “The eigenvalues and singular values of matrix sums and product, viii: displacement of indices”. In: *aequationes mathematicae* 7 (1971), pp. 219–242.
- [TV92] Vladimir G Turaev and Oleg Ya Viro. “State sum invariants of 3-manifolds and quantum 6j-symbols”. In: *Topology* 31.4 (1992), pp. 865–902.
- [TW06] Yuka U Taylor and Christopher T Woodward. “6 j symbols for and non-Euclidean tetrahedra”. In: *Selecta Mathematica* 11.3-4 (2006), p. 539.
- [VMK88] Dmitriĭ Aleksandrovich Varshalovich, Anatolij Nikolaevič Moskalev, and Valerii Kel’manovich Khersonskii. *Quantum theory of angular momentum*. World Scientific, 1988.

- [Wat99] James KG Watson. “Asymptotic approximations for certain 6-j and 9-j symbols”. In: *Journal of Physics A: Mathematical and General* 32.39 (1999), p. 6901.
- [WD09] Karl Wirth and André S Dreiding. “Edge lengths determining tetrahedrons”. In: *Elemente der Mathematik* 64.4 (2009), pp. 160–170.
- [Wey12] Hermann Weyl. “Das asymptotische Verteilungsgesetz der Eigenwerte linearer partieller Differentialgleichungen (mit einer Anwendung auf die Theorie der Hohlraumstrahlung)”. In: *Mathematische Annalen* 71.4 (1912), pp. 441–479.
- [Wie55] Helmut Wielandt. “An extremum property of sums of eigenvalues”. In: *Proceedings of the American Mathematical Society* 6.1 (1955), pp. 106–110.
- [Wig60] Eugene Paul Wigner. “Group theory and its application to the quantum mechanics of atomic spectra”. In: *American journal of physics* 28.4 (1960), pp. 408–409.
- [Wig93] Eugene P Wigner. “On the matrices which reduce the Kronecker products of representations of SR groups”. In: *The Collected Works of Eugene Paul Wigner: Part A: The Scientific Papers*. Springer, 1993, pp. 608–654.
- [Zub18] Jean-Bernard Zuber. “Horn’s problem and Harish-Chandra’s integrals. Probability density functions”. In: *Annales de l’Institut Henri Poincaré D* 5.3 (2018), pp. 309–338.

4. [TC] INTRODUCTION

Horn’s problem: The problem of determining the set of eigenvalues (a, b, c) of Hermitian matrices (A, B, C) satisfying $A + B = C$ has been investigated for over a century [Wey12; Fan49; Lid53; Wie55; TF71; TF70] and has come to be known as *Horn’s problem* after Horn who, in 1962, conjectured a solution to the problem in the form of an inductively defined, finite set of homogeneous linear inequalities. In 1998, Klyachko provided a solution to Horn’s problem [Kly98] which, when combined with a proof of the so-called saturation conjecture, due to Knutson and Tao [KT99], constitutes a proof of Horn’s original conjecture [Ful98]. These linear inequalities, now known as the *Horn inequalities* describe the convex polytope of admissible values for (a, b, c) . Recent developments on Horn’s problem include the development of the geometric method by Sherman [She15] and Berline, Vergne, and Walter [BVW18] originally pioneered by the geometric proofs due to Belkale [Bel02; Bel05], as well the characterization and computation of the induced distribution over the eigenvalues c of the sum $C = A + B$ if A and B are sampled uniformly and independently from the respective sets of Hermitian matrices with eigenvalues a and b by Coquereaux, McSwiggen, and Zuber in series of papers [CZ18; CZ18; CMZ19; Zub18]. For a pedagogical and historical introduction to Horn’s problem, we recommend the review articles of Bhatia [Bha01], Knutson [Knu00], and Fulton [Ful00] or alternatively [NP06, Appendix D]. See also the article by Daftuar and Hayden for an account of applications to topics in quantum information theory [DH05].

In parallel to Horn’s problem, variations and extensions of Horn’s problem have been explored by numerous authors. For example, Agnihotri and Woodward, Belkale, and [BK16] have studied the eigenvalues of unitary matrices and their product [AW97; Bel08; BK16], or the admissible singular values of matrices and their product [Spe05; TT71]. In 2000, it was shown by [Kly00] that *logarithms* of the admissible singular values of non-singular matrices and their product are identical to the admissible eigenvalues of Hermitian matrices and their sum, thus proving a conjecture due to Thompson [Kly00; AMW01].

The tetrahedral Horn problem: In this paper, we consider a generalization of Horn’s problem investigated previously only briefly in [CSW18]. While Horn’s problem aims to characterize the eigenvalues (a, b, c) of a triple of $n \times n$ Hermitian matrices (A, B, C) which are subjected to the constraint $A + B = C$, the generalization we consider here aims to characterize the possible eigenvalues (a, b, c, d, e, f) of a *sextuple* of $n \times n$ matrices (A, B, C, D, E, F) subject to four constraints:

{eq:tet_cond}

$$(120) \parallel A + B = C, \quad B + D = F, \quad D + C = E, \quad \text{and} \quad A + F = E.$$

These constraints arise naturally from building a tetrahedral arrangement of $n \times n$ Hermitian matrices inside the vector space of such Hermitian matrices

with edges labelled according to Fig. 2. For this reason, we refer to this eigenvalue problem as the *tetrahedral* Horn problem and we say a sextuple of eigenvalues, (a, b, c, d, e, f) , is *tetrahedral* if such matrices exist, and not tetrahedral otherwise. It is possible to view the tetrahedral Horn problem

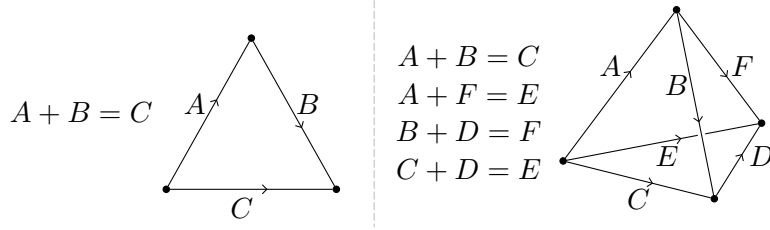


FIGURE 2. (Left) Horn's problem subjects three Hermitian matrices, (A, B, C) , to the constraint $A + B = C$ forming a triangle in the vector space of Hermitian matrices. (Right) The *tetrahedral* Horn problem subjects six Hermitian matrices, (A, B, C, D, E, F) , to a quadruple of constraints forming a tetrahedron in the vector space of Hermitian matrices.

as the conjunction of four separate instances of the original Horn original problem subject to *additional* matching constraints. Specifically, a sextuple of eigenvalues (a, b, c, d, e, f) is tetrahedral only if each of the following four triplets are trigonal:

$$(121) \quad (a, b, c), \quad (b, d, f), \quad (c, d, e), \quad (a, f, e).$$

If each of the above triplets are separately compatible in the sense of the original Horn problem, then there exists a set of twelve Hermitian matrices with the corresponding eigenvalues satisfying

$$(122) \quad A + B = C, \quad B' + D = F, \quad C' + D' = E, \quad A' + F' = E'.$$

The tetrahedral Horn problem goes beyond four separate instances of Horn's original problem in that it additionally requires the corresponding pairs of primed and unprimed Hermitian matrices above must coincide.

The case of 2×2 matrices: Beyond the superficial connection between the imposed constraints and the resulting shape formed in the vector space of Hermitian matrices (as in Fig. 2), in the special case of 2×2 Hermitian matrices, this connection is much deeper. Without loss of generality, it is always possible to restrict attention to the case of *traceless* Hermitian matrices (see Appendix A). Moreover, every 2×2 traceless Hermitian matrix can be identified by a three-dimensional vector $v = (v_1, v_2, v_3) \in \mathbb{R}^3$ using the formula:

$$(123) \quad X = \begin{pmatrix} v_3 & v_1 - iv_2 \\ v_1 + iv_2 & -v_3 \end{pmatrix},$$

in which case, the eigenvalues of X are given by

$$(124) \quad \text{eig}(X) = (\ell_x, -\ell_x),$$

where $\ell_x = \|v\|_2 = \sqrt{v_1^2 + v_2^2 + v_3^2}$ is the Euclident length of the vector v . As a consequence, the eigenvalues for a triangular arrangement of 2×2 traceless Hermitian matrices is isomorphic to the moduli space of sidelengths of Euclidean *triangles* in $\mathbb{R}^3$ [HK96; FKL10; Knu00]. This correspondence generalizes to the tetrahedral case in the sense that the space of admissible eigenvalues (a, b, c, d, e, f) for a tetrahedral arrangements of 2×2 traceless Hermitian matrices is equivalent to the moduli space of Euclidean *tetrahedra* in $\mathbb{R}^3$. This geometric knowledge about the solution space then leads to a complete solution to the tetrahedral Horn's problem for 2×2 matrices [Ric02; WD09]; specifically, the moduli space of Euclidean tetrahedra in $\mathbb{R}^3$ is described by the conjunction of the triangle inequalities for each of the ~~four~~ ^{four} faces,

$$(125) \quad |\ell_a - \ell_b| \leq \ell_c \leq \ell_a + \ell_b, \quad |\ell_b - \ell_d| \leq \ell_f \leq \ell_b + \ell_d, \quad |\ell_d - \ell_c| \leq \ell_e \leq \ell_d + \ell_c, \quad |\ell_a - \ell_f| \leq \ell_e \leq \ell_a + \ell_f,$$

together with a non-linear inequality [SJS18]:

$$(126) \quad \text{CM}(\ell_a, \ell_b, \ell_c, \ell_d, \ell_e, \ell_f) = \frac{1}{288} \det \begin{pmatrix} 0 & \ell_a^2 & \ell_c^2 & \ell_e^2 & 1 \\ \ell_a^2 & 0 & \ell_b^2 & \ell_f^2 & 1 \\ \ell_c^2 & \ell_b^2 & 0 & \ell_d^2 & 1 \\ \ell_e^2 & \ell_f^2 & \ell_d^2 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{pmatrix} \geq 0.$$

The quantity $\text{CM}(\ell_a, \ell_b, \ell_c, \ell_d, \ell_e, \ell_f)$ is known as a Cayley-Menger determinant; if $\text{CM}(\ell_a, \ell_b, \ell_c, \ell_d, \ell_e, \ell_f)$ is positive and if the triangle inequalities above are satisfied, the value of $\text{CM}(\ell_a, \ell_b, \ell_c, \ell_d, \ell_e, \ell_f)$ is equal to *square* of the volume of the corresponding tetrahedron [SJS18, Equation 5.2]². Figure 3 depicts a three-dimensional projection of the solution space described by Eq. (125) and Eq. (126). Before proceeding, we remark that even for the case of 2×2 matrices, there is a strong qualitative difference between the tetrahedral Horn problem and the original Horn problem. In particular, while the solution to the original Horn problem is completely characterized by a convex polytope (i.e., by a finite number of linear inequalities in the eigenvalues), the solution space for the tetrahedral Horn's problem is neither convex nor a polytope.

Our contributions: Regarding the tetrahedral Horn problem, we derive a new family of inequalities constraining the set of tetrahedral eigenvalues.

²The triangle inequalities $|\ell_a - \ell_b| \leq \ell_c \leq \ell_a + \ell_b$ are also equivalent to the non-negativity of the *area*, $\text{CM}(\ell_a, \ell_b, \ell_c)$, of the triangle with side-lengths ℓ_a, ℓ_b and ℓ_c where

$$\text{CM}(\ell_a, \ell_b, \ell_c)^2 = -\frac{1}{16} \det \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & \ell_a^2 & \ell_c^2 \\ 1 & \ell_a^2 & 0 & \ell_b^2 \\ 1 & \ell_c^2 & \ell_b^2 & 0 \end{pmatrix} \text{ is equivalent to Heron's formula.}$$

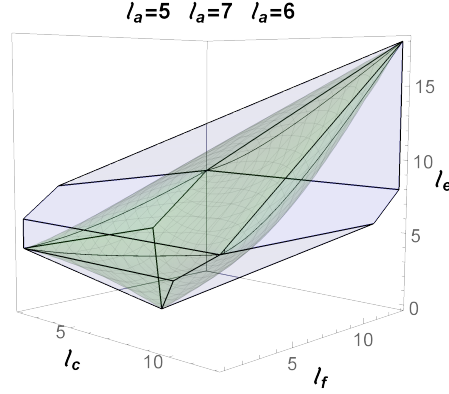


FIGURE 3. A comparison between the values for (ℓ_c, ℓ_e, ℓ_f) with fixed $(\ell_a, \ell_b, \ell_d) = (5, 7, 6)$ which merely satisfy the triangle inequalities (exterior blue polytope) and those which additionally satisfy the Cayley-Menger determinant inequality (interior green). The interior green region depicts (a slice of) the space of Euclidean tetrahedra with side lengths $(\ell_a, \ell_b, \ell_c, \ell_d, \ell_e, \ell_f)$ and thus depicts the solution to the tetrahedral Horn problem for 2×2 traceless Hermitian matrices; remark that it is neither a polytope of convex.

These inequalities are expressed in terms of Schur polynomials $s_\lambda(x)$ of the involved eigenvalues which are homogeneous symmetric polynomials indexed by Young diagrams λ . More accurately, we find our inequalities are best articulated in terms of a simple rescaling of the Schur polynomials, $\varphi_\lambda(x) := s_\lambda(x)/H_\lambda$, where H_λ is a positive integer equal to the product of the hook-lengths of all boxes in the Young diagram λ . These inequalities are presented in the following theorem.

thm:tet_iff

Theorem 14. *Let (a, b, c, d, e, f) be a sextuple of non-negative eigenvalues of length n satisfying the trace conditions:*

(eq:tet_trace_conditions)

$$\text{Tr}[a] + \text{Tr}[b] = \text{Tr}[c], \text{Tr}[b] + \text{Tr}[d] = \text{Tr}[f], \text{Tr}[c] + \text{Tr}[d] = \text{Tr}[e], \text{ and } \text{Tr}[a] + \text{Tr}[f] = \text{Tr}[e].$$

Then there exists Hermitian matrices (A, B, C, D, E, F) with satisfying $A + B = C$, $B + D = F$, $C + D = E$ and $A + F = E$ with eigenvalues (a, b, c, d, e, f) if and only if for all triples of Young diagrams (α, β, δ) ,

$$(128) \quad \varphi_\alpha(a)^{\frac{2}{3}} \varphi_\beta(b) \varphi_\delta(d)^{\frac{2}{3}} \leq \sum_{\gamma, \epsilon, \phi} \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty \varphi_\phi(f)^{\frac{1}{3}} \varphi_\gamma(c)^{\frac{1}{3}} \varphi_\epsilon(e)^{\frac{1}{3}}.$$

The expression $\begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix}$ appearing above is an operator known as the $6j$ -symbol for the group of unitary $n \times n$ matrices $U(n)$ whose significance will be explained shortly. While Theorem 14 provides a complete characterization of the set of tetrahedral eigenvalues, it does so using an infinite number of

inequalities. Nevertheless, we prove the finite subset of such inequalities of fixed degree $k = |\alpha| + |\beta| + |\delta|$ is sufficient to characterize the set of sextuples of eigenvalues, (a, b, c, d, e, f) , which are *approximately* tetrahedral in the sense that their distance, $D\left(\begin{smallmatrix} a & b & c \\ d & e & f \end{smallmatrix}\right)$, to the set of tetrahedral eigenvalues is small.

thm:approx_tet

Theorem 15. *If (a, b, c, d, e, f) are non-negative eigenvalues satisfying the trace conditions in Eq. (127) and also ?? for all triples (α, β, δ) such that $|\alpha| + |\beta| + |\delta| = k$ for some positive integer k , then*

{eq:tri_abd_cef_dist_bound}

$$(129) \quad \frac{1}{\text{Tr}[e]} D\left(\begin{smallmatrix} a & b & c \\ d & e & f \end{smallmatrix}\right) \leq O\left(n\sqrt{\frac{\log k}{k}}\right),$$

{eq:defn_tet_deviation} where $D\left(\begin{smallmatrix} a & b & c \\ d & e & f \end{smallmatrix}\right)$ is the distance measure

$$(130) \quad D\left(\begin{smallmatrix} a & b & c \\ d & e & f \end{smallmatrix}\right) := \min_{\substack{(A,B,D) \sim \\ \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d}} \|\text{eig}(A+B) - c\|_1^2 + \|\text{eig}(B+D) - f\|_1^2 + \|\text{eig}(A+B+D) - e\|_1^2.$$

Note that $D\left(\begin{smallmatrix} a & b & c \\ d & e & f \end{smallmatrix}\right) = 0$ if and only if (a, b, c, d, e, f) are tetrahedral.

Note this theorem actually implies the “if” portion of ?? as one can take $k \rightarrow \infty$. Along the way, we derive an analogous set of inequalities for the original Horn problem along with an accompanying distance bound.

6j-symbols: Given a complex reductive group G with unitary irreducible representations (irreps) V_λ labelled by highest weights λ along with tensor-product multiplicity spaces, $M_{\alpha,\beta}^\gamma := \text{Hom}_G(V_\gamma, V_\alpha \otimes V_\beta)$, the 6j-symbols are maps between these multiplicity spaces which encode the associativity constraint $(V_\alpha \otimes V_\beta) \otimes V_\gamma \cong V_\alpha \otimes (V_\beta \otimes V_\gamma)$ between a triple of irreps, V_α, V_β , and V_γ of G . Specifically, a 6j-symbol depends on six irreps, $V_\alpha, V_\beta, V_\gamma, V_\delta, V_\epsilon$, and V_ϕ , and is the linear map between multiplicity spaces of the type

$$\left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} : M_{\alpha,\beta}^\gamma \otimes M_{\delta,\gamma}^\epsilon \rightarrow M_{\alpha,\phi}^\epsilon \otimes M_{\beta,\delta}^\phi,$$

corresponding to the block-wise components of the canonical isomorphism from $\bigotimes_\gamma M_{\alpha,\beta}^\gamma \otimes M_{\delta,\gamma}^\epsilon$ to $\bigotimes_\phi M_{\alpha,\beta}^\gamma \otimes M_{\delta,\gamma}^\epsilon$. Whenever the group G is multiplicity-free, in the sense that the dimensions of all multiplicity spaces, $m_{\alpha,\beta}^\gamma := \dim M_{\alpha,\beta}^\gamma$, are either zero or one, the corresponding 6j-symbol is merely a scalar quantity.

Wigner’s 6j-symbol for SU(2): The group SU(2) is multiplicity-free and the 6j-symbols for SU(2) are, depending on the normalization and/or phase convention, are also referred to as Wigner 6j-symbols or Wigner 6j-coefficients [Wig93] or the Racah W -coefficients [Rac42]. The 6j-symbols for SU(2) play a foundational role in the quantum theory of angular momentum, modelling the recoupling theory of three angular momenta, and thus they

are used frequently in calculations involving the scattering of multiple particles in nuclear, atomic and molecular physics [VMK88; Edm96; Aqu+12]. Beyond the direct application to the theory of angular momenta, the $6j$ -symbols for $SU(2)$ also constitute a primitive object in the theory of spin networks [Pen71; BS10], representing a minimal non-trivial closed spin network. General spin networks are networks of trivalent vertices and connections labelled by irreps (spins) of $SU(2)$ which provide a discrete/combinatorial and manifestly invariant model with applications to gauge theories on graphs or manifolds [Bae96], invariants of knots and manifolds [Res87; TV92], topological quantum field theory [DJN93], elementary gates in spin network models for quantum computing [MR05], quantum change-point discrimination [AH11], and various aspects of proposed models for quantum gravity [Bae99; BC98; Bar03; Rov04; Mou84].

Throughout this paper, by the *asymptotics* of $6j$ -symbols, we are referring to their behaviour in the regime wherein the six highest weights, $\lambda \in \{\alpha, \beta, \gamma, \delta, \epsilon, \phi\}$, are becoming larger and larger in a suitable sense. The asymptotics of the $6j$ -symbol for $SU(2)$, wherein the irreps of dimension $\ell + 1$ may be identified either by integers $\ell \in \{0, 1, \dots\}$ or half-integers $j = \frac{\ell}{2} \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$, were initially investigated by Wigner [Wig60, Chapter 27]. In the context of the quantum theory of angular momenta, the limit of large ℓ , or equivalently large spins j , may also be interpreted as the limit wherein Planck's constant tends to zero, $\hbar \propto \frac{1}{k} \rightarrow 0$, which suppresses the effects of quantizing angular momentum; therefore, these asymptotic limits are also known as *semiclassical* limits. Given three particles with spins $\frac{a}{2}$, $\frac{b}{2}$ and $\frac{d}{2}$ and total spin $\frac{e}{2}$, the $6j$ -symbol, $\left\{ \begin{smallmatrix} a & b & c \\ d & e & f \end{smallmatrix} \right\}$, can be interpreted as the quantum mechanical amplitude to observe the first and second particles to have combined spin $\frac{e}{2}$ given the second and third have combined spin $\frac{f}{2}$ [Wig60; Rob99]. Wigner considered the probability for this event to occur in the asymptotic limit of large spins, i.e., the square of the $6j$ -symbol, and conjectured the estimate:

$$\left\{ \begin{smallmatrix} a & b & c \\ d & e & f \end{smallmatrix} \right\}^2 \approx \frac{1}{3\pi V}$$

where V is the volume of the Euclidean tetrahedron with side-lengths (a, b, c, d, e, f) . Building upon the work of Wigner, Ponzano and Regge conjectured a formula for the amplitude of the $6j$ -symbol itself in the limit of large spins [PR68]. Later, using techniques from geometric quantization theory, Roberts rigorously proved an asymptotic formula which is equivalent to the conjecture of Ponzano and Regge (up to minor differences) [Rob99].

Theorem 16 (Roberts). *If (a, b, c, d, e, f) are integer side-lengths of a Euclidean tetrahedron with volume V and corresponding exterior dihedral angles³ $(\theta_a, \theta_b, \theta_c, \theta_d, \theta_e, \theta_f)$, then*

$$\left\{ \begin{smallmatrix} ka & kb & kc \\ kd & ke & kf \end{smallmatrix} \right\} \stackrel{k \rightarrow \infty}{\sim} \sqrt{\frac{2}{3\pi V k^3}} \cos \left(\sum_{\ell} (k\ell + 1) \frac{\theta_{\ell}}{2} + \frac{\pi}{4} \right),$$

where the sum is over the six sides ℓ . Otherwise, if there does not exist a Euclidean tetrahedron with side-lengths (a, b, c, d, e, f) , then $\begin{Bmatrix} ka & kb & kc \\ kd & ke & kf \end{Bmatrix}$ decays to zero at an exponential rate as $k \rightarrow \infty$.

A proof of the Ponzano-Regge asymptotic formula was also obtained by Chen, Ismail, and Simeonov [CIS99] (with a correction by Watson [Wat99]) using estimates from the theory of hypergeometric functions. Additional proofs of this asymptotic correspondence between $6j$ -symbols for $SU(2)$ and Euclidean tetrahedra have emerged in recent years including the elementary proof by Gurau which applies Stirling's approximation directly to Racah's single-sum formula [Gur08] as well as group integral approach to evaluate so-called simple spin networks as Feynman graphs [FK00; FL03]. See also the paper by Bartlett and Nzaganya for an accessible presentation of these kinds of calculations [BN23]. Analogous asymptotic formulas have also been obtained by Taylor and Woodward for the $6j$ -symbol associated to the quantum group $U_q(\mathfrak{sl}_2)$ for a particular k -dependent parameter q [TW06]. In this generalized setting, the asymptotics calculated by Taylor and Woodward encode (i) the existence of a spherical tetrahedron on the unit sphere with prescribed edge-lengths whenever $q = \exp(i\pi/r(k))$ is an $r(k)$ th root of unity or (ii) the existence of a hyperbolic tetrahedron with prescribed edge-lengths when $q = \exp(\pi/r(k))$ is positive real (here $r(k) = k(r-2) + 2$ for $r > 2$ a fixed integer) [TW06].

More generally, that the asymptotics of representation-theoretic quantities (such as $6j$ -symbols) are controlled by the existence of some critical geometric object is explained by the theory of geometric quantization [Rob02]. Geometric quantization, broadly speaking, associates classical phase spaces modelled by symplectic manifolds to quantum phase spaces modelled by Hilbert spaces [Rob02; GS82; Kir04]. In particular, Kirillov's orbit principle [Kir04] proposes an association between (i) the coadjoint orbits of a compact Lie group G passing through an integral weight λ to (ii) the corresponding unitary irreps of G of highest weight λ . Furthermore, products of coadjoint orbits correspond to tensor products of irreps and moreover symplectic quotients of the resulting products of orbits are identified with invariant vectors in the resulting tensor products of irreps [Rob02; GS82]. For the aforementioned case of $SU(2)$, the coadjoint orbits can be identified the $SO(3)$ rotations of vectors in Euclidean $\mathbb{R}^3$ (i.e., spheres of fixed radius), and thus the asymptotics of representation-theoretic quantities for $SU(2)$ are controlled by the existence of corresponding geometric objects in Euclidean $\mathbb{R}^3$ with prescribed edge-lengths [Rob02].

$6j$ -symbols for $U(n)$: One application of the techniques developed in this paper is to provide estimates for the asymptotics of the $6j$ symbols for the group $U(n)$ (or equivalently for $SU(n)$) wherein the polynomial irreps

³For example, the exterior dihedral angle θ_a is the outside angle between the two faces of the tetrahedron sharing the common edge with side-length a .

V_λ^n are labelled by highest-weights which can also be identified with Young diagrams λ . One of the primary difficulties in proving statements about $6j$ symbols for $U(n)$ for $n \geq 3$ is the presence of non-trivial multiplicity spaces, meaning the dimensions of tensor-product multiplicity spaces for $U(n)$, also known as Littlewood-Richardson coefficients, can be greater than one. Because of these difficulties, over the years, considerable attention has been placed on the study of $U(n)$ (or $U_q(\mathfrak{sl}_n)$) $6j$ symbols which are multiplicity-free in the sense that the dimensions of each of the multiplicity spaces is at most one [NPZ13; AMS20; Alc+23]. Indeed, as stated by Coquereaux, “apart from very few special cases (low dimension of irreps and no multiplicity), ... we are not aware of any published general result.” [Coq23].

Our approach bypasses the difficulties presented by multiplicities by focusing only on the *norms* of the $6j$ symbol for $U(n)$. Following the procedure provided by geometric quantization, one expects the asymptotics of the $6j$ -symbol for $U(n)$ to be controlled by existence of some kind of tetrahedral object in space of coadjoint orbits for $U(n)$. In particular, the coadjoint orbits of $U(n)$ can be identified with the orbits of $n \times n$ Hermitian matrices, X , under the conjugation action, $X \mapsto U^{-1}XU$, by unitary $n \times n$ matrices $U \in U(n)$; in other words, coadjoint orbits are associated to sets of $n \times n$ Hermitian matrices sharing a common set of eigenvalues $x = (x_1 \geq \dots \geq x_n)$. Using our methods we are able to prove the following inverse polynomial lower bound.

cor:asym_6j

Corollary 17. *If (a, b, c, d, e, f) are the eigenvalues of $n \times n$ of positive semidefinite matrices (A, B, C, D, E, F) satisfying $A + B = C$, $B + D = F$, $C + D = E$, and $A + F = E$, then there exists a sequence of Young diagrams $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$ (with $|\epsilon_k| = k$) converging in proportion to the eigenvalues (a, b, c, d, e, f) such that*

$$(131) \quad \left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\| \geq \frac{1}{\text{poly}(k)}.$$

Otherwise, if (a, b, c, d, e, f) are not the eigenvalues of any such matrices, then for any sequence of Young diagrams converging in proportion to (a, b, c, d, e, f) , we have exponential decay in the value of $\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|$.

Before continuing, it is worth emphasizing the similarities and differences between our result and the asymptotic formula proven by Roberts above [Rob99]. First and foremost, our result goes beyond that of Roberts in the sense that it applies to the higher-rank Lie group $U(n)$ for $n \geq 3$. Moreover, our result, even when restricted to the case for $SU(2)$, goes beyond the statement proven by Roberts in the sense our result is applicable to *irrational* eigenvalues/edge-lengths, whereas the former is only applicable to *integer*, or with a minor re-scaling, *rational* eigenvalues/edge-lengths. like the Pozano-Regge formula, our result is different in that it merely promises the *existence* of a sextuple of sequences of Young diagrams, $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$, converging to the eigenvalues (a, b, c, d, e, f) , whereas the result of Roberts provides a

concrete statement about the particular sequence of *scaled* Young diagrams $(k\alpha, k\beta, k\gamma, k\delta, k\epsilon, k\phi)$. Moreover, our result provides a polynomial lower bound on the value of the $\left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|$ and does not provide an asymptotic formula which would serve as the proper generalization of the Ponzano and Regge formula above [PR68]. The Ponzano and Regge formula for $SU(2)$ was traditionally obtained using the method of *stationary phases* which relies on both an integral formula for the $6j$ -symbol with exponential integrand along with a geometric understanding of the *isolated* critical points for the exponential phase [PR68; Gur08]. In the case of $SU(2)$ with generic edge-lengths, there is a formulation involving exactly *two* isolated critical points corresponding the $SO(3)$ -orbit $[\tau]$ of a Euclidean tetrahedron τ with given edge-lengths and (ii) the $SO(3)$ -orbit $[\tau']$ of the Euclidean tetrahedron τ' which is the mirror-image of the tetrahedron τ with the same fixed side lengths. Generalizing this type of criticality analysis to $U(n)$ for $n \geq 3$ appears to be a significant challenge, and even if an analysis was performed, one would still need to compute a corresponding Hessian for each critical point and evaluate the total contribution [Rob02]. Perhaps a different approach for deriving the Ponzano-Regge formula, such as those based on recursion relations or estimates of hypergeometric functions appearing in [CIS99; TW06; Art23; Art24] would better suited for the case where $n \geq 3$.

Eigenvalue estimation: Our approach to proving the above theorems relies on the technique of *eigenvalue estimation*, a probabilistic technique developed in the context of quantum information theory for estimating the eigenvalues of positive semidefinite matrices which model the state of quantum systems [KW01; CM06; OW16; HM02; CHW07; Hay17]. The eigenvalue estimation technique relies heavily on the concept of Schur-Weyl duality which describes the decomposition of the k -th tensor power of the n -dimensional complex vector space $\mathbb{C}^n$, which supports representations of both the symmetric group S_k by permutation of tensor factors and the group of unitary matrices $U(n)$ (or more generally invertible complex matrices $GL(n, \mathbb{C})$), into irreducible invariant subspaces indexed by Young diagrams $\lambda = (\lambda_1, \dots, \lambda_n)$ with k boxes and at most n rows. Given a positive semidefinite matrix X with eigenvalues $x = (x_1, \dots, x_n)$, one can analogously decompose the k -th tensor power of X into the same irreducible invariant subspaces indexed by λ and then investigate the aptly named Schur-Weyl distribution, $p_x(\lambda) := \text{Tr}[P_\lambda X^{\otimes k}] / \text{Tr}[X^{\otimes k}]$, which measures the relative proportion of the decomposed pieces of $X^{\otimes k}$ and depends only on the eigenvalues x of X . The technique of eigenvalue estimation relies on the fact that the Schur-Weyl distribution concentrates strongly around those Young diagrams λ which are proportional to eigenvalues x and that this concentration strengthens as the sizes of the Young diagrams grow to infinity [KW01; CM06; OW16]. Applying the eigenvalue estimation technique to the original Horn problem was first done by Christandl [Chr08]. In this paper, we

develop a technique called *eigenvalue separation* which relates the similarity between two Schur-Weyl distributions, $p_x(\lambda)$ and $p_y(\lambda)$, to the similarity of the underlying eigenvalues, x and y , and apply this technique to tetrahedral Horn problem and obtain ???. In addition, our approach produces new results for the original Horn problem, analogous to ???, beyond those obtained in [Chr08].

Quantum marginals & the $6j$ -symbol for the symmetric group S_k : Our method for proving Corollary 17 is closely related to techniques used in an earlier paper by Christandl, Şahinoğlu, and Walter which derived analogous bounds on the $6j$ symbol for the symmetric group S_k [CŞW18]. In that paper, the authors considered the problem of determining whether there exists a density matrix ρ_{ABC} with partial traces $\rho_{AB}, \rho_{BC}, \rho_A, \rho_B$, and ρ_C with a given sextuple of eigenvalues [CŞW18]. Such problems arise naturally in the context of quantum information theory and are typically referred to as quantum marginals problems [CŞW18; Fra22]. In particular, they proved the $6j$ symbols for the symmetric group S_k (referred to as the recoupling coefficients) in the limit as $k \rightarrow \infty$ either decay exponentially or can be bounded from below by a polynomial factor depending on whether there exists a density matrix with the prescribed marginal spectra [CŞW18]. Moreover, they also demonstrated how the tetrahedral Horn problem may be viewed as a special case of the quantum marginals problem and, in a complementary manner, furthermore demonstrated how, by restricting to the subgroup $S_k \subset U(k)$ of permutation matrices and applying Nishiyama’s restriction theorem [Nis00], the $6j$ symbols for unitary group $U(n)$ may be viewed as a special case of the $6j$ symbols for the symmetric group S_k [CŞW18]. This latter result obtained by [CŞW18] is a generalizes how the Littlewood-Richardson coefficients can be cast as a special case of the Kronecker coefficients [Lit58; Mur55; IP23]. While the tetrahedral Horn problem may be viewed as a special case of the quantum marginals problem, the approach taken by this paper differs from [CŞW18] as it tackles the tetrahedral Horn problem directly, and consequently, we obtain ??? which provides stronger conditions for tetrahedral eigenvalues which go beyond the conditions previously obtained in [CŞW18]. We note that this direct approach was initiated, for the $n = 2$ case, by Backens [Bac10].

Multiple Horn problems: More recently, Alekseev et al. have investigated a few variants of the tetrahedral Horn problem and are referred to as *multiple* Horn problems in [Ale+25]. Specifically, Alekseev et al. focus on two variants, the *multiplicative multiple Horn problem* concerning the singular values of products of invertible matrices, and the *tropical multiple Horn problem* concerning singular values of concatenations of weighted planar networks [Ale+25]. The tetrahedral Horn problem we consider here is termed the *additive* multiple Horn problem in the language of [Ale+25].

Other inequalities: It is worth highlighting that there exists related results constraining the eigenvalues of sums of Hermitian matrices which are already readily applicable to the tetrahedral Horn problem. In addition to the aforementioned Horn’s inequalities [Hor62], which constraint the eigenvalues of any triple of Hermitian matrices $(A, B, A + B)$, there exists a myriad of inequalities which constrain the eigenvalues of the quadruple of Hermitian matrices $(Y, X + Y, Y + Z, X + Y + Z)$ whenever $X, Y, Z \geq 0$ are positive semidefinite matrices (see for example [BS15; TCL11; Sag13; BGS08; FG13; LS14; Sra]). These inequalities are readily applicable to the tetrahedral Horn problem under the identification $(X = A, Y = B \text{ and } Z = D)$ wherein they constrain, respectively, the eigenvalues (b, c, e, f) of the matrices (B, C, E, F) while ignoring the eigenvalues of A and D . More scarce are inequalities which additionally incorporate the eigenvalues (a, d) of the matrices (A, D) such as Minkowski-like determinant inequality [NS23, Section 6.2] (originally due to Courtade[Cou16; htt]), or applications of Ptolemy’s inequality [Sch52] or the Cayley-Menger determinant inequality [SJS18, Equation 5.2] to the space of $n \times n$ Hermitian matrices equipped with the Hilbert-Schmit inner product. While each of these previously known inequalities can be viewed as necessary conditions for the tetrahedral Horn problem, to the best of our knowledge, they remain insufficient for the tetrahedral Horn problem when $n \geq 3$. Our contribution, therefore, is to provide a family of necessary and sufficient inequalities in the form of ?? which is applicable to all dimensions n .

Organization of the paper:

TODO

ACKNOWLEDGMENTS

TCF thanks Suvrit Sra, Harold Nieuwboer, Marco Fanizza and Jack Davis for helpful discussions. MC and TCF acknowledge financial support from the European Research Council (ERC Grant Agreement No. 818761), VILLUM FONDEN via the QMATH Centre of Excellence (Grant No. 10059) and the Novo Nordisk Foundation (grant NNF20OC0059939 ‘Quantum for Life’). They also thank the National Center for Competence in Research SwissMAP of the Swiss National Science Foundation and the Section of Mathematics at the University of Geneva for their hospitality.

TC: THE FOLLOWING IS CURRENTLY BEING RE-REWRITTEN INTO PREVIOUS SECTIONS

5. [TC] INEQUALITIES FOR TETRAHEDRAL EIGENVALUES sec:ineqsgfromeqchornweyt

5.1. [tc] **Inequalities for the tetrahedral Horn problem.** In Appendix B, we saw how the quantity $\varphi_\gamma^{\alpha\beta}(X, Y)$ (??) could be used to produce inequalities applicable to the original Horn problem. In this section, we generalize this analysis to produce inequalities applicable to the tetrahedral Horn problem using a generalization of $\varphi_\gamma^{\alpha\beta}(X, Y)$ which depends on three matrices.

Definition 18. Let X, Y, Z be $n \times n$ matrices. For any sextuple $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ of Young diagrams, let

$$(132) \quad \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) := \text{Tr}[(P_{\alpha\beta}^\gamma \otimes \text{id}_{V_\delta^n}) P_{\alpha\beta\delta}^\epsilon (\text{id}_{V_\alpha^n} \otimes P_{\beta\delta}^\phi) \Phi_\alpha(X) \otimes \Phi_\beta(Y) \otimes \Phi_\delta(Z)].$$

Remark that in our definition of $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$, we permit the sizes of the involved Young diagrams to be arbitrary. However, there are numerous conditions on the sizes of the Young diagrams $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ which are necessary (but not always sufficient) for $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ to be non-zero; specifically,

{eq:yd_sizes_tet}

$$(133) \quad |\alpha| + |\beta| = |\gamma|, \quad |\beta| + |\delta| = |\phi|, \quad |\alpha| + |\phi| = |\epsilon|, \quad |\gamma| + |\delta| = |\epsilon|.$$

If we additionally let

$$(134) \quad \Omega_{\alpha\beta\delta}^{X,Y,Z} := \Phi_\alpha(X) \otimes \Phi_\beta(Y) \otimes \Phi_\delta(Z).$$

then we can compactly express $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ as

{eq:varphi_compact_tet}

$$(135) \quad \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) = \text{Tr}[\Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Pi_{\alpha\beta\delta}^{\epsilon;\phi} \Omega_{\alpha\beta\delta}^{X,Y,Z}].$$

Unlike the quantity $\varphi_\gamma^{\alpha\beta}(X, Y) = \text{Tr}[P_{\alpha\beta}^\gamma \Phi_\alpha(X) \otimes \Phi_\beta(Y)]$ involving two matrices X, Y (see ??), the quantity defined here, $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$, is not necessarily non-negative or even real valued even under the assumption that (X, Y, Z) are positive semidefinite. Nevertheless, summing over either γ or ϕ restores non-negativity because it eliminates one of the potentially non-commuting projection operators $\Pi_{\alpha\beta\delta}^{\epsilon;\gamma}$ or $\Pi_{\alpha\beta\delta}^{\epsilon;\phi}$. We denote these partial sums of $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ over γ and ϕ respectively as

{eq:psum_defns}

$$(136) \quad$$

$$\varphi_{\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) := \sum_{\gamma} \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) = \text{Tr}[\Pi_{\alpha\beta\delta}^{\epsilon;\phi} \Omega_{\alpha\beta\delta}^{X,Y,Z}] = \text{Tr}[P_{\gamma\delta}^\epsilon (\Phi_\gamma^{\alpha\beta}(X, Y) \otimes \Phi_\delta(Z))],$$

$$\varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z) := \sum_{\phi} \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) = \text{Tr}[\Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Omega_{\alpha\beta\delta}^{X,Y,Z}] = \text{Tr}[P_{\alpha\phi}^\epsilon (\Phi_\alpha(X) \otimes \Phi_\phi^{\beta\delta}(Y, Z))],$$

$$\begin{array}{c}
\begin{array}{ccccccc}
& & \xrightarrow{\sum_{\gamma}} & \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) & \xrightarrow{\sum_{\epsilon}} & \varphi_{\alpha}(X)\varphi_{\phi}^{\beta\delta}(Y, Z) & \xrightarrow{\sum_{\beta\delta}} \varphi_{\alpha}(X)\varphi_{\phi}(Y+Z) \\
\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) & \xrightarrow{\sum_{\alpha\beta\delta\gamma\phi}} \varphi_{\epsilon}(X+Y+Z) & \xleftarrow{\sum_{\alpha\beta\delta}} \varphi_{\epsilon}^{\alpha\beta\delta}(X, Y, Z) & \xrightarrow{\sum_{\epsilon}} \varphi_{\alpha}(X)\varphi_{\beta}(Y)\varphi_{\delta}(Z) & \xleftarrow{\sum_{\phi}} \varphi_{\alpha}(X)\varphi_{\beta}(Y)\varphi_{\delta}(Z) & \xleftarrow{\sum_{\phi}} \varphi_{\alpha}(X)\varphi_{\beta}(Y)\varphi_{\delta}(Z) & \\
& & \xrightarrow{\sum_{\phi}} & \varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z) & \xrightarrow{\sum_{\epsilon}} & \varphi_{\gamma}^{\alpha\beta}(X, Y)\varphi_{\delta}(Z) & \xrightarrow{\sum_{\alpha\beta}} \varphi_{\gamma}(X+Y)\varphi_{\delta}(Z)
\end{array}
\end{array}$$

FIGURE 4. Various summation identities satisfied by $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$

where the sum over γ is implicitly such that $|\gamma| = |\alpha| + |\beta|$ and the sum over ϕ is implicitly such that $|\phi| = |\beta| + |\delta|$. Although $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ is not necessarily real or positive, the above expression makes it evident that

$$(137) \quad \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \geq 0, \quad \varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z) \geq 0.$$

The various summation identities satisfied by Fig. 4 are summarized in Fig. 4.

The next series of results concerns upper bounds on the absolute value of $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$. These upper bounds follow from the following general lemma applied to the compact expression for $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ appearing in Eq. (135).

Proposition 19. *Let $\Omega \geq 0$ be a positive semidefinite operator and let Π_1, Π_2 be Hermitian projection operators acting on a common vector space. Then*

$$(138) \quad |\text{Tr}(\Pi_1 \Pi_2 \Omega)| \leq \|\Pi_1 \Pi_2\|_{\infty} \text{Tr}(\Pi_1 \Omega)^{\frac{1}{2}} \text{Tr}(\Pi_2 \Omega)^{\frac{1}{2}}.$$

Proof. As $\Omega \geq 0$ is positive semidefinite, let $\Omega^{\frac{1}{2}} \geq 0$ denote its square root. Then let $L = \Omega^{\frac{1}{2}} \Pi_2 \Pi_1$ and $R = \Omega^{\frac{1}{2}} \Pi_1$ so that

$$(139) \quad \langle L, R \rangle = \text{Tr}(L^{\dagger} R) = \text{Tr}(\Pi_1 \Pi_2 \Omega \Pi_1) = \text{Tr}(\Pi_1 \Pi_2 \Omega),$$

where the last equality uses the assumption that $\Pi_1^2 = \Pi_1$. An application of the Cauchy-Schwarz inequality for the Hilbert-Schmidt inner product $\langle L, R \rangle$ then implies

$$(140) \quad |\text{Tr}(\Pi_1 \Pi_2 \Omega)|^2 = |\langle L, R \rangle|^2 \leq \text{Tr}[L^{\dagger} L] \text{Tr}[R^{\dagger} R] = \text{Tr}[\Pi_1 \Pi_2 \Omega \Pi_2 \Pi_1] \text{Tr}[\Pi_1 \Omega \Pi_1] = \text{Tr}[\Pi_1 \Pi_2 \Omega \Pi_2] \text{Tr}[\Pi_1 \Omega].$$

Now let $\{v_j\}_j$ be an orthonormal eigenbasis for $\Pi_2 \Omega \Pi_2$ with eigenvalue $\omega_j \geq 0$. Then, because it was assumed that $\Pi_2^2 = \Pi_2$, we have

$$(141) \quad \text{Tr}[\Pi_1 \Pi_2 \Omega \Pi_2] = \text{Tr}[(\Pi_2 \Pi_1^2 \Pi_2)(\Pi_2 \Omega \Pi_2)] = \sum_j \omega_j \langle \Pi_1 \Pi_2 v_j, \Pi_1 \Pi_2 v_j \rangle \leq \sum_j \omega_j \|\Pi_1 \Pi_2\|_{\infty}^2 = \text{Tr}[\Pi_2 \Omega] \|\Pi_1 \Pi_2\|_{\infty}^2.$$

The claim then follows from combining these inequalities and taking the square root of both sides.

□
lem:varphi_tet_main_ineq

Lemma 20. *Let (X, Y, Z) be positive semidefinite $n \times n$ matrices. Then for all sextuples of Young diagrams $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$,*

$$(142) \quad \left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right| \leq \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_{\infty} \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)^{\frac{1}{2}} \varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z)^{\frac{1}{2}}.$$

Proof. Under the assumption that $X, Y, Z \geq 0$, we have that $\Omega_{\alpha\beta\delta}^{X,Y,Z} = \Phi_{\alpha}(X) \otimes \Phi_{\beta}(Y) \otimes \Phi_{\delta}(Z) \geq 0$ is positive semidefinite also. Now applying Proposition 19 to the compact expression for $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ in Eq. (135) yields

$$(143) \quad \left| \text{Tr}[\Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Pi_{\alpha\beta\delta}^{\epsilon;\phi} \Omega_{\alpha\beta\delta}^{X,Y,Z}] \right| \leq \left\| \Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Pi_{\alpha\beta\delta}^{\epsilon;\phi} \right\|_{\infty} \text{Tr}[\Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Omega_{\alpha\beta\delta}^{X,Y,Z}]^{\frac{1}{2}} \text{Tr}[\Pi_{\alpha\beta\delta}^{\epsilon;\phi} \Omega_{\alpha\beta\delta}^{X,Y,Z}]^{\frac{1}{2}}$$

which is equivalent to the claim due to ??.

□

Our next proposition enables us to place bounds on $\varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ and $\varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z)$ which depend only the eigenvalues of $X, Y, X+Y, Z, X+Y+Z$, and $Y+Z$ and not their eigenvectors.

prop:tet_psum_ineq

Proposition 21. *Let $X, Y, Z \geq 0$ be positive semidefinite $n \times n$ matrices. Then we have bounds on the partial sums,*

$$(144) \quad \begin{aligned} \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) &\leq \varphi_{\alpha}(X) \varphi_{\phi}(Y+Z), \\ \varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z) &\leq \varphi_{\gamma}(X+Y) \varphi_{\delta}(Z), \end{aligned}$$

eq:psum_yc_uniform Well as bounds on $\varphi_{\epsilon}^{\alpha\beta\delta}(X, Y, Z)$:

$$(145) \quad \max\{\varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z), \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)\} \leq \varphi_{\epsilon}^{\alpha\beta\delta}(X, Y, Z) \leq \min\{\varphi_{\epsilon}(X+Y+Z), \varphi_{\alpha}(X) \varphi_{\beta}(Y) \varphi_{\delta}(Z)\}.$$

Proof. Each of these bounds are consequences of the summation identities for $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ and the fact that the partial sums $\varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ and $\varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z)$ are non-negative real-valued whenever $X, Y, Z \geq 0$ are positive semidefinite. These summation identities are summarized in Fig. 4. To begin, the quantity $\varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ (defined in Eq. (136)) can be summed

eq:psum_yf_over_ye over ϵ to obtain:

$$(146) \quad \sum_{\epsilon} \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) = \sum_{\epsilon} \text{Tr}[P_{\alpha\phi}^{\epsilon}(\Phi_{\alpha}(X) \otimes \Phi_{\phi}^{\beta\delta}(Y, Z))] = \text{Tr}[\Phi_{\alpha}(X) \otimes \Phi_{\phi}^{\beta\delta}(Y, Z)] = \varphi_{\alpha}(X) \varphi_{\phi}^{\beta\delta}(Y, Z).$$

Further summing over β and δ and applying Proposition 34 produces

$$(147) \quad \sum_{\epsilon\beta\delta} \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) = \varphi_{\alpha}(X) \sum_{\beta\delta} \varphi_{\phi}^{\beta\delta}(Y, Z) = \varphi_{\alpha}(X) \varphi_{\phi}(Y+Z).$$

The other summation identities are analogous. Now using the non-negativity of $\varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$, we obtain the first claimed inequality in Eq. (144). The second claimed inequality in Eq. (144) follows analogously from summing $\varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z)$ over ϵ, α and β . The remaining claimed inequalities in Eq. (145)

are the result of similar summation identities. For example, summing the quantity $\varphi_{\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ over ϕ and applying Eq. (25) produces

{eq:psum_yf_over_yf}
(148)

$$\begin{aligned} \sum_{\phi} \varphi_{\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) &= \sum_{\phi} \text{Tr}[P_{\alpha\phi}^{\epsilon}(\Phi_{\alpha}(X) \otimes \Phi_{\phi}^{\beta\delta}(Y, Z))], \\ &= \sum_{\phi} \text{Tr}[P_{\alpha\beta\delta}^{\epsilon}(\text{id}_{\alpha} \otimes P_{\beta\delta}^{\phi})(\Phi_{\alpha}(X) \otimes \Phi_{\beta}(Y) \otimes \Phi_{\delta}(Z))], \\ &= \text{Tr}[P_{\alpha\beta\delta}^{\epsilon}(\Phi_{\alpha}(X) \otimes \Phi_{\beta}(Y) \otimes \Phi_{\delta}(Z))], \\ &= \varphi_{\epsilon}^{\alpha\beta\delta}(X, Y, Z). \end{aligned}$$

Therefore, from the non-negativity of $\varphi_{\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$, we obtain $\varphi_{\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \leq \varphi_{\epsilon}^{\alpha\beta\delta}(X, Y, Z)$ which establishes one half of the lower bound on $\varphi_{\epsilon}^{\alpha\beta\delta}(X, Y, Z)$ in Eq. (145); the other half of the lower bound follows analogously from summing $\varphi_{\gamma\epsilon}^{\alpha\beta\delta}(X, Y, Z)$ over γ . Finally, for the upper bounds on $\varphi_{\epsilon}^{\alpha\beta\delta}(X, Y, Z)$ in Eq. (145), we remark that Proposition 34 can be extended to handle more than two matrices by repeated application of Proposition 34; specifically, by starting with Eq. (146) and further summing over ϕ and applying Eq. (148) and Proposition 34, we obtain

$$\begin{aligned} \sum_{\epsilon} \varphi_{\epsilon}^{\alpha\beta\delta}(X, Y, Z) &= \sum_{\epsilon\phi} \varphi_{\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z), \\ (149) \quad &= \sum_{\phi} \varphi_{\alpha}(X) \varphi_{\phi}^{\beta\delta}(Y, Z), \\ &= \varphi_{\alpha}(X) \varphi_{\beta}(Y) \varphi_{\delta}(Z), \end{aligned}$$

which is three matrix analogue of one half of Proposition 34 and establishes the bound $\varphi_{\epsilon}^{\alpha\beta\delta}(X, Y, Z) \leq \varphi_{\alpha}(X) \varphi_{\beta}(Y) \varphi_{\delta}(Z)$ because $\varphi_{\epsilon}^{\alpha\beta\delta}(X, Y, Z) \geq 0$. For the other half of the three matrix analogue of Proposition 34, we can again use Eq. (148) and apply Proposition 34 twice to obtain

$$\begin{aligned} \sum_{\alpha\beta\delta} \varphi_{\epsilon}^{\alpha\beta\delta}(X, Y, Z) &= \sum_{\alpha\beta\delta\phi} \varphi_{\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z), \\ &= \sum_{\alpha\beta\delta\phi} \text{Tr}[P_{\alpha\phi}^{\epsilon}(\Phi_{\alpha}(X) \otimes \Phi_{\phi}^{\beta\delta}(Y, Z))], \\ &= \sum_{\alpha\phi} \text{Tr}[P_{\alpha\phi}^{\epsilon}(\Phi_{\alpha}(X) \otimes \sum_{\beta\delta} \Phi_{\phi}^{\beta\delta}(Y, Z))], \\ (150) \quad &= \sum_{\alpha\phi} \text{Tr}[P_{\alpha\phi}^{\epsilon}(\Phi_{\alpha}(X) \otimes \Phi_{\phi}(Y + Z))], \\ &= \sum_{\alpha\phi} \varphi_{\epsilon}^{\alpha\phi}(X, Y + Z), \\ &= \varphi_{\epsilon}(X + Y + Z), \end{aligned}$$

which in turn establishes the remaining inequality $\varphi_\epsilon^{\alpha\beta\delta}(X, Y, Z) \leq \varphi_\epsilon(X + Y + Z)$. $\square$

We can now immediately prove the “only if” portion of the claim of ??
cor: tet_eigenvalue_cubic

Corollary 22. *Let $X, Y, Z \geq 0$ be positive semidefinite $n \times n$ matrices. Then for all triples of Young diagrams (α, β, δ) ,*

$$(151) \quad \varphi_\alpha(X)^{\frac{2}{3}} \varphi_\beta(Y) \varphi_\delta(Z)^{\frac{2}{3}} \leq \sum_{\gamma, \epsilon, \phi} \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty \varphi_\phi(Y+Z)^{\frac{1}{3}} \varphi_\gamma(X+Y)^{\frac{1}{3}} \varphi_\epsilon(X+Y+Z)^{\frac{1}{3}}.$$

Proof. The claim follows by summing the inequality in Lemma 20 over triples (γ, ϵ, ϕ) and relaxing both the sides of the resulting inequality. For the lower bound, we have $\varphi_\alpha(X) \varphi_\beta(Y) \varphi_\delta(Z) = \sum_{\gamma \in \phi} \varphi_{\gamma \in \phi}^{\alpha\beta\delta}(X, Y, Z)$ which implies

$$(152) \quad \varphi_\alpha(X) \varphi_\beta(Y) \varphi_\delta(Z) \leq \sum_{\gamma \in \phi} \left| \varphi_{\gamma \in \phi}^{\alpha\beta\delta}(X, Y, Z) \right|.$$

Then for the upper bound, Proposition 21 implies

$$(153) \quad \varphi_{-\epsilon \phi}^{\alpha\beta\delta}(X, Y, Z) \varphi_{\gamma \in -}^{\alpha\beta\delta}(X, Y, Z) \leq \min\{\varphi_\alpha(X) \varphi_\phi(Y+Z) \varphi_\gamma(X+Y) \varphi_\delta(Z), \varphi_\epsilon(X+Y+Z)^2, \varphi_\alpha(X)^2 \varphi_\beta(Y)^2 \varphi_\delta(Z)^2\}.$$

Here we elect to remove the minimization by relaxing this inequality to the following:

$$(154) \quad \varphi_{-\epsilon \phi}^{\alpha\beta\delta}(X, Y, Z) \varphi_{\gamma \in -}^{\alpha\beta\delta}(X, Y, Z) \leq [\varphi_\alpha(X) \varphi_\phi(Y+Z) \varphi_\gamma(X+Y) \varphi_\delta(Z) \varphi_\epsilon(X+Y+Z)]^{\frac{2}{3}}.$$

Then the claimed result is obtained by applying Lemma 20 and then dividing by a common factor of $(\varphi_\alpha(X) \varphi_\delta(Z))^{\frac{1}{3}}$ (note that whenever $\varphi_\alpha(X) \varphi_\delta(Z) = 0$, the resulting inequality holds trivially). $\square$

The proof of the “if” portion of ?? can be found in Section 6.1 which proves the distance bound of Theorem 15 (this implies the “if” portion of ?? after sending $k \rightarrow \infty$).

TC: THE FOLLOWING SECTION IS CURRENTLY BEING RE-REWRITTEN INTO PREVIOUS SECTIONS

6. [TC] DISTANCE BOUNDS VIA EIGENVALUE CONCENTRATION

6.1. [tc] **Distance bound for the tetrahedral Horn problem.** The purpose of this section is to prove our main distance bound, Theorem 15. Note that Theorem 15 then immediately implies the “if” portion of the claim of ?? by taking $k \rightarrow \infty$. The “only if” portion of ?? was proven in Corollary 22.

Proof of Theorem 15. In the first stage, we fix an arbitrary triple (α, β, δ) with total size $|\alpha| + |\beta| + |\delta| = k$ and derive multiple bounds on the product $\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d)$. Then in the second stage, we sum over all such triples (α, β, δ) and obtain the bound on $D \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}$.

We begin by bounding $\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d)$ in two separate ways and then afterward combine them to apply the eigenvalue separation lemma (specifically Eq. (82)). The first bound on $\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d)$ is obtained from relaxing the assumption of the claim that (a, b, c, d, e, f) satisfies ?? for all triples of Young diagrams (α, β, δ) . By multiplying both sides of ?? by

(eq:abd_multiplied)
(155)

$$\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) \leq \sum_{\gamma, \epsilon, \phi} \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty (\varphi_\alpha(a)\varphi_\phi(f)\varphi_\gamma(c)\varphi_\delta(d)\varphi_\epsilon(e))^{\frac{1}{3}}.$$

Now let $(\gamma^*, \epsilon^*, \phi^*)$ denote a triple of Young diagrams which maximizes the summand above. Note that the summand is only non-zero if the size constraints $|\alpha| + |\beta| + |\delta| = |\alpha| + |\phi| = |\gamma| + |\delta| = |\epsilon| = k$ are satisfied. As there are at most $(k+1)^{3(n-1)}$ such triples satisfying the size constraints (remember α, β, δ are fixed), we conclude that

(eq:first_abd_bound)
(156)

$$\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) \leq (k+1)^{3(n-1)} \left\| \begin{Bmatrix} \alpha & \beta & \gamma^* \\ \delta & \epsilon^* & \phi^* \end{Bmatrix} \right\|_\infty (\varphi_\alpha(a)\varphi_{\phi^*}(f)\varphi_{\gamma^*}(c)\varphi_\delta(d)\varphi_{\epsilon^*}(e))^{\frac{1}{3}}.$$

The second bound is obtained by considering ?? which states that for any triple (γ, ϵ, ϕ) we have

(157)

$$\left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_2^2 \varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) = \frac{\dim(V_\alpha^n \otimes V_\beta^n \otimes V_\delta^n)}{\dim(V_\epsilon^n)} \mathbb{E}_{(X,Y,Z) \sim \mathcal{O}_x \times \mathcal{O}_y \times \mathcal{O}_z} \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z),$$

To obtain our desired bound on $\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d)$ we make a series of relaxations. First we bound the dimensions of the representations by $\frac{\dim(V_\alpha^n \otimes V_\beta^n \otimes V_\delta^n)}{\dim(V_\epsilon^n)} \leq \dim(V_\alpha^n \otimes V_\beta^n \otimes V_\delta^n) \leq (k+1)^{3\binom{n}{2}}$. Then we upper bound the average of $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ over $(X, Y, Z) \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d$ with the maximum of the

`{eq:max_abs_value_bound}` absolute value. Together this gives

$$(158) \quad \left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_2^2 \varphi_\alpha(a) \varphi_\beta(b) \varphi_\delta(d) \leq (k+1)^{3\binom{n}{2}} \max_{(X,Y,Z) \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d} \left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right|,$$

Furthermore, applying Lemma 20 and Proposition 21 to the absolute value

$$(159) \quad \left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right| \text{ on the right-hand side above produces}$$

$$\left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right| \leq \left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_\infty (\varphi_\alpha(X) \varphi_\phi(Y+Z) \varphi_\gamma(X+Y) \varphi_\delta(Z) \varphi_\epsilon(X+Y+Z))^{\frac{1}{3}}.$$

Applying Eq. (159) to Eq. (158) gives

$$(160) \quad \left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_2^2 \varphi_\alpha(a) \varphi_\beta(b) \varphi_\delta(d) \leq (k+1)^{3\binom{n}{2}} \left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_\infty \times \dots$$

$$\dots \times \max_{(X,Y,Z) \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d} (\varphi_\alpha(X) \varphi_\phi(Y+Z) \varphi_\gamma(X+Y) \varphi_\delta(Z) \varphi_\epsilon(X+Y+Z))^{\frac{1}{3}},$$

Now we choose (γ, ϵ, f) in the above bound to be equal to the optimal choice from $(\gamma^*, \epsilon^*, \phi^*)$ from Eq. (156) and multiply the result with Eq. (156) and then take a square root to obtain

$$(161) \quad \varphi_\alpha(a) \varphi_\beta(b) \varphi_\delta(d) \leq (k+1)^{\frac{3}{2}\binom{n}{2} + \frac{3}{2}(n-1)} \times \dots$$

$$\dots \times \max_{(X,Y,Z) \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d} (\varphi_\alpha(a) \varphi_\alpha(X) \varphi_{\phi^*}(f) \varphi_{\phi^*}(Y+Z) \varphi_{\gamma^*}(c) \varphi_{\gamma^*}(X+Y) \varphi_\delta(d) \varphi_\delta(Z) \varphi_{\epsilon^*}(e) \varphi_{\epsilon^*}(X+Y+Z))^{\frac{1}{3}}.$$

Note we have eliminated the $6j$ symbols by using the bound $\left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_\infty^2 \leq$

$\left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_2^2$. We are now in a position to apply the eigenvalue separation lemma (specifically Eq. (82)). For each Young diagram $\lambda \in \{\alpha, \delta, \gamma, \epsilon, \phi\}$ we apply Eq. (82) (noting $|\lambda| \leq k$ for each) to conclude

$$(162) \quad \sqrt{\varphi_\alpha(a) \varphi_\alpha(X)} \leq (k+1)^{\binom{n}{2}} \frac{\text{BC}(a, \text{eig}(X))^{|a|}}{|\alpha|!}$$

$$(163) \quad \sqrt{\varphi_\delta(d) \varphi_\delta(Z)} \leq (k+1)^{\binom{n}{2}} \frac{\text{BC}(d, \text{eig}(Z))^{|d|}}{|\delta|!}$$

$$(164) \quad \sqrt{\varphi_\gamma(c) \varphi_\gamma(X+Y)} \leq (k+1)^{\binom{n}{2}} \frac{\text{BC}(c, \text{eig}(X+Y))^{|c|}}{|\gamma|!},$$

$$(165) \quad \sqrt{\varphi_\phi(f) \varphi_\phi(Y+Z)} \leq (k+1)^{\binom{n}{2}} \frac{\text{BC}(f, \text{eig}(Y+Z))^{|f|}}{|\phi|!},$$

$$(166) \quad \sqrt{\varphi_\epsilon(e) \varphi_\epsilon(X+Y+Z)} \leq (k+1)^{\binom{n}{2}} \frac{\text{BC}(e, \text{eig}(X+Y+Z))^{|e|}}{|\epsilon|!}.$$

At this stage, we can combine the bounds for α and ϕ to obtain an expression involving the same power of k :

(167)

$$\sqrt{\varphi_\alpha(a)\varphi_\alpha(X)}\sqrt{\varphi_\phi(f)\varphi_\phi(Y+Z)} \leq (k+1)^{n(n-1)} \frac{\text{BC}(a, \text{eig}(X))^{| \alpha |}}{| \alpha |!} \frac{\text{BC}(f, \text{eig}(Y+Z))^{| \phi |}}{| \phi |!},$$

(168)

$$\leq (k+1)^{n(n-1)} \frac{[\text{BC}(a, \text{eig}(X)) + \text{BC}(f, \text{eig}(Y+Z))]^k}{k!},$$

(169)

$$= (k+1)^{n(n-1)} \frac{\text{BC}(a \oplus f, \text{eig}(X \oplus Y + Z))^k}{k!},$$

because $| \alpha | + | \phi | = | \epsilon | = k$. A similar bound also follows from combining γ with δ . Define the maximum product Bhattacharyya coefficient as

(170)

$$\text{BC} \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} := \max_{\substack{(X,Y,Z) \in \\ \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d}} \text{BC}(a \oplus f, \text{eig}(X \oplus Y + Z)) \text{BC}(c \oplus d, \text{eig}(X + Y \oplus Z)) \text{BC}(e, \text{eig}(X + Y + Z)).$$

From Eq. (161) and the above Bhattacharyya coefficient bounds we obtain

$$(171) \quad \varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) \leq (k+1)^{\frac{3}{2}\binom{n}{2} + \frac{3}{2}(n-1) + \frac{3}{3}n(n-1)} \frac{1}{k!} \text{BC} \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}^{\frac{k}{3}}$$

Now that we have established an upper bound on $\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d)$ which is uniform in the triple (α, β, δ) , the next step is to sum this expression over all triples (α, β, δ) such that $| \alpha | + | \beta | + | \delta | = k$. Note that summing $\varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d)$ over (α, β, δ) with total size $| \alpha | + | \beta | + | \delta | = k$ yields the analogue of Eq. (233):

$$(172) \quad \sum_{\substack{\alpha, \beta, \delta \\ | \alpha | + | \beta | + | \delta | = k}} \varphi_\alpha(a)\varphi_\beta(b)\varphi_\delta(d) = \frac{(\text{Tr}[a] + \text{Tr}[b] + \text{Tr}[d])^k}{k!} = \frac{\text{Tr}[e]^k}{k!}$$

where we used that the trace conditions hold, i.e., $\text{Tr}[e] = \text{Tr}[a] + \text{Tr}[b] + \text{Tr}[d]$.

As there are at most $(k+1)^{3n}$ triples Young diagrams (α, β, δ) with total size $| \alpha | + | \beta | + | \delta | = k$ being summed here, we finally conclude

$$(173) \quad \frac{\text{Tr}[e]^k}{k!} \leq (k+1)^{\frac{3}{2}\binom{n}{2} + \frac{3}{2}(n-1) + \frac{3}{3}n(n-1) + 3n} \frac{1}{k!} \text{BC} \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}^{\frac{k}{3}}$$

or equivalently

$$(174) \quad \frac{1}{\text{Tr}[e]} \text{BC} \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}^{\frac{1}{3}} \geq \left(\frac{1}{(k+1)^{\frac{7n^2+11n+6}{4}}} \right)^{\frac{1}{k}}.$$

Finally, using ??, we can substitute the maximum normalized product of Bhattacharyya coefficients here with minimum normalized sum of distances (as in Eq. (130)) to obtain Eq. (129) which completes the proof. $\square$

6.2. [tc] Asymptotics of the $6j$ symbols. In order to prove Corollary 17 is will be useful to have a bound on the quantity $\left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right|$ which directly relates the sextuple of Young diagrams $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ to the eigenvalues of $(X, Y, X + Y, Z, X + Y + Z, Y + Z)$.

lem:asym_sixj_helper

Lemma 23. *To begin let $X, Y, Z \geq 0$ be positive semidefinite $n \times n$ matrices and let (a, b, c, d, e, f) be the eigenvalues of the matrices $(X, Y, X + Y, Z, X + Y + Z, Y + Z)$. Then for all sextuples of Young diagrams, $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$, with sizes satisfying*

{eq:y_d_k_sizes}

$$(175) \quad k = |\epsilon| = |\alpha| + |\beta| + |\delta| = |\alpha| + |\phi| = |\gamma| + |\delta|,$$

we have (assuming $\text{Tr}[e] \neq 0$) the following bounds

{eq:hex_6j_bound}

$$(176) \quad \frac{k!}{\text{Tr}[e]^k} \left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right| \leq \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_{\infty},$$

{eq:hex_rel_ent_bound}

$$(177) \quad \frac{k!}{\text{Tr}[e]^k} \left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right| \leq (k+1)^{n(n-1)} \exp \left(-\frac{k}{4} R_{abc\delta\epsilon\phi}^{\alpha\beta\gamma\delta\epsilon\phi} \right),$$

where

(178)

$$R_{abc\delta\epsilon\phi}^{\alpha\beta\gamma\delta\epsilon\phi} = \frac{1}{4} \left[D \left(\frac{\alpha \oplus \beta \oplus \delta}{k} \left\| \frac{a \oplus b \oplus d}{\text{Tr}[e]} \right\| \right) + D \left(\frac{\gamma \oplus \delta}{k} \left\| \frac{c \oplus d}{\text{Tr}[e]} \right\| \right) + D \left(\frac{\alpha \oplus \phi}{k} \left\| \frac{a \oplus f}{\text{Tr}[e]} \right\| \right) + D \left(\frac{\epsilon}{k} \left\| \frac{e}{\text{Tr}[e]} \right\| \right) \right].$$

{eq:abs_quad_bound}

Proof. A combination of Lemma 20 and Proposition 21 implies the bound

(179)

$$\left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right| \leq \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_{\infty} (\varphi_{\alpha}(a) \varphi_{\beta}(b) \varphi_{\delta}(d) \times \varphi_{\gamma}(c) \varphi_{\delta}(d) \times \varphi_{\epsilon}(e) \times \varphi_{\alpha}(a) \varphi_{\phi}(f))^{\frac{1}{4}}.$$

To prove Eq. (176), note that each of $\varphi_{\alpha}(a) \varphi_{\beta}(b) \varphi_{\delta}(d)$, $\varphi_{\gamma}(c) \varphi_{\delta}(d)$, $\varphi_{\epsilon}(e)$, and $\varphi_{\alpha}(a) \varphi_{\phi}(f)$ is upper bounded by $\frac{\text{Tr}[e]^k}{k!}$ because they each sum to $\frac{\text{Tr}[e]^k}{k!}$ when summing over the corresponding sets of Young diagrams with sizes fixed by Eq. (175) (see also Eq. (233)). Applying these upper bounds to Eq. (179) (and multiplying by $\frac{k!}{\text{Tr}[e]^k}$) gives Eq. (176).

For the proof of Eq. (177), we use the multinomial distribution bound from Proposition 8 to obtain

(180)

$$\varphi_{\alpha}(a) \varphi_{\beta}(b) \varphi_{\delta}(d) \leq (k+1)^{3\binom{n}{2}} \frac{a_1^{\alpha_1}}{\alpha_1!} \dots \frac{a_n^{\alpha_n}}{\alpha_n!} \frac{b_1^{\beta_1}}{\beta_1!} \dots \frac{b_n^{\beta_n}}{\beta_n!} \frac{d_1^{\delta_1}}{\delta_1!} \dots \frac{d_n^{\delta_n}}{\delta_n!} = (k+1)^{3\binom{n}{2}} \frac{\text{Tr}[e]^k}{k!} m_{\alpha \oplus \beta \oplus \delta}(a \oplus b \oplus d).$$

Then applying the relative entropy bound from the proof of Proposition 10 yields

(181)

$$\frac{k!}{\text{Tr}[e]^k} \varphi_{\alpha}(a) \varphi_{\beta}(b) \varphi_{\delta}(d) \leq (k+1)^{3\binom{n}{2}} \exp \left[-k D \left(\frac{\alpha \oplus \beta \oplus \delta}{k} \left\| \frac{a \oplus b \oplus d}{\text{Tr}[e]} \right\| \right) \right].$$

Analogously, we obtain

$$(182) \quad \frac{k!}{\text{Tr}[e]^k} \varphi_\gamma(c) \varphi_\delta(d) \leq (k+1)^{2\binom{n}{2}} \exp \left[-kD \left(\frac{\gamma \oplus \delta}{k} \left\| \frac{c \oplus d}{\text{Tr}[e]} \right\| \right) \right],$$

$$(183) \quad \frac{k!}{\text{Tr}[e]^k} \varphi_\alpha(a) \varphi_\phi(f) \leq (k+1)^{2\binom{n}{2}} \exp \left[-kD \left(\frac{\alpha \oplus \phi}{k} \left\| \frac{a \oplus f}{\text{Tr}[e]} \right\| \right) \right],$$

$$(184) \quad \frac{k!}{\text{Tr}[e]^k} \varphi_\epsilon(e) \leq (k+1)^{\binom{n}{2}} \exp \left[-kD \left(\frac{\epsilon}{k} \left\| \frac{e}{\text{Tr}[e]} \right\| \right) \right].$$

Applying these bounds to Eq. (179) and using $\left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty \leq 1$, we obtain Eq. (177) $\square$

Proof of Corollary 17. First we prove the claimed inverse polynomial lower bound, and then afterwards prove the exponential decay.

Assume (a, b, c, d, e, f) are the eigenvalues of positive semidefinite matrices (A, B, C, D, E, F) satisfying the tetrahedral constraints. Summing $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(A, B, D)$ over all sextuples of Young diagrams satisfying Eq. (175) produces

$$(185) \quad \sum_{\substack{\alpha, \beta, \delta \\ \gamma, \epsilon, \phi \\ |\epsilon|=k}} \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(A, B, D) = \frac{\text{Tr}[e]^k}{k!}.$$

Therefore, the sextuple which maximizes the absolute value of $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(A, B, D)$ over this k -dependent domain of summation, which we denote by $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$, must satisfy

$$(186) \quad \frac{k!}{\text{Tr}[e]^k} \left| \varphi_{\gamma_k \epsilon_k \phi_k}^{\alpha_k \beta_k \delta_k}(A, B, D) \right| \geq \frac{1}{(k+1)^{6 \text{rank}[e]}}$$

where $(k+1)^{6 \text{rank}[e]}$ is a crude upper bound on the number of terms in the above summation which are non-zero as there are at most $(k+1)^r$ Young diagrams with size at most k and depth at most r (note that each of $\varphi_\alpha(a), \varphi_\beta(b), \varphi_\delta(d), \varphi_\phi(f), \varphi_\gamma(c)$ and $\varphi_\epsilon(e)$ are respectively only non-zero if the depth of the corresponding Young diagram is at most the rank of the corresponding set of eigenvalues and $\text{rank}[e]$ is always the maximum possible rank). Therefore, applying Eq. (176) in Lemma 23 gives

`{eq:n_independent_inverse_poly}`

$$(187) \quad \left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_\infty \geq \frac{1}{(k+1)^{6 \text{rank}[e]}},$$

while Eq. (177) in Lemma 23 gives

$$(188) \quad \exp \left(-k R_{abcdef}^{\alpha_k \beta_k \gamma_k \delta_k \epsilon_k \phi_k} \right) \geq \frac{1}{(k+1)^{n(n-1)}} \frac{1}{(k+1)^{6 \text{rank}[e]}}.$$

Thus we have demonstrated the existence of a sequence of sextuples of Young diagrams for which (i) $\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_\infty$ satisfies the claimed inverse polynomial lower bound, and also (ii) $R_{abcdef}^{\alpha_k \beta_k \gamma_k \delta_k \epsilon_k \phi_k} \rightarrow 0$ which implies the convergence claim:

$$(189) \quad \frac{\alpha_k}{k} \rightarrow \frac{a}{\text{Tr}[e]}, \quad \frac{\beta_k}{k} \rightarrow \frac{b}{\text{Tr}[e]}, \quad \frac{\gamma_k}{k} \rightarrow \frac{c}{\text{Tr}[e]}, \quad \frac{\delta_k}{k} \rightarrow \frac{d}{\text{Tr}[e]}, \quad \frac{\epsilon_k}{k} \rightarrow \frac{e}{\text{Tr}[e]}, \quad \frac{\phi_k}{k} \rightarrow \frac{f}{\text{Tr}[e]}.$$

We now prove the exponential decay on $\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_\infty$ when $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$ converges to some non-tetrahedral eigenvalues. It will be slightly easier to prove the exponential decay of $\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_2$ which is actually a bit stronger than necessary. Without loss of generality, we can assume the Young diagrams $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$ satisfy the size constraints Eq. (175) as before (since otherwise $\begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} = 0$). Then from ??, we conclude that for *any* eigenvalues (a, b, d) ,

$$(190) \quad \left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_2^2 \varphi_{\alpha_k}(a) \varphi_{\beta_k}(b) \varphi_{\delta_k}(d) = \frac{\dim(V_{\alpha_k}^n \otimes V_{\beta_k}^n \otimes V_{\delta_k}^n)}{\dim(V_{\epsilon_k}^n)} \mathbb{E}_{(X,Y,Z) \sim \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d} \varphi_{\gamma_k \epsilon_k \phi_k}^{\alpha_k \beta_k \delta_k}(X, Y, Z).$$

As in the proof of Theorem 15 we first we bound the dimensions of the representations by $\frac{\dim(V_{\alpha_k}^n \otimes V_{\beta_k}^n \otimes V_{\delta_k}^n)}{\dim(V_{\epsilon_k}^n)} \leq \dim(V_{\alpha_k}^n \otimes V_{\beta_k}^n \otimes V_{\delta_k}^n) \leq (k+1)^{3\binom{n}{2}}$.

Then we upper bound the average of $\varphi_{\gamma_k \epsilon_k \phi_k}^{\alpha_k \beta_k \delta_k}(X, Y, Z)$ with the maximum of the absolute value. Together this gives

$$(191) \quad \left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_2^2 \varphi_{\alpha_k}(a) \varphi_{\beta_k}(b) \varphi_{\delta_k}(d) \leq (k+1)^{3\binom{n}{2}} \max_{(X,Y,Z) \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d} \left| \varphi_{\gamma_k \epsilon_k \phi_k}^{\alpha_k \beta_k \delta_k}(X, Y, Z) \right|.$$

As this bound holds for all eigenvalues (a, b, d) , we can freely choose, for each k , a particular triple of eigenvalues (a^*, b^*, d^*) which ensures

$$(192) \quad \frac{k!}{\text{Tr}[e^*]^k} \varphi_{\alpha_k}(a^*) \varphi_{\beta_k}(b^*) \varphi_{\delta_k}(d^*) \geq \frac{1}{(k+1)^{3n}}.$$

where we are denoting the sum $\text{Tr}[a^*] + \text{Tr}[b^*] + \text{Tr}[d^*]$ with $\text{Tr}[e^*]$ by a slight abuse of notation as no eigenvalues e^* have been chosen here. That such a choice can always be made follows from the normalization condition

$$(193) \quad \sum_{\substack{\alpha, \beta, \delta \\ |\alpha| + |\beta| + |\delta| = k}} \varphi_\alpha(a) \varphi_\beta(b) \varphi_\delta(d) = \frac{(\text{Tr}[a] + \text{Tr}[b] + \text{Tr}[d])^k}{k!},$$

and noting that there are at most $(k+1)^{3n}$ such triples (α, β, δ) of Young diagrams in the sum (indeed it suffices to choose (a^*, b^*, d^*) to maximize

$\varphi_{\alpha_k}(a)\varphi_{\beta_k}(b)\varphi_{\delta_k}(d)$). After this choice is made, we obtain the upper bound (194)

$$\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_2^2 \leq (k+1)^{3\binom{n}{2}+3n} \frac{\text{Tr}[e^*]^k}{k!} \max_{\substack{(X,Y,Z) \in \\ \mathcal{O}_{a^*} \times \mathcal{O}_{b^*} \times \mathcal{O}_{d^*}}} \left| \varphi_{\gamma_k \epsilon_k \phi_k}^{\alpha_k \beta_k \delta_k}(X, Y, Z) \right|.$$

Applying Eq. (177) from Lemma 23 here produces (195)

$$\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_2^2 \leq (k+1)^{3\binom{n}{2}+3n+n(n-1)} \exp \left(-k \min_{\substack{(X,Y,Z) \in \\ \mathcal{O}_{a^*} \times \mathcal{O}_{b^*} \times \mathcal{O}_{d^*}}} R_{a^* b^* c^* d^* e^* f^*}^{\alpha_k \beta_k \gamma_k \delta_k \epsilon_k \phi_k} \right).$$

where $c^* = \exp(X + Y)$, $f^* = \text{eig}(Y + Z)$, and $e^* = \text{eig}(X + Y + Z)$.

Finally, since it was assumed that $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$ are converging to non-tetrahedral eigenvalues, there must exist a sufficiently large K such that for all $k \geq K$ and all tetrahedral eigenvalues (a', b', c', d', e', f') , we have a separation $R_{a' b' c' d' e' f'}^{\alpha_k \beta_k \gamma_k \delta_k \epsilon_k \phi_k} \geq r$ for some positive constant $r > 0$. Since $(a^*, b^*, c^*, d^*, e^*, f^*)$ are tetrahedral eigenvalues, we can conclude for such sufficiently large $k \geq K$, $R_{a^* b^* c^* d^* e^* f^*}^{\alpha_k \beta_k \gamma_k \delta_k \epsilon_k \phi_k} \geq r$ and therefore

$$(196) \quad \left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_2^2 \leq (k+1)^{3\binom{n}{2}+3n+n(n-1)} \exp(-kr),$$

which proves the desired exponential decay. $\square$

APPENDIX A. SYMMETRIES OF TETRAHEDRAL EIGENVALUES sec:Symmetries

There are numerous types of symmetries present in the statement of the tetrahedral Horn problem. The purpose of this section is to describe the scaling, inversion, tetrahedral, and shifting symmetries of the problem. prop:scaling

Proposition 24 (Scaling). *Let (a, b, c, d, e, f) be a sextuple of candidate eigenvalues and let $s \in (0, \infty)$ be a positive real number. Then the scaled eigenvalues, (sa, sb, sc, sd, se, sf) , are tetrahedral if and only if the unscaled eigenvalues (a, b, c, d, e, f) are tetrahedral.*

Proof. The statement follows from the fact that $\text{eig}(sX) = s \text{eig}(X)$ holds for all Hermitian matrices X and positive scalars s and that each of the constraints in Eq. (120) are homogeneous in the six Hermitian matrices (A, B, C, D, E, F) , e.g., we have $sA + sB - sC = 0$ if and only if $A + B - C = 0$ because $s \neq 0$. $\square$

One may wonder whether it is possible to extend the scaling symmetry to negative values for s . Indeed, this can be accomplished, provided one accounts for the fact that negating a Hermitian matrix both negates and reverses the sorting order of its eigenvalues. prop:inversion

Proposition 25 (Inversion). *Let (a, b, c, d, e, f) be a sextuple of candidate eigenvalues. Given any list of non-increasing eigenvalues $x = (x_1 \geq \dots \geq$*

x_n), let x^* denote the inverted list of resorted and negated eigenvalues: $x^* = (-x_n \geq \dots \geq -x_1)$. Then (a, b, c, d, e, f) are tetrahedral if and only if $(a^*, b^*, c^*, d^*, e^*, f^*)$ are tetrahedral.

Proof. The statement holds because $\text{eig}(-X) = x^*$ whenever $\text{eig}(X) = x$ and moreover the sextuple of matrices (A, B, C, D, E, F) satisfies Eq. (120) if and only if the sextuple of negated matrices $(-A, -B, -C, -D, -E, -F)$ satisfies Eq. (120). $\square$

The inversion symmetry Proposition 25 is just one element belonging to a group of 48 symmetries consisting of inversions of eigenvalues, $x \rightarrow x^*$, or permutations of elements in the sextuple (a, b, c, d, e, f) . prop:tet-symmetry

Proposition 26 (Oriented). *The sextuple (a, b, c, d, e, f) of eigenvalues are tetrahedral if and only if the following sextuples of eigenvalues are tetrahedral:*

{eq:gens_of_tet_sym}

$$(197) \quad (a \ f \ e \ d^* \ c \ b) \quad (c \ e^* \ d^* \ f \ b \ a^*)$$

Moreover, these two operations generate a finite group of order 48 whose action, grouped by conjugacy classes, produces:

{eq:all_of_tet_sym}

$$(198) \quad \begin{array}{cccc} (a \ b \ c \ d \ e \ f) & (a^* \ c \ b \ d \ f \ e) & (a \ c^* \ b^* \ d^* \ f^* \ e^*) & (a^* \ b^* \ c^* \ d^* \ e^* \ f^*) \\ (d \ b \ f \ a \ e \ c) & (b^* \ a^* \ c^* \ e \ d \ f) & (b \ a \ c \ e^* \ d^* \ f^*) & (d^* \ b^* \ f^* \ a^* \ e^* \ c^*) \\ (d^* \ e \ c \ a^* \ b \ f) & (f^* \ b \ d^* \ c^* \ e^* \ a^*) & (f \ b^* \ d \ c \ e \ a) & (d \ e^* \ c^* \ a \ b^* \ f^*) \\ (a \ e^* \ f^* \ d \ b^* \ c^*) & (e \ d^* \ c \ b^* \ a \ f^*) & (e^* \ d \ c^* \ b \ a^* \ f) & (a^* \ e \ f \ d^* \ b \ c) \\ (b \ f^* \ d^* \ e \ c \ a) & (c \ b^* \ a \ f \ e \ d) & (c^* \ b \ a^* \ f^* \ e^* \ d^*) & (b^* \ f \ d \ e^* \ c^* \ a^*) \\ (c \ a^* \ b \ f^* \ d^* \ e^*) & (a \ f \ e \ d^* \ c \ b) & (a^* \ f^* \ e^* \ d \ c^* \ b^*) & (c^* \ a \ b^* \ f \ d \ e) \\ (e \ c^* \ d \ b \ f \ a^*) & & & (e^* \ c \ d^* \ b^* \ f^* \ a) \\ (f \ a \ e \ c^* \ d \ b^*) & (b \ d \ f \ e^* \ a^* \ c^*) & (b^* \ d^* \ f^* \ e \ a \ c) & (f^* \ a^* \ e^* \ c \ d^* \ b) \\ (f^* \ d \ b^* \ c \ a \ e) & (d \ f^* \ b^* \ a^* \ c^* \ e^*) & (d^* \ f \ b \ a \ c \ e) & (f \ d^* \ b \ c^* \ a^* \ e^*) \\ (e^* \ f \ a^* \ b^* \ c^* \ d) & (f \ e^* \ a^* \ c \ b \ d^*) & (f^* \ e \ a \ c^* \ b^* \ d) & (e \ f^* \ a \ b \ c \ d^*) \\ (c^* \ d^* \ e^* \ f \ a^* \ b) & (e^* \ a \ f^* \ b \ d^* \ c) & (e \ a^* \ f \ b^* \ d \ c^*) & (c \ d \ e \ f^* \ a \ b^*) \\ (b^* \ c \ a \ e^* \ f^* \ d^*) & (d^* \ c^* \ e^* \ a \ f^* \ b^*) & (d \ c \ e \ a^* \ f \ b) & (b \ c^* \ a^* \ e \ f \ d) \\ & (c^* \ e \ d \ f^* \ b^* \ a) & (c \ e^* \ d^* \ f \ b \ a^*) & \end{array}$$

Proof. To begin, notice that each of the operations on (a, b, c, d, e, f) appearing in Eq. (197) are invertible (indeed the first operation has order two, while the second operation has order four). The claim that these two operations generate a finite group of order 48 can be checked explicitly using computer software. Now suppose (a, b, c, d, e, f) are the eigenvalues of a sextuple of Hermitian matrices (A, B, C, D, E, F) satisfying Eq. (120). Then, for example, the sextuple $(A', B', C', D', E', F') = (A, F, E, -D, C, B)$ (which has the eigenvalues (a, f, e, d^*, c, b)) is also tetrahedral because

{eq:signed_perm_rows}

(199)

$$\begin{aligned} A' + B' - C' &= (A + F - E) = 0, \\ B' + D' - F' &= -(B + D - F) = 0, \\ C' + D' - E' &= -(C + D - E) = 0, \\ A' + F' - E' &= (A + B - C) = 0. \end{aligned}$$

Moreover, as the operation mapping (a, b, c, d, e, f) to (a^*, c, b, d, f, e) is invertible, one also has that if (a, b, c, d, e, f) is not tetrahedral, then neither is (a^*, c, b, d, f, e) . An analogous argument holds for the operation mapping (a, b, c, d, e, f) to (c, e^*, d^*, f, b, a^*) with (A', B', C', D', E', F') given by $(C, -E, -D, F, B, -A)$. $\square$

Remark 27. The group G of order 48 appearing in Proposition 26 corresponds to the subgroup of all possible signed permutations of 6 elements generated by permutations and negations of the elements of the sextuple (A, B, C, D, E, F) which preserves the set of linear constraints in Eq. (120). The group G is isomorphic to both the direct product $S_2 \times S_4$ and to the signed permutation group of three elements, i.e., $(S_2 \times S_2 \times S_2) \rtimes S_3$.

The first isomorphism, $G \simeq S_2 \times S_4$, can be understood as S_2 acting by the inversion of all eigenvalues,

$$(200) \quad (a, b, c, d, e, f) \leftrightarrow (a^*, b^*, c^*, d^*, e^*, f^*),$$

while S_4 acts by permutation of the *vertices* of the tetrahedron in Fig. 2: each vertex permutation induces a permutation of the six directed edges with the possibility of inverting the orientation of a subset of the edges (such edge orientation reversals can be accounted for a corresponding inversion of eigenvalues).

The second isomorphism, $G \simeq (S_2 \times S_2 \times S_2) \rtimes S_3$, can be understood by considering the triplet $(A, D), (B, E), (C, F)$ of non-adjacent pairs of edges in Fig. 2. Accordingly, it is helpful to arrange the sextuple (a, b, c, d, e, f) into a 2×3 grid using the following identification:

$$(201) \quad (a \ b \ c \ d \ e \ f) \cong \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}.$$

Then one can see that each S_2 factor in the direct product $S_2 \times S_2 \times S_2$ acts faithfully on the sextuple by both negating a certain subset of the columns and permuting the upper and lower rows leaving just one column unchanged.

$$(202) \quad \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \leftrightarrow \begin{pmatrix} a & e^* & f^* \\ d & b^* & c^* \end{pmatrix}, \quad \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \leftrightarrow \begin{pmatrix} d & b & f \\ a & e & c \end{pmatrix}, \quad \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \leftrightarrow \begin{pmatrix} d^* & e & c \\ a^* & b & f \end{pmatrix}.$$

The symmetric group S_3 of order $3! = 6$ acts instead by permuting the columns and negating a subset of the entries in the top row, producing a subgroup of order 6 of the form:

$$(203) \quad \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \quad \begin{pmatrix} b & c^* & a^* \\ e & f & d \end{pmatrix} \quad \begin{pmatrix} c^* & a & b^* \\ f & d & e \end{pmatrix} \\ \begin{pmatrix} a^* & c & b \\ d & f & e \end{pmatrix} \quad \begin{pmatrix} b^* & a^* & c^* \\ e & d & f \end{pmatrix} \quad \begin{pmatrix} c & b^* & a \\ f & e & d \end{pmatrix}$$

We further remark that these symmetries also corresponding to symmetries of the Wigner $6j$ operator $\left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\}$ in ?? with the involution $\lambda \mapsto \lambda^*$ corresponding taking the dual representation $V_\lambda^n \mapsto (V_\lambda^n)^*$. In the case where $n = 2$, the involution map becomes the identity map (as every irreducible representation of $SU(2)$ is self-dual, or the reversal of a pair $(x, -x)$

of traceless eigenvalues is trivial), and thus the symmetry group collapses to a group of order $|S_4| = 4! = 24$.

prop:shifting

Proposition 28 (Shifting symmetry). *Let (a, b, c, d, e, f) be a sextuple of candidate eigenvalues and let $x, y, z \in \mathbb{R}$ be fixed. Define shifted eigenvalues $(\tilde{a}, \tilde{b}, \tilde{c}, \tilde{d}, \tilde{e}, \tilde{f})$ by*

feq:shifted_tet_eigenvalues}

(204)

$$\begin{aligned} \tilde{a}_i &:= a_i + x, & \tilde{b}_i &:= b_i + y, & \tilde{c}_i &:= c_i + x + y, \\ \tilde{d}_i &:= d_i + z, & \tilde{e}_i &:= e_i + x + y + z, & \tilde{f}_i &:= f_i + y + z. \end{aligned}$$

Then shifted eigenvalues $(\tilde{a}, \tilde{b}, \tilde{c}, \tilde{d}, \tilde{e}, \tilde{f})$ are tetrahedral if and only if the unshifted eigenvalues (a, b, c, d, e, f) are tetrahedral.

Proof. Given a Hermitian matrix A with eigenvalues $\text{eig}(A) = a$ and a constant $w \in \mathbb{R}$, the eigenvalues of $A + wI$ (where I is the identity matrix) are

$$(205) \quad \text{eig}(A + wI) = (a_1 + w \geq \cdots \geq a_n + w).$$

The statement then follows because the sextuple of matrices $(\tilde{A}, \tilde{B}, \tilde{C}, \tilde{D}, \tilde{E}, \tilde{F})$, where

$$(206) \quad \begin{aligned} \tilde{A} &:= A + xI, & \tilde{B} &:= B + yI, & \tilde{C} &:= C + (x + y)I, \\ \tilde{D} &:= D + zI, & \tilde{E} &:= E + (x + y + z)I, & \tilde{F} &:= F + (y + z)I, \end{aligned}$$

have the shifted eigenvalues $(\tilde{a}, \tilde{b}, \tilde{c}, \tilde{d}, \tilde{e}, \tilde{f})$ and satisfy the tetrahedral constraints in Eq. (120) if and only if (A, B, C, D, E, F) satisfy the tetrahedral constraints in Eq. (120). $\square$

As a corollary of the shifting symmetries (Proposition 28) one concludes that to solve the tetrahedral Horn problem, in a certain sense, it suffices to solve the tetrahedral Horn problem merely for traceless matrices (Corollary 29) or, alternatively, for positive definite or positive semidefinite matrices (Corollary 30).

cor:traceless

Corollary 29 (Traceless eigenvalues). *Let (a, b, c, d, e, f) be a sextuple of candidate eigenvalues and for any list of n eigenvalues $x = (x_1 \geq \cdots \geq x_n)$, let $x^0 := x - \text{Tr}[x]/n$ denote the traceless list of eigenvalues obtained by shifting x by $-\text{Tr}[x]/n$. Then (a, b, c, d, e, f) are tetrahedral if and only if (i) their traces $(\text{Tr}[a], \text{Tr}[b], \text{Tr}[c], \text{Tr}[d], \text{Tr}[e], \text{Tr}[f])$ are tetrahedral (they satisfy Eq. (127)) and (ii) $(a^0, b^0, c^0, d^0, e^0, f^0)$ are tetrahedral.*

Proof. This “only if” portion of the statement follows from both the tetrahedral trace conditions (Eq. (127)) and applications of the shifting symmetry Proposition 28 with (x, y, z) values

$$(207) \quad x = -n^{-1} \text{Tr}(A), \quad y = -n^{-1} \text{Tr}(B) \quad z = -n^{-1} \text{Tr}(D).$$

The “if” portion of the statement follows from the shifting symmetry applied to the negated (x, y, z) values. $\square$

The next corollary of the shifting symmetry proves that to solve the tetrahedral Horn problem it suffices to solve the tetrahedral Horn problem for non-negative (or positive) eigenvalues only. Indeed, one can simply take the shifting parameters (x, y, z) appearing in Proposition 28 to be sufficiently large. The following corollary establishes, in a certain sense, the minimum shifting required to ensure non-negative (or positive) eigenvalues for all six Hermitian matrices. cor:psd

Corollary 30 (Non-negativity). *If (a, b, c, d, e, f) are a tetrahedral sextuple of eigenvalues, then necessarily the smallest eigenvalues satisfy*

$$(208) \quad c_n \geq a_n + b_n, \quad f_n \geq b_n + d_n, \quad e_n \geq d_n + c_n, \quad e_n \geq a_n + f_n.$$

Moreover, if (a, b, c, d, e, f) are a sextuple of candidate eigenvalues satisfying Eq. (208), then for any $(x, y, z) \in \mathbb{R}^3$ satisfying

$$(209) \quad x \geq -a_n, \quad y \geq -b_n, \quad z \geq -d_n,$$

the resulting shifted eigenvalues $(\tilde{a}, \tilde{b}, \tilde{c}, \tilde{d}, \tilde{e}, \tilde{f})$ (given by Eq. (204)) are all non-negative.

Proof. That tetrahedral eigenvalues satisfy Eq. (208) follows from four applications of the well-known inequality $c_n \geq a_n + b_n$ for the original Horn problem where $C = A + B$ [Bha01]. As a reminder, the smallest eigenvalue x_n of a matrix X is characterized by a minimization: $x_n = \min_{\|v\|=1} \langle v, Xv \rangle$ which produces

$$(210) \quad c_n = \min_{\|v\|=1} \langle v, Cv \rangle = \min_{\|v\|=1} (\langle v, Av \rangle + \langle v, Bv \rangle) \geq \min_{\|v\|=1} \langle v, Av \rangle + \min_{\|w\|=1} \langle w, Bw \rangle = a_n + b_n.$$

Four applications of this inequality produces Eq. (208). Now if (x, y, z) satisfy Eq. (209), we have for all non-negativity of the shifted eigenvalues, e.g.,

$$(211) \quad \forall j : \quad \tilde{a}_j = a_j + x \geq a_n + x \geq 0,$$

Similar statements hold for b, c, d, e and f . □

As a consequence of Corollary 30, we will, without loss of generality, henceforth only consider candidate eigenvalues (a, b, c, d, e, f) which are non-negative in the sense that they are associated to positive semidefinite Hermitian matrices.

Remark 31. Although this section listed a handful of symmetries of the tetrahedral Horn problem, there remains the possibility of additional symmetries which are unknown at this time. For example, in the case of 2×2 Hermitian matrices, an additional class of symmetries on the set of edge-lengths (a, b, c, e, d, f) of tetrahedra, known as the *Regge symmetries*, hold [AI19; Boa07; PR68, Theorem 1.1]. If there exists a tetrahedron with edge lengths (a, b, c, d, e, f) , then there also exists a tetrahedron with edge lengths

$$(212) \quad (s_1 - a, s_1 - b, c, s_1 - d, s_1 - e, f) \quad \text{where} \quad s_1 = \frac{a + b + d + e}{2}.$$

Combining this symmetry with the permutational symmetries given by Proposition 26 (noting eigenvalues of traceless 2×2 Hermitian matrices are invariant under negation) one concludes the co-existence of a family of six tetrahedra with edge lengths given by [VMK88, Chapter 9]:

(213)

$$\begin{aligned} (a, b, c, d, e, f) & \quad (s_1 - a, s_1 - b, c, s_1 - d, s_1 - e, f) \\ (a, s_3 - b, s_3 - c, d, s_3 - e, s_3 - f) & \quad (s_2 - a, b, s_2 - c, s_2 - d, e, s_2 - f) \\ (s_2 - d, s_1 - e, s_3 - f, s_2 - a, s_1 - b, s_3 - c) & \quad (s_1 - d, s_3 - e, s_2 - f, s_1 - a, s_3 - b, s_2 - c) \end{aligned}$$

where

$$(214) \quad s_1 = \frac{a + b + d + e}{2}, \quad s_2 = \frac{a + c + d + f}{2}, \quad s_3 = \frac{b + c + e + f}{2}.$$

Remark 32. The Regge symmetry of a tetrahedron can be understood as a consequence of an earlier discovery of a surprising symmetry of Wigner $6j$ symbol for $SU(2)$ originally due to Regge [Reg59]. The paper due to Boalch provides a comprehensive and modern treatment of the Regge symmetry for the $6j$ symbol for $SU(2)$ by viewing it as a consequence of a symmetry of multiplicity-free $6j$ -symbols for $SU(3)$ [Boa07]. That the Regge symmetry of a tetrahedron emerges from the Regge symmetry of the associated $6j$ symbols can be viewed as a consequence of the semiclassical limit of the latter [PR68; Rob99]. It has also been shown that the Regge symmetry of Euclidean tetrahedra is actually a scissors-congruence preserving both the volume and Dehn invariants [AI19]. At the moment, it remains unclear if analogous symmetries for the generalized Wigner $6j$ symbol exist for $n \geq 3$ (although there are conjectured symmetries for the $6j$ symbols for $U_q(\mathfrak{sl}_n)$ which would follow from the eigenvalue hypothesis [MS18; LS23]). It could also be possible for additional symmetries to be present in the tetrahedral Horn problem for $n \geq 3$ which are *not* present in the $6j$ symbol for $U(n)$.

APPENDIX B. RESULTS FOR THE ORIGINAL HORN PROBLEM sec: new_ineq_horn_orig

The purpose of this section is to describe how our methods can also be used to prove the following result which pertains to the original Horn problem. thm: tri_iff

Theorem 33. *Let $(a, b, c) \in (\mathbb{R}_+^n)^3$ be a triple satisfying the trace conditions. Then $(a, b, c) \in \text{Horn}^+(n)$ if and only if for all pairs of positive weights $(\alpha, \beta) \in (\Lambda_{++}^*)^2$ the following inequality is satisfied*

{eq: tri_iff}

$$(215) \quad \varphi_\alpha(a) \varphi_\beta(b) \leq \sum_{c_{\alpha\beta}^\gamma > 0} \varphi_\gamma(c).$$

Furthermore, if these inequalities are satisfied for all pairs $(\alpha, \beta) \in (\Lambda_{++}^)^2$ with $|\alpha| + |\beta| = k$, then*

{eq: tri_dist_bound}

$$(216) \quad \frac{D(a, b; c)}{\text{Tr}[c]} \leq O\left(n \sqrt{\frac{\log k}{k}}\right).$$

where $D(a, b; c) := \min_{(A, B) \in \mathcal{O}_a \times \mathcal{O}_b} \|\text{eig}(A + B) - c\|_1$.

The proof of this theorem is divided into two parts. First we prove the necessity of Eq. (215) in Appendix B.1. Then afterwards we prove the distance bound in Eq. (216) which in turn implies the sufficiency of Eq. (215) upon taking the limit as $k \rightarrow \infty$. sec:tri_necess

B.1. Necessary conditions. The following proposition follows directly from the binomial theorem (??) and pertains to sums of the quantity $\varphi_\gamma^{\alpha\beta}(X, Y)$. prop:disummation

Proposition 34. *Let X, Y be $n \times n$ matrices. Then the following identities hold:*

$$(217) \quad \forall \alpha, \beta : \quad \sum_{\gamma} \varphi_\gamma^{\alpha\beta}(X, Y) = \varphi_\alpha(X) \varphi_\beta(Y),$$

$$(218) \quad \forall \gamma : \quad \sum_{\alpha, \beta} \varphi_\gamma^{\alpha\beta}(X, Y) = \varphi_\gamma(X + Y).$$

Proof. The first identity involving a summation over α and β is simply the result of taking the trace of both sides of ??. For the second identity, involving a summation over γ , follows from the resolution of the identity

$$(219) \quad \sum_{\gamma \vdash |\alpha| + |\beta|} P_{\alpha\beta}^\gamma = \text{id}_{V_\alpha^n \otimes V_\beta^n}:$$

$$\varphi_\alpha(X) \varphi_\beta(Y) = \text{Tr}[\Phi_\alpha(X) \otimes \Phi_\beta(Y)] = \sum_{\gamma} \text{Tr}[P_{\alpha\beta}^\gamma(\Phi_\alpha(X) \otimes \Phi_\beta(Y))] = \sum_{\gamma} \varphi_\gamma^{\alpha\beta}(X, Y),$$

where the last equality is just the definition of $\varphi_\gamma^{\alpha\beta}(X, Y)$. Once again, the condition that $|\gamma| = |\alpha| + |\beta|$ is implicitly present. $\square$

Remark 35. One might consider what happens to the operator $\Phi_\gamma^{\alpha\beta}(X, Y)$ when one sums over γ instead of (α, β) . However, an analogous identity concerning a sum over γ of the operator $\Phi_\gamma^{\alpha\beta}(X, Y)$ does not immediately follow because the tensor product $\Phi_\alpha(X) \otimes \Phi_\beta(Y)$ is not necessarily block diagonalizable into γ -isotypic subspaces in any obvious way. One exception occurs when $X = Y$, in which case,

$$(220) \quad \pi_\alpha(X) \otimes \pi_\beta(X) = \sum_{\gamma} \iota_{\alpha\beta}^\gamma(\pi_\gamma(X) \otimes \text{id}_{C_{\alpha\beta}^\gamma}) \iota_{\alpha\beta}^{\gamma;\dagger},$$

which is the operator version of the character product formula $\chi_\alpha(X) \chi_\beta(X) = \sum_{\gamma} c_{\alpha\beta}^\gamma \chi_\gamma(X)$.

Under the assumption that the matrices X and Y are positive semidefinite, we can conclude that the real number $\varphi_\gamma^{\alpha\beta}(X, Y)$ is also non-negative because $\varphi_\gamma^{\alpha\beta}(X, Y) = \text{Tr}[P_{\alpha\beta}^\gamma(\Phi_\alpha(X) \otimes \Phi_\beta(Y))]$ where $P_{\alpha\beta}^\gamma$ is a projection operator and $\Phi_\alpha(X), \Phi_\beta(Y) \geq 0$ because $X, Y \geq 0$. Moreover, we note that $\varphi_\gamma^{\alpha\beta}(X, Y) = 0$ whenever the Littlewood-Richardson coefficient $c_{\alpha\beta}^\gamma$ vanishes. These observations imply the following necessary condition for compatibility of eigenvalues for the original Horn problem.

prop:coupling_condition

Proposition 36. *Let (a, b, c) be a triple of non-negative eigenvalues. Then there exists positive semidefinite $n \times n$ matrices (A, B, C) satisfying $A + B = C$ with eigenvalues (a, b, c) only if for every triple of Young diagrams (α, β, γ) there exists a non-negative $\Gamma_{\alpha\beta}^\gamma \geq 0$ such that $\Gamma_{\alpha\beta}^\gamma = 0$ whenever $c_{\alpha\beta}^\gamma = 0$ and*

(221)

$$\forall \alpha, \beta : \sum_{\gamma} \Gamma_{\alpha\beta}^\gamma = \varphi_\alpha(a) \varphi_\beta(b), \quad \text{and} \quad \forall \gamma : \sum_{\alpha, \beta} \Gamma_{\alpha\beta}^\gamma = \varphi_\gamma(c).$$

Proof. The claim follows directly from Proposition 34 after identifying $\Gamma_{\alpha\beta}^\gamma = \varphi_\gamma^{\alpha\beta}(A, B) \geq 0$. □

cor:row_col_horn_ineqs

Corollary 37. *Let (a, b, c) be the eigenvalues of positive semidefinite $n \times n$ matrices (A, B, C) satisfying $A + B = C$. Then,*

(222)

$$\forall \alpha, \beta : \varphi_\alpha(a) \varphi_\beta(b) \leq \sum_{\substack{\gamma \\ c_{\alpha\beta}^\gamma > 0}} \varphi_\gamma(c), \quad \text{and} \quad \forall \gamma : \varphi_\gamma(c) \leq \sum_{\substack{\alpha, \beta \\ c_{\alpha\beta}^\gamma > 0}} \varphi_\alpha(a) \varphi_\beta(b).$$

Proof. The claim follows from Proposition 36 and noting that whenever $\Gamma_{\alpha\beta}^\gamma > 0$ (only when $c_{\alpha\beta}^\gamma > 0$) we have the bound $\Gamma_{\alpha\beta}^\gamma \leq \min\{\varphi_\alpha(a) \varphi_\beta(b), \varphi_\gamma(c)\}$ for all triples (α, β, γ) . □

Remark 38. Using Proposition 36, it is possible to prove even stronger inequalities than those appearing in Corollary 37 by examining the set of triples of Young diagrams (α, β, γ) where $c_{\alpha\beta}^\gamma \neq 0$. We illustrate this point with an example. Let $k = |\alpha| + |\beta| = |\gamma| = 4$, then the following is the table of values for $c_{\alpha\beta}^\gamma$.

$c_{\alpha\beta}^\gamma$	α	β	γ	α	β	γ	α	β	γ	α	β	γ	α	β	γ	α	β	γ
	1	0	0	0	0	0	1	0	0	1	0	0	0	1	0	0	0	0
	0	1	0	0	0	0	1	1	0	1	1	1	0	1	1	0	0	0
	0	0	1	0	0	0	0	1	0	1	0	0	1	0	1	0	0	0
	0	0	0	1	0	0	0	1	1	0	1	1	1	0	1	1	0	0
	0	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	0	1

Using this table, one sees the inequalities in Corollary 37 are associated to single columns or single rows, e.g., along the $(\alpha, \beta) = (\alpha, \beta)$, we obtain

(223)

$$\varphi_\alpha(X) \varphi_\beta(Y) \leq \varphi_{\alpha\beta}(X + Y) + \varphi_{\beta\alpha}(X + Y).$$

However, using Proposition 36, a stronger inequality is obtained by including all columns covered by the rows associated to the right-hand side of the above

inequality, e.g.,

(224)

$$\varphi_{\boxplus}(X) + \varphi_{\boxminus}(X) + \varphi_{\boxtimes}(X)\varphi_{\boxplus}(Y) + \varphi_{\boxtimes}(X)\varphi_{\boxminus}(Y) + \varphi_{\boxplus}(Y) + \varphi_{\boxminus}(Y) \leq \varphi_{\boxplus}(X+Y) + \varphi_{\boxminus}(X+Y).$$

In general, Proposition 36 describes a linear system of inequality constraints in the variables $\Gamma_{\alpha\beta}^\gamma, \varphi_\alpha(a)\varphi_\beta(b)$, and $\varphi_\gamma(c)$ which, in principle, be solved for $\varphi_\alpha(a)\varphi_\beta(b)$ and $\varphi_\gamma(c)$ by standard linear quantifier elimination techniques such as Fourier-Motzkin elimination [DE+72]. While these additional inequalities are certainly interesting to explore, Theorem 33 establishes that for the purposes of resolving the original Horn problem, it is sufficient to restrict our attention to inequalities of the form Corollary 37.

`sec:dist_bound_for_orig_horn`

B.2. Distance bound & sufficiency. Here we prove the distance bound stated in Theorem 33. To begin we fix an *arbitrary* pair (α, β) with total size $|\alpha| + |\beta| = k$ and derive multiple bounds on the product $\varphi_\alpha(a)\varphi_\beta(b)$. Afterwards, we will sum over all such pairs and then obtain the claimed bound on $D(a, b; c)$.

First we bound $\varphi_\alpha(a)\varphi_\beta(b)$ in two ways and then apply the eigenvalue separation lemma (specifically Eq. (82)). The first bound on $\varphi_\alpha(a)\varphi_\beta(b)$ is obtained from relaxing Eq. (215) by considering a Young diagram γ^* of size k which maximizes $\varphi_{\gamma^*}(c)$ among those γ such that $c_{\alpha\beta}^\gamma > 0$ for the given (a, b, c) . As there are at most $(k+1)^{n-1}$ Young diagrams of size k , we conclude

`{eq:first_ab_bound}`

$$(225) \quad \varphi_\alpha(a)\varphi_\beta(b) \leq (k+1)^{n-1} \varphi_{\gamma^*}(c).$$

The second bound on $\varphi_\alpha(a)\varphi_\beta(b)$ is obtained by considering ?? which states that for any γ such that $c_{\alpha\beta}^\gamma > 0$,

$$(226) \quad \varphi_\alpha(a)\varphi_\beta(b) = \frac{\dim V_\alpha^n \otimes V_\beta^n}{c_{\alpha\beta}^\gamma \dim V_\gamma^n} \mathbb{E}_{(X,Y) \sim \mathcal{O}_x \times \mathcal{O}_y} \varphi_\gamma^{\alpha\beta}(X, Y).$$

To relax this into a usable bound we (i) bound the dimensions by $\frac{\dim V_\alpha^n \otimes V_\beta^n}{c_{\alpha\beta}^\gamma \dim V_\gamma^n} \leq \dim(V_\alpha^n \otimes V_\beta^n) \leq (k+1)^{2\binom{n}{2}}$, then (ii) bound the average over $X \in \mathcal{O}_x$ and $Y \in \mathcal{O}_y$ with a maximum, then (iii) apply Corollary 37 to get $\varphi_\gamma^{\alpha\beta}(X, Y) \leq \varphi_\gamma(X+Y)$ and then finally (iv) specialize to the optimal $\gamma \mapsto \gamma^*$ to produce

`{eq:second_ab_bound}`

$$(227) \quad \varphi_\alpha(a)\varphi_\beta(b) \leq (k+1)^{2\binom{n}{2}} \max_{(X,Y) \in \mathcal{O}_a \times \mathcal{O}_b} \varphi_{\gamma^*}(X+Y).$$

Then we combine the two bounds on $\varphi_\alpha(a)\varphi_\beta(b)$ from Eq. (225) and Eq. (227) and apply the eigenvalue separation bound Eq. (82) to obtain

(228)

$$\begin{aligned} \varphi_\alpha(a)\varphi_\beta(b) &\leq \sqrt{(k+1)^{2\binom{n}{2}}(k+1)^{n-1} \max_{\substack{(X,Y) \in \\ \mathcal{O}_a \times \mathcal{O}_b}} \varphi_{\gamma^*}(X+Y)\varphi_{\gamma^*}(c)}, \\ (229) \quad &\leq \sqrt{(k+1)^{2\binom{n}{2}}(k+1)^{n-1}} (k+1)^{\binom{n}{2}} \max_{\substack{(X,Y) \in \\ \mathcal{O}_a \times \mathcal{O}_b}} \frac{\text{BC}(\text{eig}(X+Y), c)^k}{k!}, \end{aligned}$$

`{eq:ab_intermediate_bound}`

$$(230) \quad = (k+1)^{2\binom{n}{2} + \frac{n-1}{2}} \frac{\text{BC}(a, b; c)^k}{k!},$$

where $\text{BC}(a, b; c)$ is defined as the maximum Bhattacharyya coefficient

$$(231) \quad \text{BC}(a, b; c) := \max_{\substack{(X,Y) \in \\ \mathcal{O}_a \times \mathcal{O}_b}} \text{BC}(\text{eig}(X+Y), c).$$

Now we sum both sides of Eq. (230) over *all* pairs (α, β) with total size $|\alpha| + |\beta| = k$ to obtain

$$(232) \quad \frac{\text{Tr}[c]^k}{k!} \leq (k+1)^{2\binom{n}{2} + \frac{n-1}{2} + 2n} \frac{\text{BC}(a, b; c)^k}{k!},$$

where the left-hand side follows from the trace condition $\text{Tr}[c] = \text{Tr}[a] + \text{Tr}[b]$ and

`{eq:binomial_trace}`

$$(233) \quad \sum_{\ell=0}^k \sum_{\substack{\alpha \vdash \ell \\ \beta \vdash k-\ell}} \varphi_\alpha(a)\varphi_\beta(b) = \sum_{\ell=0}^k \frac{\text{Tr}[a]^\ell}{\ell!} \frac{\text{Tr}[b]^{k-\ell}}{(k-\ell)!} = \frac{(\text{Tr}[a] + \text{Tr}[b])^k}{k!},$$

whereas the extra factor of $2n$ in the exponent on the right-hand side is obtained because there are at most $(k+1)^n$ Young diagrams with n rows and size at most k and we are summing over a pair (α, β) with total size $|\alpha| + |\beta| = k$. This establishes a lower bound on $\text{BC}(a, b; c)$ of the form

$$(234) \quad \frac{\text{BC}(a, b; c)}{\text{Tr}[c]} \geq \left(\frac{1}{(k+1)^{\frac{2n^2+3n-1}{2}}} \right)^{\frac{1}{k}}$$

Finally, using [??](#), we substitute the maximum normalized Bhattacharyya coefficient $\frac{\text{BC}(a, b; c)}{\text{Tr}[c]}$ with a minimum normalized distance to obtain Eq. (216) which completes the proof.

APPENDIX C. THE LARGE n LIMIT & AN ENTROPIC INEQUALITY `sec:large_n`

The purpose of this section is to show how to leverage the inverse polynomial bound on $\left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|$ for tetrahedral (a, b, c, d, e, f) provided by Corollary 17 to derive inequalities on the entropies of the eigenvalues. Our strategy will be to prove an upper bound on the $6j$ symbol which depends only on

the dimensions of the underlying irreducible representations and the multiplicity spaces between them. By combining this upper bound with the lower bound provided by Corollary 17 (specifically the n -independent bound in Eq. (187)), we will prove the following.

thm:entropic_ineq

eq:entropic_ineq

Theorem 39. *Let (a, b, c, d, e, f) be tetrahedral eigenvalues. Then*

$$(235) \quad s(c) + s(f) - s(b) - s(e) \geq 0,$$

where

eq:little_s_defn

$$(236) \quad s(x) := - \sum_{i=1}^n x_i \log x_i.$$

We begin with our upper bound on the $6j$ symbol.

lem:sixj_ssa_bound

Lemma 40. *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of Young diagrams. Then*

$$(237) \quad \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_2^2 \leq \left(c_{\alpha, \phi}^{\epsilon} c_{\beta, \delta}^{\phi} c_{\alpha, \beta}^{\gamma} c_{\gamma, \delta}^{\epsilon} \right)^{\frac{1}{2}} \frac{\dim(V_{\gamma}^n) \dim(V_{\phi}^n)}{\dim(V_{\beta}^n) \dim(V_{\epsilon}^n)}.$$

Proof. Here we provide a diagrammatic proof of the claimed inequality. For the sake of clarity, we abbreviate the irreducible representation labels, V_{λ}^n , and omit the multiplicity space labels on the intermediate steps.

(238)

The first equality follows from ???. The inequality follows from an application of the Cauchy-Schwarz inequality with respect to the annotated contraction. The final equality follows from applying Schur's lemma to components of the diagram mapping V_{β}^n to itself on the left and V_{ϵ}^n to itself on the right. Note that the factors of $\dim(V_{\epsilon}^n)^2$ appearing on left-most and right-most sides of the above inequality cancel producing the claimed inequality. $\square$

An immediate application of Lemma 40 is to combine it with Eq. (187). If (a, b, c, d, e, f) are tetrahedral eigenvalues, then by Corollary 17 there exists

eq:combined_bound

a converging sequence of Young diagrams $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$ such that

$$(239) \quad \frac{1}{q(k)} \leq \left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_2^2 \leq \left(c_{\alpha_k, \phi_k}^{\epsilon_k} c_{\beta_k, \delta_k}^{\phi_k} c_{\alpha_k, \beta_k}^{\gamma_k} c_{\gamma_k, \delta_k}^{\epsilon_k} \right)^{\frac{1}{2}} \frac{\dim(V_{\gamma_k}^n) \dim(V_{\phi_k}^n)}{\dim(V_{\beta_k}^n) \dim(V_{\epsilon_k}^n)}.$$

where $q(k)$ is some polynomial in k independent of the dimension n (see Eq. (187)). Remarkably, the Littlewood-Richardson coefficients $c_{\alpha_k, \phi_k}^{\epsilon_k}, c_{\beta_k, \delta_k}^{\phi_k}, c_{\alpha_k, \beta_k}^{\gamma_k}$,

and $c_{\gamma_k, \delta_k}^{\epsilon_k}$ are also *independent* of the dimension parameter n defining the group $U(n)$ (assuming n is at least as large as the maximum depth of any of the involved Young diagrams. This observation provides us with an opportunity to optimize the above bound by varying the value of n .

In particular, it will be useful to better understand how the dimension, $\dim(V_\lambda^n)$, of the irreducible representation V_λ^n of $U(n)$ grows as n becomes much larger than the number of rows of λ . In general, for fixed λ , the function $n \mapsto \dim(V_\lambda^n)$ can grow at an exponential rate with increasing n . One can find estimates for $\dim(V_\lambda^n)$ in the regime where n is large compared to $|\lambda|$ and also the last row $\lambda_{\min}$ is large compared to the depth of λ in paper by Itzykson and Nauenberg [IN66, Section III]. Here we use an elegant presentation of this scaling, which relates the asymptotics of $\dim(V_\lambda^n)$ to the value of $\dim(W_\lambda)$ (see [Har23, Lemma 2] or [CHW07, Equations (25, 26)]).

Proposition 41. *Let λ be a Young diagram with depth at most n . Then*

$$(240) \quad \frac{\dim(V_\lambda^n)}{n^{|\lambda|}} = \frac{\dim(W_\lambda)}{|\lambda|!} \prod_{(i,j) \in \lambda} \left(1 + \frac{j-i}{n}\right).$$

Proof. The statement follows from comparing the the Weyl dimension formula for $\dim(V_\lambda^n)$ given in Eq. (241) with the Frobenius dimension formula for $\dim(W_\lambda)$ given in [GW00, Corollaries 9.1.5]:

$$(241) \quad \dim(W_\lambda) = \frac{|\lambda|! \Delta(\lambda + \rho)}{\prod_{i=1}^n (\lambda + \rho)_i!}.$$

That this formula for $\dim(W_\lambda)$ does not depend on n for $n \geq \ell(\lambda)$ and coincides with the Hook-length formula given earlier is proved by [GW00, Corollary 9.1.6]. For additional details, see also [Har23, Appendix A]. $\square$

From Proposition 41 we obtain the following corollary.

Corollary 42. *Let $\gamma, \phi, \beta, \epsilon$ be a quadruple of Young diagrams such that*

$$(242) \quad |\gamma| + |\phi| = |\beta| + |\epsilon|.$$

Then

$$(243) \quad \lim_{n \rightarrow \infty} \frac{\dim(V_\gamma^n) \dim(V_\phi^n)}{\dim(V_\beta^n) \dim(V_\epsilon^n)} = \frac{|\beta|! |\epsilon|! \dim(W_\gamma) \dim(W_\phi)}{|\gamma|! |\phi|! \dim(W_\beta) \dim(W_\epsilon)} = \frac{H_\beta H_\epsilon}{H_\gamma H_\phi}$$

Proof. From Proposition 41 one evidently has for any Young diagram λ the limit

$$(244) \quad \lim_{n \rightarrow \infty} \frac{\dim(V_\lambda^n)}{n^{|\lambda|}} = \frac{\dim(W_\lambda)}{|\lambda|!} = \frac{1}{H_\lambda}$$

where H_λ is the product of the hook lengths in λ as defined by Eq. (42). The claim then follows because the missing prefactor, $\frac{n^{|\beta|} n^{|\epsilon|}}{n^{|\gamma|} n^{|\phi|}}$, is equal to one by Eq. (242). $\square$

We are now in a position to prove the main result of this section.

Proof of Theorem 39. As (a, b, c, d, e, f) are tetrahedral, the inverse polynomial bound on $\left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|$ for tetrahedral (a, b, c, d, e, f) provided by Corollary 17 implies it must at least be non-zero and thus Eq. (242) holds. Therefore, by Corollary 42 and the discussion surrounding the bound in Eq. (239) we conclude that along the convergent sequence of Young diagrams,

$$\text{\textcolor{blue}{(eq:hook_ratio_limit)}} \quad (245) \quad \lim_{k \rightarrow \infty} \frac{1}{k} \log \left(\frac{H_{\beta_k} H_{\epsilon_k}}{H_{\gamma_k} H_{\phi_k}} \right) \geq 0$$

provided the limit. To prove the limit exists, an application of Stirling's approximation, $\log n! \approx n \log n - n$ [Rob55], to the approximation formula $H_\lambda \approx \prod_{i=1}^n \lambda_i!$ produces for any constant η the formula

$$(246) \quad \frac{1}{\eta} \log H_\lambda \approx (\log \eta - 1)|\lambda| + \sum_{i=1}^n \frac{\lambda_i}{\eta} \log \frac{\lambda_i}{\eta}$$

Because $|\gamma| + |\phi| = |\beta| + |\epsilon|$ (Eq. (242)), the $(\log \eta - 1)$ terms will cancel in Eq. (245). Then using the fact that that $(\beta_k, \gamma_k, \epsilon_k, \phi_k)$ converge in proportion to the eigenvalues (b, c, e, f) (with proportionality constant $\eta = \frac{k}{\text{Tr}[e]}$), we conclude that Eq. (245) has a limit and moreover it equals the left-hand side of Eq. (235). $\square$